

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12385318
B2
August 12, 2025
Fink; Nicholas et al.

Electronic door system

Abstract

An electronic door system includes a door frame, a door hingedly coupled to the door frame, an electrically-controllable component coupled to or disposed within the door, a door wiring harness, and a cover. The door defines a channel extending along at least a portion of a peripheral edge thereof. The door wiring harness includes a wire extending along and within the channel. The wire is coupled to the electrically-controllable component. The cover extends along the peripheral edge and covers the channel.

Inventors: Fink; Nicholas (Maumee, OH), Russell; May (Maumee, OH), Finley; Brett (Maumee, OH), Weil; Jacob (Maumee, OH), Schneider; Daniel (Maumee, OH), Butcher; Keith (Maumee, OH), Harvey; Logan (Maumee, OH), Knoll; Brandon (Maumee, OH), Waite; Andrew (Maumee, OH), Kowalski; Ben (Maumee, OH), Starling; Ryan (Maumee, OH), Wang; Frances (Maumee, OH), Gonzalez; Joseph (Maumee, OH), Hezlep; David (Maumee, OH), Orchosky; Steven (Maumee, OH), Saegusa; Hidekazu (Maumee, OH), Smith; Videl (Maumee, OH), Klein; Bill (Maumee, OH), Allmandinger; Adam (Maumee, OH), Quirin; Tim (Maumee, OH), Klein; Jon (Maumee, OH), Burleson; John (Maumee, OH), Anderson; Kevin (Maumee, OH), Frackelton; Brian (Maumee, OH), Whitaker; Todd (Maumee, OH), Polly; George (Maumee, OH), Chang; Arlber (Maumee, OH), Fielding; Derek (Maumee, OH)

Applicant: Therma-Tru Corporation (Maumee, OH)

Family ID: 89381576

Assignee: Therma-Tru Corporation (Maumee, OH)

Appl. No.: 18/579838

Filed (or PCT Filed): June 30, 2023

PCT No.:

PCT/US2023/026767

PCT Pub. No.:

WO2024/006540

PCT Pub. Date:

January 04, 2024

Prior Publication Data

Document Identifier

Publication Date

US 20250116154 A1

Apr. 10, 2025

Related U.S. Application Data

us-provisional-application US 63430280 20221205

us-provisional-application US 63357709 20220701

Publication Classification

Int. Cl.: E06B7/28 (20060101); E05F15/71 (20150101); E05F15/73 (20150101)

U.S. Cl.:

CPC E06B7/28 (20130101); E05F15/71 (20150115); E05F15/73 (20150115); E05Y2900/132 (20130101)

Field of Classification Search

CPC: E06B (7/28); E06B (9/24); E06B (1/528); E06B (1/52); E06B (3/7001); E05F (15/71); E05F (15/73); E05Y (2900/132)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3793500	12/1973	Gerber	N/A	N/A
4332429	12/1981	Frick et al.	N/A	N/A
5063711	12/1990	Huelin	49/504	E06B 1/18
5398175	12/1994	Pea	N/A	N/A
5541585	12/1995	Duhame et al.	N/A	N/A
5586895	12/1995	Zehrung	N/A	N/A
6049287	12/1999	Yulkowski	N/A	N/A
6058635	12/1999	Morris	52/28	G09F 13/04
6064316	12/1999	Glick et al.	N/A	N/A
7125136	12/2005	Dedic	362/152	G08B 7/062
8539717	12/2012	Speyer et al.	N/A	N/A
8772970	12/2013	Lambrou	N/A	N/A
9265115	12/2015	Dunn	N/A	F21S 9/022
9644399	12/2016	Johnson et al.	N/A	N/A
9683391	12/2016	Johnson et al.	N/A	N/A

9685015	12/2016	Johnson	N/A	N/A
9685017	12/2016	Johnson	N/A	N/A
9691198	12/2016	Cheng et al.	N/A	N/A
9691203	12/2016	Shen	N/A	N/A
9702168	12/2016	Jadallah et al.	N/A	N/A
9704320	12/2016	Johnson et al.	N/A	N/A
9727328	12/2016	Johnson	N/A	N/A
9761073	12/2016	Yu et al.	N/A	N/A
9761074	12/2016	Cheng et al.	N/A	N/A
9767632	12/2016	Johnson	N/A	N/A
9916746	12/2017	Johnson et al.	N/A	N/A
10017963	12/2017	Johnson et al.	N/A	N/A
10304273	12/2018	Johnson et al.	N/A	N/A
10328900	12/2018	Yakovenko et al.	N/A	N/A
10361880	12/2018	Marcinkowski et al.	N/A	N/A
10388094	12/2018	Johnson	N/A	N/A
10443266	12/2018	Johnson et al.	N/A	N/A
10445999	12/2018	Johnson et al.	N/A	N/A
10616721	12/2019	Theurer et al.	N/A	N/A
10691953	12/2019	Johnson et al.	N/A	N/A
10927591	12/2020	Brychell	N/A	F21V 33/006
11268300	12/2021	Langenberg et al.	N/A	N/A
11739583	12/2022	Sorice et al.	N/A	N/A
2004/0085205	12/2003	Yeh	N/A	N/A
2005/0068629	12/2004	Fernando et al.	N/A	N/A
2006/0007005	12/2005	Yui et al.	N/A	N/A
2006/0010793	12/2005	Martino	N/A	N/A
2006/0156361	12/2005	Wang et al.	N/A	N/A
2006/0164205	12/2005	Buckingham et al.	N/A	N/A
2008/0013303	12/2007	Guarino	N/A	N/A
2009/0313790	12/2008	Schau	N/A	N/A
2010/0283579	12/2009	Kraus et al.	N/A	N/A
2012/0280783	12/2011	Gerhardt et al.	N/A	N/A
2013/0186001	12/2012	Cui et al.	N/A	N/A
2015/0027178	12/2014	Scalisi	N/A	N/A
2016/0049025	12/2015	Johnson	N/A	N/A
2016/0105650	12/2015	Carter	N/A	N/A
2016/0239001	12/2015	Chin et al.	N/A	N/A
2016/0275781	12/2015	Nold	N/A	N/A
2016/0284181	12/2015	Johnson	N/A	N/A
2016/0319569	12/2015	Johnson et al.	N/A	N/A
2016/0319571	12/2015	Johnson	N/A	N/A
2017/0040827	12/2016	Weber	N/A	N/A
2017/0143136	12/2016	Gierens	N/A	B32B 17/10541
2017/0193724	12/2016	Johnson et al.	N/A	N/A
2017/0228603	12/2016	Johnson	N/A	N/A
2017/0265124	12/2016	Seemann et al.	N/A	N/A
2017/0284129	12/2016	King	N/A	N/A

2018/0073274	12/2017	Johnson et al.	N/A	N/A
2018/0088431	12/2017	Holt et al.	N/A	N/A
2018/0135337	12/2017	Johnson et al.	N/A	N/A
2018/0179786	12/2017	Johnson	N/A	N/A
2018/0189117	12/2017	Shrivastava et al.	N/A	N/A
2018/0268675	12/2017	Johnson et al.	N/A	N/A
2018/0283049	12/2017	Shiner et al.	N/A	N/A
2018/0364662	12/2017	Meganathan et al.	N/A	N/A
2019/0019364	12/2018	Cheng et al.	N/A	N/A
2019/0130686	12/2018	Cheng et al.	N/A	N/A
2019/0130687	12/2018	Johnson	N/A	N/A
2019/0130712	12/2018	Johnson et al.	N/A	N/A
2020/0236509	12/2019	Theurer et al.	N/A	N/A
2020/0250946	12/2019	Johnson et al.	N/A	N/A
2021/0035394	12/2020	Anderson et al.	N/A	N/A
2021/0207419	12/2020	Sorice et al.	N/A	N/A
2021/0207420	12/2020	Sorice et al.	N/A	N/A
2021/0207421	12/2020	Sorice et al.	N/A	N/A
2022/0108228	12/2021	Salter et al.	N/A	N/A
2022/0170623	12/2021	Hutton	N/A	F21V 29/70
2022/0263297	12/2021	Bodurka	N/A	E06B 7/28

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103223896	12/2012	CN	N/A
106711688	12/2016	CN	N/A
206283038	12/2016	CN	N/A
206283075	12/2016	CN	N/A
206865208	12/2017	CN	N/A
10-102907	12/1997	JP	N/A
1020020088996	12/2001	KR	N/A
101632944	12/2015	KR	N/A
WO-2007/082959	12/2006	WO	N/A
WO-2019/102502	12/2018	WO	N/A

OTHER PUBLICATIONS

Final Office Action on U.S. Appl. No. 16/530,957 DTD Jul. 22, 2022, 13 pages. cited by applicant
International Preliminary Report on Patentability on PCT PCT/US2019/045010 dated Feb. 18, 2021, 7 pages. cited by applicant

International Search Report and Written Opinion dated Oct. 20, 2023, from PCT/US2023/026767, 21 pages. cited by applicant

International Search Report and Written Opinion received for PCT application No. PCT/US2019/045010, 11 pages. cited by applicant

Non-Final Office Action on U.S. Appl. No. 16/530,957 dated Dec. 15, 2021, 12 pages. cited by applicant

Non-Final Office Action on U.S. Appl. No. 16/530,960 DTD Jun. 16, 2020, 20 pages. cited by applicant

Notice of Allowance on U.S. Appl. No. 16/530,957 DTD Oct. 12, 2022, 9 pages. cited by applicant
US Office Action on U.S. Appl. No. 17/072,950 dated Aug. 18, 2021, 15 pages. cited by applicant

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a National Stage filing of International Application No. PCT/US2023/026767, filed Jun. 30, 2023, which claims the benefit and priority of (a) U.S. Provisional Patent Application No. 63/357,709, filed Jul. 1, 2022, and (b) U.S. Provisional Patent Application No. 63/430,280, filed Dec. 5, 2022, both of which are incorporated herein by reference in their entireties.

BACKGROUND

(1) The present disclosure relates generally to systems for monitoring and controlling entry into a structure. More specifically, the present disclosure relates to a system for electronically controlling a door.

SUMMARY

(2) One embodiment relates to an electronic door system. The electronic door system includes a door frame, a door hingedly coupled to the door frame, an electrically-controllable component coupled to or disposed within the door, a door wiring harness, and a cover. The door defines a channel extending along at least a portion of a peripheral edge thereof. The door wiring harness includes a wire extending along and within the channel. The wire is coupled to the electrically-controllable component. The cover extends along the peripheral edge and covers the channel.

(3) Another embodiment relates to an electronic door system. The electronic door system includes a door frame, a frame wiring harness extending through the door frame, a light strip, a light cover, a door hingedly coupled to the door frame, an electrically-controllable component coupled to or disposed within the door, a door wiring harness coupled to the frame wiring harness and the electrically-controllable component, a channel cover, an access panel configured to be positioned in a location spaced from the door and the door frame, a power converter disposed within the access panel, a backup battery disposed within the access panel, a power management controller disposed within the access panel and coupled to the power converter and the backup battery, and an access panel wiring harness coupled to the power management controller and the frame wiring harness. The door frame includes a first side jamb, a second side jamb, a head jamb, and a sill. At least one of the first side jamb or the second side jamb defines a light channel extending substantially along an entire longitudinal length thereof. The light strip is disposed within and extends along the light channel. A portion of the light strip extends into the head jamb and is connected to the frame wiring harness. The light cover encloses the light strip within the light channel. The door defines a wire channel extending along at least a portion of a peripheral edge thereof. The door wiring harness includes one or more wires extending along and within the wire channel. The channel cover extends along the peripheral edge and covers the channel. The power converter is configured to receive AC power from a mains power source and convert the AC power to DC power.

(4) Still another embodiment relates to an electronic door system. The electronic door system includes a door frame, a door hingedly coupled to the door frame, an electronic locking mechanism coupled to the door and configured to secure the door to the frame in a closed position, one or more first sensors, at least one of (a) a second sensor or (b) a capacitive touch doorknob or handle, and a controller. The controller is configured to determine a location of the user relative to the door based on one or more first signals acquired from the one or more first sensors, determine an access intent of the user based on a second signal acquired from the at least one of (a) the second sensor or (b) the capacitive touch doorknob or handle, and make a door access decision to unlock the electronic

locking mechanism at least partially based on the location of the user indicating that the user is outside and not inside of a building within which the electronic door system is installed and the access intent indicating that the user is attempting to enter the building.

(5) This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:
- (2) FIG. 1 is a front view showing an exterior side of a door system, according to an exemplary embodiment.
- (3) FIG. 2 is a rear view showing an interior side of the door system of FIG. 1, according to an exemplary embodiment.
- (4) FIG. 3 is an exploded view of an exterior side of a door of the door system of FIG. 1, according to an exemplary embodiment.
- (5) FIG. 4 is an exploded view of an interior side of the door of FIG. 3, according to an exemplary embodiment.
- (6) FIG. 5 is a front view showing an exterior side of a subframe of the door of FIG. 3, according to an exemplary embodiment.
- (7) FIG. 6 is a rear view showing an interior side of the subframe of FIG. 5, according to an exemplary embodiment.
- (8) FIG. 7 is a rear view showing the interior side of the door system of FIG. 1 with a window of the door system in a viewing state, according to an exemplary embodiment.
- (9) FIG. 8 is another rear view showing the interior side of the door system of FIG. 1 with the window of the door system in a privacy state, according to an exemplary embodiment.
- (10) FIG. 9A is a block diagram of a control system of the door system of FIG. 1, according to an exemplary embodiment.
- (11) FIG. 9B is a block diagram of the control system of FIG. 9A illustrating the transfer of electrical energy throughout the control system, according to an exemplary embodiment.
- (12) FIGS. 10-13 are various views of power transfer assemblies of the door system of FIG. 1, according to various exemplary embodiments.
- (13) FIG. 14 is a block diagram of a controller for use within a module of the control system of FIGS. 9A and 9B, according to an exemplary embodiment.
- (14) FIG. 15 is a top view of a power supply unit of the control system of FIG. 9B, according to an exemplary embodiment.
- (15) FIG. 16 is a block diagram of the power supply unit of FIG. 15, according to an exemplary embodiment.
- (16) FIG. 17 is a top view of a window control unit of the control system of FIG. 9B, according to an exemplary embodiment.
- (17) FIG. 18 is a block diagram of the window control unit of FIG. 17, according to an exemplary embodiment.
- (18) FIG. 19 is a top view of a main control unit of the control system of FIG. 9B, according to an exemplary embodiment.
- (19) FIG. 20 is a block diagram of the main control unit of FIG. 19, according to an exemplary

embodiment.

(20) FIG. 21 is a perspective view of a deadbolt unit of the control system of FIG. 9 and a corresponding strike plate, according to an exemplary embodiment.

(21) FIG. 22 is a block diagram of the deadbolt unit of FIG. 21, according to an exemplary embodiment.

(22) FIG. 23 is a block diagram of a bridge module of the control system of FIG. 9B, according to an exemplary embodiment.

(23) FIG. 24 is a block diagram of a doorbell unit of the control system of FIG. 9B, according to an exemplary embodiment.

(24) FIG. 25 is a block diagram of a mesh network of the control system of FIG. 9B, according to an exemplary embodiment.

(25) FIG. 26 is an exploded perspective view of the door of FIG. 3 showing the main control unit of FIG. 19, according to an exemplary embodiment.

(26) FIG. 27 is a perspective view of a mounting tray of the main control unit of FIG. 19, according to an exemplary embodiment.

(27) FIG. 28 is a perspective view of an insert of the door of FIG. 3 used to mount the main control unit of FIG. 19, according to an exemplary embodiment.

(28) FIG. 29 is a side view of the insert of FIG. 29, according to an exemplary embodiment.

(29) FIG. 30 is a side view of the mounting tray of FIG. 27, according to an exemplary embodiment.

(30) FIG. 31 is an exploded perspective view of the door of FIG. 3 showing the power supply unit of FIG. 15, according to an exemplary embodiment.

(31) FIG. 32 is an exploded perspective view of the door of FIG. 3 showing the window control unit of FIG. 17, according to an exemplary embodiment.

(32) FIG. 33 is a perspective view of an insert of the door of FIG. 3 used to mount the window control unit of FIG. 17, according to an exemplary embodiment.

(33) FIG. 34 is a front perspective view showing the exterior side of the subframe of FIG. 5, according to an exemplary embodiment.

(34) FIG. 35 is a rear perspective view showing the interior side of the subframe of FIG. 5, according to an exemplary embodiment.

(35) FIG. 36 is a bottom section view of the subframe of FIG. 5, according to an exemplary embodiment.

(36) FIG. 37 is a front perspective view showing the exterior side of an upper portion of a lock stile of the subframe of FIG. 5, according to an exemplary embodiment.

(37) FIG. 38 is a front perspective view showing the exterior side of a lower portion of the lock stile of FIG. 37, according to an exemplary embodiment.

(38) FIG. 39 is a block diagram of an interactive unit of the control system of FIG. 9B, according to an exemplary embodiment.

(39) FIG. 40 is a front view showing an exterior side of the interactive unit of FIG. 39, according to an exemplary embodiment.

(40) FIG. 41 is a front partial section view showing the exterior side of the interactive unit of FIG. 39, according to an exemplary embodiment.

(41) FIG. 42 is a rear view showing an interior side of the interactive unit of FIG. 39, according to an exemplary embodiment.

(42) FIG. 43 is a rear partial section view showing the interior side of the interactive unit of FIG. 39, according to an exemplary embodiment.

(43) FIGS. 44 and 45 are exploded perspective views of the interactive unit of FIG. 39, according to an exemplary embodiment.

(44) FIGS. 46 and 47 are exploded side views of the interactive unit of FIG. 39, according to an exemplary embodiment.

(45) FIG. 48 is an exploded view of a lock control interface of the interactive unit of FIG. 39, according to an exemplary embodiment.

(46) FIG. 49 is partial side section view of the lock control interface of FIG. 48, according to an exemplary embodiment.

(47) FIGS. 50-52 are top views of a magnet arrangement of the lock control interface of FIG. 48 in various configurations, according to an exemplary embodiment.

(48) FIGS. 53-55 are rear views of a slider actuator of the lock control interface of FIG. 48 in various configurations, according to an exemplary embodiment.

(49) FIG. 56 is a bottom perspective showing the interior side of a reduced configuration of the interactive unit of FIG. 39, according to an exemplary embodiment.

(50) FIG. 57 is an exploded view showing the interior side of the interactive unit of FIG. 39, according to an exemplary embodiment.

(51) FIG. 58 is a top perspective view showing the interior side of the reduced configuration of the interactive unit shown in FIG. 39, according to an exemplary embodiment.

(52) FIG. 59 is a front view showing an exterior side of an interactive unit of the control system of FIG. 9B, according to another exemplary embodiment.

(53) FIG. 60 is a rear view showing an interior side of the interactive unit of FIG. 59, according to an exemplary embodiment.

(54) FIG. 61 is a screenshot of a graphical user interface provided by the control system of FIGS. 9A and 9B, according to an exemplary embodiment.

(55) FIG. 62 is another front view showing the exterior side of the door system of FIG. 1.

(56) FIG. 63 is another rear view showing the interior side of the door system of FIG. 1.

(57) FIG. 64 is a front view showing an exterior side of a door of the door system of FIG. 1, according to another exemplary embodiment.

(58) FIG. 65 is a top, right perspective view of the door of FIG. 64, according to an exemplary embodiment.

(59) FIG. 66 is a right side view of the door of FIG. 64, according to an exemplary embodiment.

(60) FIG. 67 is a top section view of the door of FIG. 64, according to an exemplary embodiment.

(61) FIG. 68 is a perspective view of a cover of the door of FIG. 64, according to an exemplary embodiment.

(62) FIG. 69 is a perspective view of an insert of the door of FIG. 64, according to an exemplary embodiment.

(63) FIG. 70 is a perspective view of an insert of the door of FIG. 64, according to another exemplary embodiment.

(64) FIG. 71 is a perspective view of a cover of the door of FIG. 64, according to another exemplary embodiment.

(65) FIG. 72 is a front view of the cover of FIG. 71 installed in the door of FIG. 64, according to an exemplary embodiment.

(66) FIG. 73 is a front view of a raceway of the door of FIG. 64, according to an exemplary embodiment.

(67) FIG. 74 is a perspective view of a cover of the raceway of FIG. 73, according to an exemplary embodiment.

(68) FIG. 75 is a perspective view of a body of the raceway of FIG. 73, according to an exemplary embodiment.

(69) FIG. 76 is a perspective view of a tubular cover of the door of FIG. 64, according to an exemplary embodiment.

(70) FIG. 77 is a front view of the tubular cover of FIG. 76 installed in the door of FIG. 64, according to an exemplary embodiment.

(71) FIG. 78 is a perspective view of the door of FIG. 64, according to an exemplary embodiment.

(72) FIG. 79 is a front section view of a mounting arrangement for a hinge of the door of FIG. 64,

according to an exemplary embodiment.

(73) FIG. **80** is a perspective view of a hinge insert of the mounting arrangement of FIG. **79**, according to an exemplary embodiment.

(74) FIG. **81** is a perspective view of a hinge insert of the mounting arrangement of FIG. **79**, according to another exemplary embodiment.

(75) FIG. **82** is an exploded view of a door assembly of the door of FIG. **64**, according to an exemplary embodiment.

(76) FIG. **83** is a detailed view of a support of a top rail of the door assembly of FIG. **82**, according to an exemplary embodiment.

(77) FIG. **84** is a detailed view of an electrical connector of the door assembly of FIG. **82**, according to an exemplary embodiment.

(78) FIGS. **85-88** are various detailed views of door skins of the door assembly of FIG. **82**, according to an exemplary embodiment.

(79) FIG. **89** is a cross-sectional view of the door assembly of FIG. **82**, according to an exemplary embodiment.

(80) FIGS. **90** and **91** are detailed views of the door assembly of FIG. **89**, according to an exemplary embodiment.

(81) FIG. **92** is a front view showing an exterior side of a door system, according to another exemplary embodiment.

(82) FIGS. **93** and **94** are various views of a door of the door system of FIG. **92**, according to an exemplary embodiment.

(83) FIG. **95** is a detailed view of a light strip of the door system of FIG. **92**, according to an exemplary embodiment.

(84) FIG. **96** is a detailed view of a light strip of the door system of FIG. **92**, according to another exemplary embodiment.

(85) FIG. **97** is a front view showing the exterior side of the door system of FIG. **92**, according to another exemplary embodiment.

(86) FIG. **98** is a detailed view of a light strip of the door system of FIG. **97**, according to an exemplary embodiment.

(87) FIG. **99** is a front view showing the exterior side of the door system of FIG. **92**, according to another exemplary embodiment.

(88) FIG. **100** is schematic view of a floor plan of a residence including the door system of FIGS. **1**, **92**, **97**, and/or **99**, according to an exemplary embodiment.

DETAILED DESCRIPTION

(89) Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

(90) Referring generally to the figures, an electronic door system includes a door that is hingedly coupled to a door frame. In some embodiments, the door includes a series of modules that each facilitate certain functionalities within the door. The door may include a main control module that handles general processing within the door system. The door may include a power supply module that controls the transfer, conversion, and distribution of electrical energy throughout the door. The door may include a window control module that controls the operation of an electrochromic or electrically-operated privacy window. The door may include one or more deadbolt modules that each electronically control the operation of a respective deadbolt. The door may include a bridge module that controls communication with devices outside of the door. Each of these modules may be mounted on a tray that is received within a recess defined by the door. The trays may be mounted in a forward orientation or a reverse orientation to facilitate converting the door between a

right hand configuration and a left hand configuration. The door system may further include a doorbell module positioned outside of the door and configured to control an external doorbell.

(91) In some embodiments, the modules of the door system each include a controller, such that the modules can perform their respective functions independently. The modules may communicate data wirelessly between one another, forming a wireless mesh network. The mesh network can wirelessly accept communication with new modules, such that a module can be easily added or replaced by simply supplying power to the new module.

(92) In some embodiments, the bridge module and the main control module both communicate with a user interface. The user interface may include a camera and/or a touchscreen display that displays image data from the camera. The user interface may be modular such that the display can be added or removed by the end user. When a user removes the display, the image data from the camera may be viewed remotely on a user device.

(93) Overview of Door System

(94) Referring to FIGS. 1 and 2, an electronic door system or electronic door assembly is shown as door system **10**, according to an exemplary embodiment. The door system **10** may facilitate selective access to a structure or building, such as a home, an apartment, a garage, or a commercial building, having one or more walls **12**. While the door system **10** may be discussed herein as being used with a home, it should be understood that the door system **10** may alternatively be used with any other type of structure. By way of example, the door system **10** may prevent access to an interior of the structure by unauthorized users and permit access to the interior of the structure by authorized users (e.g., homeowners, residents, invited guests, etc.). The door system **10** may provide one or more electronic systems that facilitate control over access to the interior of the structure.

(95) The door system **10** includes a stationary portion or door frame **20** coupled to or installed within the wall **12**. The door frame **20** includes a series of frame sections, shown as side jambs **22**, head jamb **24**, and sill **26**. The side jambs **22** extend substantially vertically and are laterally offset from one another. The head jamb **24** extends substantially horizontally between the top end portions of the side jambs **22**. The sill **26** extends substantially horizontally between the bottom end portions of the side jambs **22**. Together, the side jambs **22**, the head jamb **24**, and the sill **26** define a passage, aperture, or opening, shown as doorway **28** therebetween. The doorway **28** may be sized to facilitate the passage of one or more users through the doorway **28**.

(96) The door system **10** further includes a movable portion, body, door slab, or door leaf, shown as door **30**. The door **30** is movably coupled to the door frame **20** by one or more hinges, shown as hinges **32**. Specifically, the hinges **32** are each coupled to a side jamb **22** and to the door **30**. As shown, the hinges **32** pivotally couple the door **30** to the door frame **20** such that the door is pivotable or rotatable about a substantially vertical axis. In other embodiments, the door **30** is otherwise movably coupled to the door frame **20** (e.g., slidably coupled).

(97) The door **30** is selectively repositionable between (a) a blocking position, shut position, or closed position and (b) an unobstructed position or open position. In the closed position, the door **30** extends across the doorway **28**, preventing movement of a user through the doorway **28**. In the open position, the door **30** is moved out of the doorway **28**, permitting movement of the user through the doorway **28**. The door **30** may swing inward (e.g., into the structure) and/or outward (e.g., away from the structure). In some configurations, the door **30** is coupled to a first one of the side jambs **22**. In other configurations, the door **30** is coupled to the other of the side jambs **22**. In some embodiments, the door **30** can be coupled to either of the side jambs **22** according to a preference of the user. Accordingly, the door **30** can be configured as a right hand door or a left hand door.

(98) Referring to FIGS. 1-6, the door **30** includes a frame or door substructure, shown as subframe **40**. The subframe **40** includes a series of frame members or structural members including (a) a first stile or vertical member, shown as hinge stile **42**, (b) a first rail or horizontal member, shown as top

rail **44**, (c) a second stile or vertical member, shown as lock stile **46**, and (d) a second rail or horizontal member, shown as bottom rail **48**. The hinge stile **42**, the top rail **44**, the lock stile **46**, and the bottom rail **48** are fixedly coupled to one another to form the subframe **40**. The lock stile **46** extends substantially vertically and is positioned adjacent a first one of the side jambs **22**. The hinges **32** may be directly coupled to the hinge stile **42**. The lock stile **46** extends substantially vertically and is positioned adjacent the other side jamb **22**. The top rail **44** extends substantially horizontally between the hinge stile **42** and the lock stile **46** and is positioned adjacent the head jamb **24**. The bottom rail **48** extends substantially horizontally between the hinge stile **42** and the lock stile **46** and is positioned adjacent the sill **26**. In some embodiments, the subframe **40** is made from wood, such as engineered wood or hardwood. In other embodiments, the subframe **40** is made from another material, such as metal, plastic, or a composite material (e.g., fiberglass).

(99) The door further includes a first panel, front panel, or door skin, shown as exterior skin **50**, and a second panel, rear panel, or door skin, shown as interior skin **52**. The exterior skin **50** extends along a first side (e.g., a front side or exterior side) of the subframe **40**. The interior skin **52** extends along a second side (e.g., a rear side or interior side) of the subframe **40**. The exterior skin **50** and/or the interior skin **52** may be fixedly coupled to the subframe **40**. In some embodiments, the exterior skin **50** and the interior skin **52** may be removably coupled to the subframe **40** (e.g., to facilitate access to the interior of the door **30** for maintenance or assembly). In some embodiments, the exterior skin **50** and the interior skin **52** are made from a composite material, such as fiberglass. In other embodiments, the exterior skin **50** and the interior skin **52** are made from another material, such as plastic, wood, or metal. While described herein as being a door assembly including the subframe **40**, the exterior skin **50**, and the interior skin **52**, in some embodiments, the door **30** is configured as a solid core door.

(100) The door **30** further includes one or more panes of transparent material (e.g., glass, polycarbonate, etc.) or privacy windows, shown as window **60**. As shown, the window **60** is positioned adjacent the center of the door **30**, such that the window **60** is surrounded on the top, bottom, left, and right sides by the subframe **40**. The window **60** provides a line of sight through the door **30**. The window **60** may be held in place by (e.g., fixedly coupled to) the subframe **40**, the exterior skin **50**, and/or the interior skin **52**. In some embodiments, the door **30** includes a single, uninterrupted window **60** (e.g., as shown in FIG. 1). In other embodiments, the door **30** includes multiple windows **60**. In yet other embodiments, the window **60** is omitted.

(101) In some embodiments, the window **60** is operable (e.g., electrically) to vary a visibility level through the window **60**. By way of example, the window **60** may be made from and/or coated with an electrochromic material that changes opacity in response to an applied voltage. In some embodiments, the visibility level of the window **60** is varied in response to a user input, permitting the user to control the visibility level of the window **60**. This variable visibility level may facilitate a clear line of sight through the door **30** when desired by the user and obscuring the line of sight through the door **30** when the user desires additional privacy.

(102) In a first state, transparent state, or viewing state (e.g., as shown in FIG. 7), the window **60** is clear, such that a line of sight is permitted through the window **60**. Specifically, in the viewing state, the window **60** may provide (a) a first line of sight through the window **60** from the exterior side of the door **30** toward the interior side of the door **30** and/or (b) a second line of sight through the window **60** from the interior side of the door **30** toward the exterior side of the door **30**. By way of example, the viewing state may be used when a user on the interior side of the door **30** wishes to permit an individual on the exterior side of the door **30** to see through the door **30**. By way of another example, the viewing state may be used when a user on the interior side of the door **30** wishes to let light through the window **60** to brighten the inside of the home or when the user wishes to see an object on the opposite side of the door.

(103) In a second state, translucent state, opaque state, tinted state, or privacy state (e.g., as shown in FIG. 8), an opacity of the window **60** increases relative to the viewing state such that the line of

sight through the window **60** is at least partially obscured (e.g., the window **60** becomes tinted or otherwise changes opacity). In the privacy state, the window **60** may partially obscure the line of sight such that some light still passes through the window **60** (e.g., the window **60** is translucent or partially tinted). In the privacy state, the window **60** may be completely opaque (e.g., heavily tinted), preventing any line of sight through the window **60**. In some embodiments, the window **60** has multiple privacy states, each with differing levels of opacity. By way of example, the privacy state or privacy states may be utilized when a user wishes to prevent individuals on the exterior of the door **30** from looking through the window **60**, when the user wishes to limit the amount of light passing through the window **60** (e.g., to control the brightness of a room), and/or when the user wishes to prevent an individual on the interior of the door **30** from looking through the window **60** (e.g., to reduce visual distractions from passersby).

(104) In other embodiments, the window **60** is made from a material having a constant opacity (e.g., a non-variable opacity). In yet other embodiments, the window **60** permits a first line of sight through the window **60** from the interior side of the door **30** toward the exterior side of the door **30** and prevents a second line of sight through the window **60** from the exterior side of the door **30** toward the interior side of the door **30**. By way of example, the window **60** may be constructed as a two-way mirror.

(105) Referring to FIGS. **1**, **2**, **7**, and **8**, the door **30** includes an interface assembly, lockset, or handle assembly, shown as handle assembly **70**. The handle assembly **70** serves as an interface through which the user can control opening and closing of the door **30**. The handle assembly **70** includes a pair of interfaces (e.g., handles, knobs, pulls, push plates, etc.), shown as handles **72**, configured to interface with a hand of a user. Specifically, a first handle **72** is positioned on the exterior side of the door **30**, and a second handle **72** is positioned on the interior side of the door. The user may apply a pushing or pulling force onto the handle **72** to move the door **30** toward the open position or the closed position, as desired.

(106) In some embodiments, the handle assembly **70** includes a latch and/or main deadbolt that extends outward from the door **30** to be received within a corresponding pocket or pockets defined by the door frame **20** to hold the door **30** in the closed position and/or locked configuration. In some such embodiments, the latch may be retracted by moving (e.g., rotating) one or both of the handles **72** relative to the rest of the door **30** and/or the main deadbolt may be retracted by turning a deadbolt thumb turn. Once retracted, the door **30** may be moved out of the closed position. In some embodiments, components of the handle assembly **70** operate mechanically and/or electromechanically (e.g., the handle assembly **70** includes an electromechanical locking mechanism such as an electronic main deadbolt). In some embodiments, the handle assembly **70** can be operated by either handle **72**.

(107) Referring to FIGS. **3-6**, the door **30** defines a passage or recess, shown as handle bore **74**. The handle bore **74** extends through the door **30** from an exterior face to an interior face. Specifically, the handle bore **74** is defined by and extends through the exterior skin **50**, the lock stile **46**, and the interior skin **52**. The handle bore **74** may receive the handle assembly **70**, coupling the handle bore **74** to the subframe **40**. In some embodiments, the handle bore **74** is vertically positioned on the door **30** to place the handle assembly **70** at waist height for a user.

(108) Referring to FIGS. **3**, **4**, **9A**, and **9B**, the door system **10** includes an electronic control system, modular control system, or intelligent control system, shown as control system **100**. The control system **100** may provide one or more features or functions that facilitate operation of the door system **10**. The control system **100** may receive one or more inputs (e.g., sensor inputs, user commands, etc.) and provide one or more outputs (e.g., lights, sounds, movement of actuators, etc.). The control system **100** may provide certain features that are not available on other doors, such as remote operation, autonomous operation, and/or improved intrusion resistance, among other features.

(109) Referring to FIGS. **1-4** and **9B**, the control system **100** may be at least partially powered by

an external power source or external power supply, shown as power source **102**. The power source **102** is configured to supply electrical energy to power operation of the door **30** and the electronic components thereof. By way of example, the power source **102** may supply electrical energy from a power grid (e.g., a municipal power grid), generators, solar panels, wind turbines, batteries, capacitors, fuel cells, or other sources. The power source **102** may be positioned inside and/or outside of the structure having the walls **12**.

(110) The control system **100** further includes one or more power converters or power conditioners (e.g., transformers, alternating current (AC) to direct current (DC) rectifiers, DC to AC inverters, etc.), shown as power converter **104**. As shown, the power converter **104** is positioned within the wall **12** and adjacent the doorway **28**. The power converter **104** may condition the electrical energy prior to the electrical energy being transferred into the door **30**. By way of example, the power converter **104** may change the electrical energy between AC and DC, vary a voltage or current of the electrical energy, or otherwise condition or convert the electrical energy. In one example, the power converter **104** receives electrical energy from the power source **102** at 120V AC and converts the electrical energy to 24V DC. In other embodiments, the power converter **104** is omitted, and the electrical energy is delivered directly from the power source **102** (e.g., to the power transfer assembly **106**).

(111) The control system **100** further includes a power transfer device or power transfer unit, shown as power transfer assembly **106**, that is coupled to the door **30** and the door frame **20**. The power transfer assembly **106** is configured to transfer electrical energy from the power converter **104** (or directly from the power source **102**) to the door **30**. In order to facilitate operation of door **30** by a user, the power transfer assembly **106** facilitates movement of the door **30** (e.g., by bending, sliding, rotating, or otherwise moving with the door **30**). In some embodiments, the power converter **104** converts the electrical energy to a format that the power transfer assembly **106** is configured to transfer. By way of example, the power transfer assembly **106** may be rated to transfer electrical energy at 24V DC, and the power converter **104** may convert the electrical energy from the power source **102** to 24V DC prior to the electrical energy reaching the power transfer assembly **106**.

(112) In some embodiments, the power transfer assembly **106** includes an electric power transfer hinge. FIG. **10** illustrates a power transfer hinge **110** of the power transfer assembly **106**, according to an exemplary embodiment. The power transfer hinge **110** includes a pair of hinge bodies, shown as hinge leaves **112**, that are movably (e.g., pivotally) coupled to one another by a pin **114**. One of the hinge leaves **112** may be coupled to a side jamb **22**, and the other hinge leaf **112** may be coupled to the hinge stile **42**. Wires or cables, shown as wires **116**, pass through a passage or raceway defined through the hinge leaves **112**. Beneficially, the power transfer hinge **110** may completely obscure the wires **116** from view (e.g., to prevent tampering and exposure to contaminants). The wires **116** may be electrically coupled to the power converter **104**. In some embodiments, the power transfer hinge **110** is configured to transfer electrical energy at 24V DC. In other embodiments, the power transfer hinge **110** has a different mechanical arrangement.

(113) In some embodiments, the power transfer assembly **106** includes an electric power transfer cable. FIG. **11** illustrates a power transfer cable **120** of the power transfer assembly **106**, according to an exemplary embodiment. The power transfer cable **120** includes a pair of base members, shown as mounts **122**. One of the mounts **122** may be coupled to a side jamb **22**. The other mount **122** may be coupled to hinge stile **42**. A flexible conduit (e.g., a tube, a pipe, a hose, a sleeve, etc.), shown as conduit **124**, has a first end portion coupled to one mount **122** and a second, opposite end portion coupled to the other mount **122**. The conduit **124** defines a passage containing one or more wires or cables, shown as wires **126**. Beneficially, the power transfer cable **120** may completely obscure the wires **126** from view. The wires **126** may be electrically coupled to the power converter **104**. The conduit **124** may be flexible and/or flexibly mounted such that the conduit **124** can move and flex to accommodate motion of the door **30** without disconnecting the wires **126**.

(114) In some embodiments, the power transfer assembly **106** is configured to transfer energy wirelessly. One such example is shown in FIG. **12**. In this embodiment, the power transfer assembly **106** includes a pair of wireless power transfer pads, panels, or coils, shown as power transfer panels **128**. As shown, one of the power transfer panels **128** is coupled to a side jamb **22**, and the other power transfer panel **128** is coupled to the hinge stile **42**. The power transfer panels **128** are positioned such that, when the door **30** is in the closed position, the power transfer panels **128** are facing one another, in close proximity with one another, and substantially aligned with one another. The power transfer panels **128** may be configured to transfer electrical energy wirelessly (e.g., through induction). Accordingly, one of the power transfer panels **128** may be configured to receive AC electrical energy from the power converter **104** (or directly from the power source **102**). Although the power transfer may be more efficient when the door **30** is in the closed position (e.g., due to the arrangement of the power transfer panels **128**), the power transfer panels **128** may still transfer energy between one another even when the door **30** is in the open position.

(115) Referring to FIG. **13**, another embodiment of the power transfer assembly **106** in which the power transfer assembly **106** includes a pair of electrical contacts or electrical connectors, shown as connectors **129**. As shown, one of the connectors **129** is coupled to a side jamb **22**, and the other connector **129** is coupled to the hinge stile **42**. The connectors **129** are positioned such that, when the door **30** is in the closed position, the connectors **129** engage one another to form an electrical connection. The connectors **129** may disconnect from one another when the door **30** is moved to the open position. Each of the connectors **129** may include one or more electrical contacts that each are configured to engage a corresponding electrical contact of the other connector **129**. One of the connectors **129** may be electrically coupled to the power converter **104**. The connectors **129** may be configured to transfer AC or DC electrical energy. In one embodiment, one of the connectors **129** is recessed and the other connector **129** protrudes outward. In some embodiments, at least one of the connectors **129** is spring biased, and/or selectively extendable and/or retractable.

(116) Referring to FIGS. **3** and **4**, the control system **100** includes a series of modular control sections or modules: a first module or power supply module (e.g., a power management unit), shown as power supply unit **130**; a second module or window control module, shown as window control unit **132**; a third module or main control module, shown as main control unit **134**; a fourth module, communication unit, or wireless bridge, shown as bridge module **136**; and one or more fifth modules or lock modules, shown as deadbolt units **138**. Each module may be configured to perform one or more dedicated functions. The modules may communicate between one other (e.g., wirelessly). The deadbolt units **138** may be auxiliary or supplementary to a main deadbolt of the handle assembly **70**.

(117) The power supply unit **130** may control distribution of electrical energy throughout the door **30**. The window control unit **132** may control operation of the window **60** (e.g., changing the configuration of the window **60** between the viewing state and the privacy state). The main control unit **134** may control the overall operation of the door system **10**. The bridge module **136** may facilitate communication between the modules/components of the door **30** and one or more external devices (e.g., a user device, a server, a doorbell module, etc.). The deadbolt units **138** each include a lock member that selectively engages the door frame **20** to prevent movement of the door **30** out of the closed position (e.g., in addition to the main deadbolt of the handle assembly **70**, etc.).

(118) The control system **100** further includes a first user interface, a first operator interface, a first interface module, or a first interactive unit, shown as interactive unit **140**. The interactive unit **140** may act as a user interface to provide information to a user or to receive information (e.g., commands) from a user. The interactive unit **140** may be accessed from the interior side and/or the exterior side of the door **30**. In embodiments where the interactive unit **140** is accessible from both the interior side and the exterior side of the door **30**, the interactive unit **140** may permit utilizing different functions depending upon which side of the interactive unit **140** is being accessed. In some embodiments, the interactive unit **140** is in wired communication with both the main control

unit **134** and the bridge module **136**. The main control unit **134** and the bridge module **136** may include controllers that control operation of the interactive unit **140**.

(119) Referring to FIGS. **3** and **4**, the door **30** defines a series of pockets, recesses, passages, chambers, or receivers each configured to receive one or more of the modules. The recesses permit the modules to be contained within the door **30** while still being accessible (e.g., to facilitate maintenance, upgrades, assembly, etc.).

(120) The door **30** defines a first pocket, recess, passage, chamber, or receiver, shown as power supply recess **150**. Specifically, the power supply recess **150** is defined by the hinge stile **42** and extends laterally into the hinge stile **42**. The power supply recess **150** receives the power supply unit **130**. The power supply recess **150** is sealed by a cover, shown as plate **152**, that is removably coupled to the hinge stile **42**. When assembled, the power supply unit **130** is fully contained within the door **30** by the hinge stile **42** and the plate **152**. The plate **152** may prevent ingress of debris (e.g., water, dust, etc.) into the power supply recess **150**.

(121) The door **30** defines a second pocket, recess, passage, chamber, or receiver, shown as window controller recess **154**. Specifically, the window controller recess **154** is defined by the lock stile **46** and extends laterally into the lock stile **46**. The window controller recess **154** receives the window control unit **132**. The window controller recess **154** is sealed by a cover, shown as plate **156**, that is removably coupled to the lock stile **46**. When assembled, the window control unit **132** is fully contained within the door **30** by the lock stile **46** and the plate **156**. The plate **156** may prevent ingress of debris (e.g., water, dust, etc.) into the window controller recess **154**.

(122) The door **30** defines a third pocket, recess, passage, chamber, or receiver, shown as main controller recess **158**. Specifically, the main controller recess **158** is defined by the lock stile **46** and extends laterally into the lock stile **46**. The main controller recess **158** receives the main control unit **134**. The main controller recess **158** is sealed by a cover, shown as plate **160**, that is removably coupled to the lock stile **46**. When assembled, the main control unit **134** is fully contained within the door **30** by the lock stile **46** and the plate **160**. The plate **160** may prevent ingress of debris (e.g., water, dust, etc.) into the main controller recess **158**.

(123) The door **30** defines a pair of fourth pockets, recesses, passages, chambers, or receivers, shown as deadbolt recesses **162**. Specifically, the deadbolt recesses **162** are defined by the lock stile **46** and extend laterally into the lock stile **46**. In some embodiments, the deadbolt recesses **162** are additionally or alternatively defined by the top rail **44** and/or the bottom rail **48**. The deadbolt recesses **162** each receive one of the deadbolt units **138**. The deadbolt units **138** may each engage the lock stile **46** to prevent ingress of debris (e.g., water, dust, etc.) into the deadbolt recesses **162**. In other embodiments, the door **30** defines only one deadbolt recess **162** and only receives one deadbolt unit **138**. In still other embodiments, the door **30** defines three or more deadbolt recesses **162** and receives three or more deadbolt units **138**.

(124) Referring to FIGS. **5** and **6**, the door **30** defines a pair of passages or recesses, shown as interface bores **164**. The interface bores **164** extend through the door **30** from an exterior face to an interior face. Specifically, the interface bores **164** are defined by and extend through the exterior skin **50**, the lock stile **46**, and the interior skin **52**. The interactive unit **140** extends through the interface bores **164** to couple the interactive unit **140** to the subframe **40**. The interactive unit **140** extends completely through the interface bores **164**, such that the interactive unit **140** is exposed on both the interior side and the exterior side of the door **30**.

(125) Referring to FIGS. **1-4**, the door **30** further includes a first power distribution system or bus, shown as door wiring harness **170**, that distributes electrical energy between the modules within the door **30**. The door wiring harness **170** is electrically coupled to the power supply unit **130**, the window control unit **132**, the main control unit **134**, the interactive unit **140**, the bridge module **136**, and the deadbolt units **138**. In operation, electrical energy passes from the power source **102** into the power converter **104**, which conditions the electrical energy (e.g., to a different voltage or current, to DC, etc.). The converted electrical energy passes through the power transfer assembly

106 to the power supply unit **130**. The power supply unit **130** further conditions the electrical energy for use with each of the individual modules and provides the electrical energy to the door wiring harness **170**. The door wiring harness **170** then distributes the electrical energy to the window control unit **132**, the main control unit **134**, the interactive unit **140**, the bridge module **136**, and the deadbolt units **138**.

(126) Referring to FIGS. **3-6**, the door wiring harness **170** is contained within the door **30**. As shown, the subframe **40** defines one or more paths, recesses, grooves, or passages, shown as wire passage **172**, containing the door wiring harness **170**. The door wiring harness **170** passes along the wire passage **172** to reach each of the modules. The wire passage **172** is defined by the hinge stile **42**, the top rail **44**, and the lock stile **46**. The wire passage **172** is covered by the exterior skin **50** and the interior skin **52**. In some embodiments, the wire passage **172** is sufficiently covered by the exterior skin **50** and the interior skin **52** for the door wiring harness **170** to not be visible from outside of the door **30**. As shown, the wire passage **172** is arranged such that the door wiring harness **170** extends at least partway around the window **60**.

(127) Referring to FIGS. **1** and **2**, the door system **10** further includes an annunciation system or notification system, shown as doorbell system **180**. The doorbell system **180** includes an input device or user interface (e.g., a button, a switch, knob, a level, a pull cord, etc.), shown as external doorbell **182**, positioned on an exterior surface of a wall **12**. By way of example, the external doorbell **182** may be positioned adjacent a side door of a home separate from the door **30**. The external doorbell **182** is configured to receive a user input or command (e.g., a button press, etc.) and output a doorbell signal indicating that the user wishes to provide a notification inside of the home. The external doorbell **182** is operatively coupled to a doorbell control module, shown as doorbell module **184**, of the control system **100**. The doorbell module **184** may be positioned outside of the door **30** (e.g., elsewhere within the home or structure) while remaining in communication with the modules of the door **30** (e.g., wirelessly). The doorbell module **184** is operatively coupled to a notifier, shown as doorbell speaker **186** (which may be already installed within the structure to which the door system **10** is being installed). The doorbell speaker **186** may be configured to provide a notification (e.g., an auditory notification, such as a ringing bell or siren). In other embodiments, the doorbell speaker **186** is replaced by or supplemented with a device configured to provide another type of notification (e.g., a visual notification, such as a flashing light). By way of example, a lamp may be operatively coupled to the doorbell module **184**. The doorbell system **180**, and particularly, the doorbell module **184**, may be configured to facilitate retrofitting the door system **10** to a structure that already has a doorbell system installed. In some embodiments, one or more components of the doorbell system **180** are omitted. By way of example, the interactive unit **140** may include a doorbell, a speaker, etc. to perform the functionality of the doorbell system **180** described herein. By way of another example, the interactive unit **140** may include a doorbell that transmits a wireless signal to the doorbell module **184** (e.g., via a wireless mesh communication network) to activate the doorbell speaker **186** installed in the structure.

(128) A user may initiate operation of the doorbell system **180** by interacting with the external doorbell **182**, causing the external doorbell **182** to provide a doorbell signal to the doorbell module **184**. Additionally or alternatively, another component of the control system **100**, such as the main control unit **134**, may provide a doorbell signal to the doorbell module **184** (e.g., wirelessly through a mess communication network described herein). In response to receiving the doorbell signal, the doorbell module **184** may provide a command signal (e.g., an electrical current) to the doorbell speaker **186**, causing the doorbell speaker **186** to provide a notification to an occupant of the home. The notification may indicate to an occupant of the home that a user is awaiting permission to access the home.

(129) Referring to FIG. **9A**, the control system **100** may include one or more external devices that communicate with the door **30**. It should be understood that the door **30** in FIG. **9A** may represent

any module of the door **30** (e.g., the power supply unit **130**, the window control unit **132**, the main control unit **134**, the bridge module **136**, the deadbolt unit(s) **138**, the doorbell module **184**, etc.). The door **30** may communicate with the external devices through a network (e.g., a wired and/or wireless network, a local area network, a wide area network, a cellular network, etc.), shown as network **207**. In some embodiments, the control system **100** includes a network interface, shown as access point **208**, that facilitates communication between the door **30** and the network **207** (e.g., a router, etc.). By way of example, the door **30** may be part of a local area network, and the access point **208** may facilitate communication between the local area network and a wide area network (e.g., the Internet). In other embodiments, the door **30** communicates directly with the network **207**, and the access point **208** is omitted. By way of example, the network **207** may be a cellular network, and the door **30** may communicate directly with the network **207** (e.g., through a cellular receiver integrated into the bridge module **136**).

(130) As shown in FIG. **9A**, the control system **100** may communicate with one or more user devices (e.g., smartphones, tablets, laptop computers, desktop computers, key fob, etc.), shown as user devices **190**. Each user device **190** may be associated with (e.g., owned by, registered to, etc.) one or more users. A user may utilize the user device **190** to issue commands to the control system **100** and/or review information provided by the control system **100**.

(131) Referring still to FIG. **9A**, the control system **100** may additionally or alternatively include one or more external computing units or remote systems, shown as server(s) **200**. The server(s) **200** may be associated with a homeowner, a manufacturer of the door **30**, and/or a third party service. The server(s) **200** may store information and/or perform processing to facilitate operation of the door system **10**.

(132) One or more of the servers **200** may be designated as third-party servers **209**. The third-party servers **209** may be associated with (e.g., operated by) a third party. By way of example, the third party may be a service provider that offers Internet-based hosting services or cloud infrastructure. By way of example, the third-party servers **209** may be operated by Amazon Web Services, Google Cloud, Microsoft Azure, IBM Cloud, Oracle Cloud, or another cloud hosting service.

(133) As shown in FIG. **9B**, each user device **190** includes a processing circuit, shown as processor **192**, operatively coupled to a memory device, shown as memory **194**. The memory **194** may store one or more instructions that are executed by the processor **192** to perform the processes described herein. The user device **190** further includes a network interface or communication adapter, shown as communication interface **196**. The communication interface **196** is configured to facilitate communication between the user device **190** and the other components of the control system **100** (e.g., through the network **207**). The communication interface **196** may be configured to perform wired and/or wireless configuration. By way of example, the communication interface **196** may communicate using near-field communication (NFC), radio-frequency identification (RFID), Bluetooth low energy (BLE), Bluetooth, Wi-Fi, Zigbee, over a cellular network, or through another communication protocol. The communication interface **196** may communicate directly with other devices or over a local area network or a wide area network (e.g., the Internet). The user device **190** includes a user interface **198**, which communicates information to a user and/or receives information (e.g., commands) from the user. By way of example, the user interface **198** may include microphones, speakers, touchscreens, buttons, switches, knobs, or other input devices and/or output devices.

(134) As shown in FIG. **9B**, each server **200** includes a processing circuit, shown as processor **202**, operatively coupled to a memory device, shown as memory **204**. The memory **204** may store one or more instructions that are executed by the processor **202** to perform the processes described herein. The server **200** further includes a network interface or communication adapter, shown as communication interface **206**. The communication interface **206** is configured to facilitate communication between the server **200** and the other components of the control system **100** (e.g., through the network **207**). The communication interface **206** may be configured to perform wired

and/or wireless configuration. By way of example, the communication interface **206** may communicate using NFC, RFID, BLE, Bluetooth, Wi-Fi, Zigbee, over a cellular network, or through another communication protocol. The communication interface **206** may communicate directly with other devices or over a local area network or a wide area network (e.g., the Internet).

(135) Modular Control System

(136) Module Features

(137) The modules described herein may each include one or more controllers. For the sake of compact description, an exemplary controller is shown in FIG. **14** as controller **210**. In some embodiments, the controller **210** is formed on a printed circuit board. The controller **210** may represent the functional components of any of the controllers of any of the modules. As shown in FIG. **14**, the controller **210** includes a processing circuit, shown as processor **212**, operatively coupled to a memory device, shown as memory **214**. The memory **214** may store one or more instructions that are executed by the processor **212** to perform the processes described herein.

(138) The controller **210** further includes a network interface or communication adapter, shown as communication interface **216**. The communication interface **216** is configured to facilitate communication between the controller **210**, the other components of the corresponding module, and the controllers **210** of the other modules. The communication interface **216** may be configured to perform wired communication. Additionally or alternatively, the communication interface **216** may be configured to perform wireless configuration. By way of example, the communication interface **216** may communicate using NFC, RFID, Bluetooth, Wi-Fi, Zigbee, over a cellular network, or through another type of wireless communication. The communication interface **216** communicate directly with other device or over a local area network or a wide area network (e.g., the Internet). By way of example, the communication interface **216** may communicate through the network **207** and/or the access point **208**.

(139) In some embodiments, each of the modules is configured as a single, self-contained assembly. By way of example, each module may include a chassis, frame, base member, base plate, or mounting tray to which each of the components of the module are attached. By coupling the components of the module to a mounting tray, the entire module may be moved or otherwise manipulated as a single structure (e.g., for transport, for maintenance, etc.). By way of example, if the main control unit **134** malfunctions or needs to be upgraded, the main control unit **134** can be removed and replaced with another main control unit **134** with minimal effort.

(140) Power Supply Unit

(141) Referring to FIGS. **3**, **4**, **9B**, **15**, and **16**, the power supply unit **130** is shown according to an exemplary embodiment. The power supply unit **130** includes a chassis, frame, base member, or base plate, shown as mounting tray **220**, that is configured to support the components of the power supply unit **130**.

(142) The power supply unit **130** further includes an energy storage device, shown as backup battery **222**, coupled to the mounting tray **220**. The backup battery **222** may be configured to store energy and release the stored energy as electrical energy to power one or more functions of the door **30**. The backup battery **222** may include one or more batteries (e.g., lithium-ion batteries, nickel-cadmium batteries, lead-acid batteries, etc.). In other embodiments, the backup battery **222** additionally or alternatively includes a different type of energy storage device, such as capacitors or supercapacitors. The backup battery **222** may be electrically coupled to the power transfer assembly **106** and/or to the door wiring harness **170** (e.g., through the power converter **226**). The backup battery **222** may be charged by electrical energy supplied from the power source **102**. Alternatively, the backup battery **222** may be a disposable battery that is replaced when depleted. In some embodiments, the backup battery **222** is a 24V battery. In some embodiments, the backup battery **222** is a 12V battery, a 5V battery, and/or has still another voltage.

(143) The backup battery **222** may be used to power one or more operations of the door **30** when the electrical energy from the power source **102** is interrupted or otherwise insufficient to meet the

power demand of the door **30**. By way of example, the backup battery **222** may be used during a power outage that disables the power source **102** or a brownout that reduces the power supplied by the power source **102**. By way of another example, the backup battery **222** may be used in a state where the power transfer assembly **106** is unable to transfer electrical energy to the door **30** (e.g., when the door **30** is opened and connectors **129** of FIG. **13** are disengaged from one another). Accordingly, the backup battery **222** may beneficially ensure that a user can actuate or otherwise control components of the door system **10** even if the supply of external power to the door **30** is interrupted. In some embodiments, the backup battery **222** is capable of supplying the power required to operate the door **30** for at least two days. In some embodiments, the backup battery **222** only provides power to a subset of the electrical component of the door system **10** (e.g., when the state of charge thereof falls below a threshold, does not provide power to the window **60**, only provides power to one of the deadbolt units **138**, etc.), as described in more detail herein.

(144) The power supply unit **130** includes a controller or control unit, shown as battery management controller **224**, that is coupled to the mounting tray **220** and operatively coupled to the backup battery **222**. The battery management controller **224** may utilize the controller **210** of FIG. **14**. The battery management controller **224** may be configured to manage charging and discharging of the backup battery **222**. By way of example, the battery management controller **224** may control when and how much electrical energy is discharged from the backup battery **222**. By way of another example, the battery management controller **224** may monitor a charge level of the backup battery **222** (e.g., through a voltage sensor). The battery management controller **224** may manage the amount of electrical energy that is supplied to charge the backup battery **222** (e.g., to maximize the charge level of the backup battery **222** and/or to improve a health of the backup battery **222**).

(145) The power supply unit **130** further includes more power converters or power conditioners (e.g., transformers, alternating current (AC) to direct current (DC) rectifiers, DC to AC inverters, etc.), shown as power converter **226**. As shown, the power converter **226** is directly coupled to the mounting tray **220**. The power converter **226** may be electrically coupled to the power transfer assembly **106** and/or the backup battery **222** such that the power converter **226** converts or conditions a portion of the electrical energy supplied by the power transfer assembly **106** and/or the backup battery **222** for use with certain components of the door **30**. In one embodiment, the power transfer assembly **106** supplies electrical energy at 24V DC, and the power converter **226** includes a DC to DC converter that converts a portion of the 24V electrical energy to 5V DC. A first subset of components of the door **30** (e.g., the window control unit **132**, the deadbolt units **138**) operate on 24V DC power, and a second subset of the components of the door **30** (e.g., the main control unit **134**, the interactive unit **140**, and the bridge module **136**) operate on 5V DC power. Accordingly, the power converter **226** permits the use of components within the door **30** having different input power requirements. In some embodiments, the power converter **226** additionally or alternatively performs a different conversion operation on some or all of the electrical energy from the power transfer assembly **106**. By way of example, the power converter **226** may receive AC electrical energy from the power transfer assembly **106** and convert the electrical energy to DC.

(146) The power supply unit **130** includes a second controller or control unit (e.g., a power management controller or unit), shown power supply controller **228**, that is coupled to the mounting tray **220**. The power supply controller **228** may utilize the controller **210** of FIG. **14**. The power supply controller **228** is electrically coupled to the power transfer assembly **106**, the backup battery **222**, the power converter **226**, and the door wiring harness **170**. The power supply controller **228** is configured to monitor, manage, and control the distribution of electrical energy throughout the door **30**. The power supply controller **228** may include one or more sensors (e.g., voltage sensors, current sensors, etc.) that monitor electrical state of the control system **100** (e.g., how much current or voltage is being supplied at different locations throughout the control system **100**). The power supply controller **228** may control what portion of the electrical energy demand of the door **30** is satisfied by the power source **102** and what portion of the electrical energy demand

of the door **30** is satisfied by the backup battery **222** (e.g., by increasing or decreasing the flow of electrical energy supplied by the backup battery **222**). The power supply controller **228** may control the amount of electrical energy that is supplied directly (e.g., at 24V DC) and the amount of electrical energy that is converted by the power converter **226** (e.g., to 5V DC). In some embodiments, the power supply controller **228** and the battery management controller **224** are integrated into a single controller.

(147) Window Control Unit

(148) Referring to FIGS. **17** and **18**, the window control unit **132** is shown according to an exemplary embodiment. The window control unit **132** includes a chassis, frame, base member, or base plate, shown as mounting tray **230**, that is configured to support the components of the window control unit **132**. The window control unit **132** further includes a controller or control unit, shown as window controller **232**. The window controller **232** may utilize the controller **210** of FIG. **14**. The window controller **232** is electrically coupled to the power supply unit **130** and the window **60** (e.g., through the door wiring harness **170**). The window controller **232** is configured to selectively supply electrical energy to the window **60** to reconfigure the window **60** between the viewing state and the privacy state. By way of example, the window controller **232** may apply a first voltage (e.g., 0V) to the window **60** to configure the window **60** into the viewing state and apply a second voltage to the window (e.g., 24V) to configure the window **60** into the privacy state. In some embodiments, the window controller **232** applies a range of different voltages to the window **60** in the privacy state to permit control over a degree of tinting or opacity of the window **60** in the privacy state (e.g., to change between a low tint or opacity setting, a medium tint or opacity setting, and a high tint or opacity setting).

(149) Main Control Unit

(150) Referring to FIGS. **19** and **20**, the main control unit **134** is shown according to an exemplary embodiment. The main control unit **134** includes a chassis, frame, base member, or base plate, shown as mounting tray **240**, that is configured to support the components of the main control unit **134**. The main control unit **134** further includes a controller or control unit, shown as main controller **242**. The main controller **242** may utilize the controller **210** of FIG. **14**. The main controller **242** is configured to control the overall operation of the door system **10** (e.g., by receiving information from or providing information to the other modules of the door system **10**).

(151) The main controller **242** may provide information (e.g., commands, data, etc.) to each of the other modules to control operation of the door system **10**. By way of example, the main controller **242** may provide commands to control operation of the power supply unit **130**, the window control unit **132**, the interactive unit **140**, the bridge module **136**, the deadbolt units **138**, and/or the doorbell module **184**. The main controller **242** may receive information (e.g., user commands, sensor data, operational states, etc.) from the power supply unit **130**, the window control unit **132**, the interactive unit **140**, the bridge module **136**, the deadbolt units **138**, and/or the doorbell module **184**. The main controller **242** may receive information from and/or provide information to the user devices **190** and/or the servers **200** (e.g., through the bridge module **136**).

(152) The main controller **242** may make determinations regarding the operation of the door system **10** based on received information and provide one or more commands based on the results of the determinations. By way of example, the main controller **242** may receive a doorbell signal from the interactive unit **140**, the doorbell signal indicating that a user wishes to provide a notification within a home. In response to receiving the doorbell signal, the main controller **242** may send a command to the doorbell module **184** indicating that the doorbell module **184** should activate the doorbell speaker **186** to provide a notification.

(153) The main control unit **134** includes a power converter or power conditioner, shown as power regulator **244**, that is coupled to the mounting tray **240**. The power regulator **244** may receive electrical energy (e.g., from the power supply unit **130**) and condition the electrical energy for use within the main control unit **134**. By way of example, the power regulator **244** may convert 24V

DC electrical energy to 12V DC electrical energy for use by the main control unit **134**. By way of another example, the power regulator **244** may convert AC electrical energy to DC electrical energy for use by the main control unit **134**.

(154) The main control unit **134** further includes one or more switches, contactors, transistors, or relays, shown as relays **246**, that are coupled to the mounting tray **240**. In some embodiments, the relays **246** are metal-oxide-semiconductor field-effect transistors (MOSFETs). The relays **246** may be operatively coupled to the main controller **242** such that operation of the relays **246** is controlled by the main control unit **134**. The relays **246** may be configured to selectively electrically couple or decouple two components in response to receiving a signal from the main control unit **134**. By way of example, each relay **246** may be positioned to selectively electrically couple one or more of the deadbolt units **138** to the power supply unit **130**. In such an example, activation of the relay **246** may cause a latch or deadbolt of the deadbolt unit **138** to be extended or retracted.

(155) In some embodiments, the main control unit **134** includes a memory device, memory, or data storage, shown as media storage **248**. The media storage **248** may be used to store image data (e.g., still images), video data (e.g., video clips), and/or audio data (e.g., standalone audio or audio corresponding to the video data). By way of example, video, still images, and/or audio may be recorded by the door **30** (e.g., by the camera **520**, via a microphone, etc.) during operation of the door **30**. The main control unit **134** may store some or all of the recorded video data, image data, and audio data in the media storage **248** for later playback (e.g., by the user device **190**) or for transmission and/or analysis (e.g., by the main controller **242**, by the server **200**, etc.).

(156) Deadbolt Unit

(157) Referring to FIGS. **21** and **22**, one of the deadbolt units **138** is shown according to an exemplary embodiment. Each of the deadbolt units **138** may have similar components in a similar arrangement. In other embodiments, the door **30** includes more or fewer than two deadbolt units **138** (e.g., one, three, four, etc.). As shown, each deadbolt unit **138** includes a chassis, frame, base member, or base plate, shown as plate **250**, that is configured to support the components of the deadbolt unit **138**. The plate **250** may engage the lock stile **46** (and/or the top rail **44** and/or the bottom rail **48**) to form a seal and prevent ingress of contaminants into the corresponding deadbolt recess **162**. In some embodiments, the plate **250** is removably coupled to the lock stile **46** (and/or the top rail **44** and/or the bottom rail **48**) (e.g., to facilitate removal or replacement of the deadbolt unit **138**).

(158) The deadbolt unit **138** includes a movable member or latch, shown as deadbolt **252**, that is movably coupled to the plate **250**. As shown, the deadbolt **252** extends through an aperture defined by the plate **250**. The deadbolt **252** is repositionable relative to the plate **250** between an extended position, shown in FIG. **21**, and a retracted position in which the deadbolt **252** is retracted into the plate **250**. Accompanying each deadbolt unit **138** is a corresponding plate, shown as strike plate **254**. The strike plate **254** may be fixedly coupled to one of the side jambs **22** (and/or the head jamb **24** and/or the sill **26**). The strike plate **254** defines an aperture, shown as deadbolt aperture **256**, that is sized to receive the deadbolt **252**. The strike plate **254** is positioned such that the deadbolt aperture **256** receives the deadbolt **252** when the door **30** is in the closed position and the deadbolt **252** is in the extended position. This condition may be referred to as a locked state of the door **30**. In the locked state, engagement between the strike plate **254** and the deadbolt **252** prevents the door **30** from being moved out of or opened from the closed position. When the deadbolt **252** is in the retracted position, the deadbolt **252** disengages from the strike plate **254**, and the door **30** can swing freely without interference from the deadbolt unit **138**. This condition may be referred to as an unlocked state of the door **30**. In other embodiments, the strike plate **254** is omitted, and the side jamb **22** (and/or the head jamb **24** and/or the sill **26**) defines the deadbolt aperture **256**.

(159) The deadbolt unit **138** further includes a linear actuator (e.g., an electric motor, a solenoid, etc.), shown as deadbolt actuator **260**, that is coupled to the plate **250** and to the deadbolt **252**. The deadbolt actuator **260** is configured to move the deadbolt **252** relative to the plate **250** between the

extended position and the retracted position. Accordingly, the deadbolt actuator **260** may be used to reconfigure the door **30** between the locked state and the unlocked state.

(160) In some embodiments, the deadbolt actuator **260** is an electric actuator. By way of example, the deadbolt unit **138** may include a solenoid that is activated in response to receiving electrical energy. The solenoid may generate a magnetic force that extends or retracts the deadbolt **252**. By way of example, the deadbolt **252** may be biased into the extended position by a biasing member of the deadbolt actuator **260**, such as a compression spring. The solenoid may be activated to overcome the biasing force of the biasing member and retract the deadbolt **252**. In other embodiments, the deadbolt actuator **260** utilizes a different type of electric actuator, such as an electric motor. In other embodiments, the deadbolt actuator **260** is a different type of actuator, such as a hydraulic cylinder or pneumatic cylinder.

(161) In other deadbolt arrangements, a deadbolt is positioned adjacent a mechanical knob that transfers a mechanical force supplied by a user to reposition the deadbolt. Accordingly, the deadbolt must be positioned adjacent to this knob in order to function. This limits the range of locations where the deadbolt can be placed without introducing difficulty in accessing the knob. In contrast, the deadbolt unit **138** can be placed anywhere along the perimeter of the door **30** due to the deadbolt actuator **260**. As the deadbolt actuator **260** is electrically actuated, the deadbolt actuator **260** is not required to be positioned near a user interface. This permits the deadbolt actuators **260** to be spaced such that the door **30** can include multiple deadbolt units **138**, increasing the security of the door **30**. As shown, two deadbolt units **138** are positioned along the lock stile **46** vertically spaced apart from one another. Additionally or alternatively, the door **30** could include (a) a deadbolt unit **138** positioned along the top rail **44** and configured to engage the head jamb **24**, (b) a deadbolt unit **138** positioned along the hinge stile **42** and configured to engage the corresponding side jamb **22**, and/or (c) a deadbolt unit **138** positioned along the bottom rail **48** and configured to engage the sill **26**.

(162) In other embodiments, the door **30** includes a deadbolt **252** that is mechanically actuated by a user in addition to a deadbolt **252** that is actuated by a deadbolt actuator **260**. A mechanically-actuated deadbolt arrangement may include an interface, such as a knob or slider, that is mechanically linked to a deadbolt **252** by a lever, linkage, or other mechanical coupler. Movement of the interface by a user may cause a corresponding movement of the deadbolt **252** due to the application of a mechanical force by the interface. A sensor may detect movement of the mechanically-actuated deadbolt **252**, and deadbolt actuator **260** may electrically actuate a second deadbolt **252** in tandem with the mechanically-actuated deadbolt **252**.

(163) Each deadbolt unit **138** further includes a sensor (e.g., a button, a switch, a limit switch, etc.), shown as door close sensor **262**, coupled to the plate **250**. The door close sensor **262** includes an interface, rod, or protrusion, shown as button **264**, that is movable relative to the plate **250**. The button **264** is positioned to engage the strike plate **254** when the door **30** is in the closed position. In this position, the button **264** is forced to retract into the plate **250**. The button **264** is biased to extend (e.g., by a biasing member such as a compression spring), such that when the door **30** is in the open position, the button **264** comes out of contact with the strike plate **254** and extends back out of the plate **250**. When the button **264** is retracted, the door close sensor **262** supplies a door closed signal indicating that the door **30** is in the closed position. When the button **264** is extended, the door close sensor **262** supplies a door open signal indicating that the door **30** is in the open position. Accordingly, the door close sensor **262** provides information indicating whether or not the door **30** is closed. In other embodiments, the door close sensor **262** is otherwise constructed. By way of example, the door close sensor **262** may include a Hall effect sensor that is configured to detect the presence of a magnet coupled to the door frame **20**.

(164) Each deadbolt unit **138** further includes a controller or control unit, shown as deadbolt controller **266**. The deadbolt controller **266** may utilize the controller **210** of FIG. **14**. The deadbolt controller **266** is operatively coupled to the deadbolt actuator **260** and the door close sensor **262**.

The deadbolt controller **266** may provide an actuation signal to command the deadbolt actuator **260** to extend or retract the deadbolt **252**. The deadbolt controller **266** may receive the door close signal and the door open signal from the door close sensor **262** and determine, based on the received signal, if the door **30** is in the open position or the closed position.

(165) In some embodiments, the deadbolt controller **266** is in communication (e.g., wirelessly) with the main controller **242**. By way of example, the deadbolt controller **266** may provide the actuation signal to the deadbolt actuator **260** in response to receiving a command from the main controller **242**. By way of another example, the deadbolt controller **266** may provide the door close signal and the door open signal to the main controller **242**. In other embodiments, the deadbolt controller **266** is omitted, and the deadbolt actuator **260** and the door close sensor **262** are in direct communication with the main controller **242** (e.g., through the door wiring harness **170**). By way of example, one of the relays **246** may be used to provide the actuation signal to the deadbolt actuator **260**.

(166) Bridge Module

(167) Referring to FIG. **23**, the bridge module **136** is shown according to an exemplary embodiment. The bridge module **136** includes a controller or control unit, shown as bridge controller **270**. The bridge controller **270** may utilize the controller **210** of FIG. **14**. The bridge controller **270** may act as a wireless bridge to control or otherwise facilitate communication between (i) the modules of the door system **10** (e.g., the power supply unit **130**, the window control unit **132**, the main control unit **134**, the interactive unit **140**, the deadbolt units **138**, the doorbell module **184**, etc.) and (ii) the user devices **190** and/or the servers **200**. Further details regarding the wireless bridge functionality of the bridge module **136** may be found in U.S. Patent Publication No. 2022/0319264, filed Jun. 17, 2022, which is incorporated herein by reference in its entirety.

(168) In some embodiments, the bridge module **136** further includes a communication interface **272** operatively coupled to the bridge controller **270**. The communication interface **216** of the bridge controller **270** and the communication interface **272** may be configured to communicate using different communication protocols. By way of example, the communication interface **216** may communicate using a first communication protocol that is utilized by the other modules (e.g., the power supply unit **130**, the window control unit **132**, the main control unit **134**, the interactive unit **140**, the deadbolt units **138**, etc.). The communication interface **272** may communicate using a second communication protocol utilized by other external devices (e.g., the user devices **190**, the servers **200**, etc.). In some embodiments, the first communication protocol is a short-range communication protocol, and the second communication protocol is a long-range communication protocol). Possible communication protocols usable by the communication interface **216** and/or the communication interface **272** include, but are not limited to: Bluetooth, BLE, NFC, RFID, Wi-Fi, cellular, infrared, radiofrequency, and/or ZigBee.

(169) Doorbell Module

(170) Referring to FIGS. **9B** and **24**, the doorbell module **184** is shown according to an exemplary embodiment. The doorbell module **184** includes a controller or control unit, shown as doorbell controller **280**. The doorbell controller **280** may utilize the controller **210** of FIG. **14**. The doorbell controller **280** may be operatively coupled to the external doorbell **182** and the doorbell speaker **186**. The doorbell controller **280** is configured to facilitate operation of the doorbell system **180**. By way of example, the doorbell controller **280** may receive a doorbell signal from the external doorbell **182** and/or a doorbell signal from the door **30** (e.g., from the main control unit **134** in response to a user interacting with the interactive unit **140**). In response to receiving a doorbell signal, the doorbell controller **280** may command the doorbell speaker **186** to provide a notification (e.g., an audible chime or alarm, a flashing light, etc.). In some embodiments, the external doorbell **182** and/or the doorbell speaker **186** are existing elements in a home or structure that are installed separately from the door **30**. The doorbell controller **280** may facilitate connecting the external doorbell **182** and/or the doorbell speaker **186** to the control system **100** such that the doorbell system **180** is seamlessly integrated into operation of the door **30**.

(171) The doorbell module **184** may be positioned separate from the door **30**. Accordingly, the doorbell module **184** may include another power source or power supply, shown as power source **282**. The power source **282** is operatively coupled to the doorbell controller **280** and configured to supply electrical energy to power operation of the doorbell module **184**, the external doorbell **182**, and/or the doorbell speaker **186**. In some embodiments, the power source **282** is external to the doorbell module **184**. By way of example, the power source **282** may supply electrical energy from a power grid (e.g., a municipal power grid), generators, solar panels, wind turbines, fuel cells, or other external sources. In some embodiments, the power source **282** is internal to the doorbell module **184**. By way of example, the power source **282** may include batteries, capacitors, or other internal sources.

(172) Module Mesh Network

(173) Referring to FIG. 25, the modules (e.g., the power supply unit **130**, the window control unit **132**, the main control unit **134**, the bridge module **136**, the deadbolt unit(s) **138**, the doorbell module **184**, etc.) communicate with one another to form an Internet of Things (IoT) wireless mesh network, shown as mesh network **300**. In FIG. 25, exemplary of some possible lines of communication between the modules are illustrated as dashed lines. By way of example, a dashed line between the main control unit **134** and a deadbolt unit **138** may indicate that the main control unit **134** is communicating directly with the deadbolt unit **138**.

(174) The mesh network **300** may communicate data between any of the modules. In some embodiments, the modules communicate wirelessly. The modules may communicate using various wireless communication protocols. In some such embodiments, the modules communicate using the MQTT communication protocol. In other embodiments, the modules communicate using another communication protocol. By way of example, the modules may communicate using NFC, RFID, BLE, Bluetooth, Wi-Fi, Zigbee, over a cellular network, or through another communication protocol.

(175) As shown in FIG. 25, each of the modules can communicate directly with any of the other modules. By way of example, the main control unit **134** may communicate directly with the power supply unit **130**, the window control unit **132**, the bridge module **136**, the deadbolt unit(s) **138**, and/or the doorbell module **184**. By way of another example, the bridge module **136** may communicate directly with the power supply unit **130**, the window control unit **132**, the main control unit **134**, the deadbolt unit(s) **138**, and/or the doorbell module **184**.

(176) In some embodiments, the modules communicate indirectly through one another. By way of example, the main control unit **134** may communicate with a deadbolt unit **138** through the bridge module **136**. In such an example, the main control unit **134** may transfer a packet of information (e.g., a command to extend a deadbolt **252**) to the bridge module **136**. The bridge module **136** may then transfer the packet to the deadbolt unit **138**.

(177) Communicating indirectly through another module may be more advantageous than direct communication under certain circumstances. By way of example, a first module may attempt to transfer a packet of information to a second module. However, the first module may determine that the strength of the connection between the first module and the second module is insufficient to transfer the information. For example, the first module may determine that a transfer speed of the connection between the first module and the second module is below a predetermined threshold speed. The connection between the first module and the second module may deteriorate due to a source of interference being positioned between the first module and the second module (e.g., a metallic portion of the door **30**, external interference, etc.). In response to such a determination, the first module may identify a third module with which a stronger connection can be formed. After identifying the third module, the first module may transfer the packet to the third module along with a command for the third module to transfer the packet to the second module. In response to receiving the command, the third module may transfer the packet to the second module.

(178) In some embodiments, each of the modules acts independently to determine how to

communicate data to other modules of the mesh network **300**. By way of example, when the main control unit **134** needs to transfer a packet of information to the window control unit **132**, the main control unit **134** may determine if the packet should be transferred directly to the window control unit **132** or indirectly through another module. In other embodiments, one of the modules (e.g., the main control unit **134**, the bridge module **136**, etc.) acts as a master controller, and the other modules act as slave controllers. The master controller may communicate with each of the slave controllers to identify what information needs to be communicated, and the master controller may dictate how each of the slave controllers communicate the information between one another.

(179) In some embodiments, the bridge module **136** is configured to communicate with one or more external devices (e.g., a user device **190**, a server **200**, etc.). In some embodiments, the modules communicate with one another using a first communication protocol (e.g., a short-range communication protocol), and the bridge module **136** communicates with the external devices using a second communication protocol (e.g., a long-range communication protocol). In this way, the bridge module **136** acts as a bridge between the mesh network **300** and the external devices. The external devices may communicate between one another (e.g., user device **190** to server **200**, server **200** to user device **190**, user device **190** to user device **190**, server **200** to server **200**, etc.).

(180) In one example, the modules communicate wirelessly with one another within the door **30** using the MQTT communication protocol. The bridge module **136** may communicate with a user device **190** over the Internet (e.g., through a Wi-Fi or cellular connection). When communication between one of the modules and an external device is desired, the module may transfer a packet of information to the bridge module **136** over the mesh network **300**. The bridge module **136** may then transfer the packet to the user device **190**. Alternatively, the bridge module **136** may transfer the packet to a server **200**, which may subsequently transfer the packet to the user device **190**.

(181) The mesh network **300** may provide a variety of advantages relative to other control system arrangements. When assembling the door **30**, the control system **100** requires only a door wiring harness **170** that provides power. Because the modules communicate wirelessly with one another, wired data transfer connections between the modules are not necessary. Eliminating the need for wired data connections reduces the complexity of the assembly process of the door **30**, reduces the number of components within the door **30**, and eliminates potential points of failure.

(182) When a module is added to the door **30**, the module may connect to the mesh network **300** without a user having to connect the module to a wired data connection. By way of example, when supplied with power (e.g., from a connection to the door wiring harness **170**, from an external power source, etc.), a module may automatically begin seeking to pair with a nearby mesh network **300**. The modules of the mesh network **300** may search to determine if any nearby modules are attempting to pair. By way of example, the modules of the mesh network **300** may perform this search periodically (e.g., once per minute, once per second, etc.). By way of another example, the modules of the mesh network **300** may perform this search in response to a command from the interactive unit **140**, a user device **190**, or a server **200**.

(183) When a module of the mesh network **300** identifies a new module that is seeking to pair with the mesh network **300**, the new module may be added automatically. Alternatively, the mesh network **300** may require an authorized user or device to approve addition of the new module to the mesh network **300**. By way of example, the mesh network **300** may provide a first packet of information (e.g., a request packet) to a user device **190** or the interactive unit **140** that causes the user device **190** or the interactive unit **140** to display an authentication prompt. In response to the user device **190** receiving a command from a user to approve the authentication, the user device **190** may transmit a second packet (e.g., an approval packet) back to the mesh network **300**. In response to the interactive unit **140** receiving a command (e.g., a key code, a biometric input, a fob swipe, etc.) from the user, the interactive unit **140** may provide the second packet back to the mesh network **300**. In response to a module of the mesh network **300** receiving the approval packet, the new module may be added to the mesh network **300**.

(184) New modules may be quickly and easily added to the mesh network **300** by an end user (e.g., a homeowner). By way of example, if a user determines that a module of the mesh network **300** is malfunctioning, the malfunctioning module may be disconnected from the door wiring harness **170** and removed from the door **30**. A replacement module may be connected to the door wiring harness **170**, inserted into the door **30**, and paired with the mesh network **300**. By way of another example, the end user may desire to upgrade a module of the door **30** (e.g., to provide new features that were not originally present in the door **30**, to permit the door system **10** to communicate over a new communication protocol, etc.). The end user may acquire (e.g., purchase) a module having the desired functionality. The upgraded module may be connected to the door wiring harness **170**, inserted into the door **30**, and paired with the mesh network **300**. By way of another example, an end user may desire to add a doorbell module **184** to the mesh network **300**. The user may acquire the doorbell module **184**, connect the doorbell module **184** to an external doorbell **182** and/or a doorbell speaker **186** within their home, and wirelessly pair the doorbell module **184** with the mesh network **300**. In some embodiments, the process of adding a module to the mesh network **300** or replacing an existing module may be performed without specialized tools, permitting the end user to maintain or upgrade the door system **10** without requiring specialized training or assistance from a service professional.

(185) Module Mounting

(186) Referring to FIGS. **26-30**, the mounting arrangement of the main control unit **134** within the door **30** is shown according to an exemplary embodiment. As discussed with reference to FIGS. **3-6**, the main control unit **134** is inserted into the main controller recess **158** defined by the lock stile **46**, and the main controller recess **158** is sealed by the plate **160**. By way of example, the plate **160** may be positioned to extend across the main controller recess **158**, and the plate **160** may be secured to the lock stile **46** by one or more fasteners. In some embodiments, the plate **160** forms a seal against the lock stile **46** to prevent the ingress of debris or moisture into the main controller recess **158**. The components of the main control unit **134** are supported by the mounting tray **240**, such that the entire main control unit **134** can slide into and out of the main controller recess **158** at once. To further support the main control unit **134**, the door **30** includes a support or insert, shown as insert **350**, that is positioned within the main controller recess **158**. The insert **350** may be fixedly coupled to the lock stile **46**. The mounting tray **240** may slidably engage the insert **350** to slidably and removably couple the mounting tray **240**, and therefore the main control unit **134**, to the lock stile **46**.

(187) Referring to FIGS. **27** and **30**, the mounting tray **240** is shown according to an exemplary embodiment. For ease of description, a coordinate system is defined relative to the mounting tray **240** in FIGS. **27** and **30**. The coordinate system includes a lateral X axis, a vertical Y axis, and a longitudinal Z axis.

(188) The mounting tray **240** includes a base, panel, or main body, shown as baseplate **360**. The baseplate **360** is generally flat and rectangular. As shown, the baseplate **360** extends within a lateral and vertical plane. The baseplate **360** defines a series of apertures or passages, shown as mounting holes **362**, extending longitudinally through the baseplate **360**. The mounting holes **362** may each be configured to receive a fastener to couple a component (e.g., the main controller **242**, the power regulator **244**, the relays **246**, etc.) to the baseplate **360**. The baseplate **360** includes a pair of opposing edges (e.g., a top edge and a bottom edge), shown as rails **364**. The rails **364** extend laterally along the baseplate **360** from a front side of the mounting tray **240** to a rear side of the mounting tray **240**. In some embodiments, the rails **364** are substantially parallel to one another. The rails **364** may slidably engage the insert **350** to constrain motion of the mounting tray **240** relative to the insert **350**.

(189) The mounting tray **240** further includes a pair of flanges, protrusions, or walls, shown as flange **370** and flange **380**, each extending in the same direction longitudinally outward from the baseplate **360**. The flange **370** is positioned on a left side of the baseplate **360** and extends

substantially vertically. The flange **380** is positioned on a right side of the baseplate **360** and extends substantially vertically. In some embodiments, the flange **370**, the baseplate **360**, and the flange **380** are formed as a single, continuous piece. By way of example, the flange **370** and the flange **380** may be bent portions of a piece of sheet metal containing the baseplate **360**.

(190) The flange **370** extends within a substantially vertical and longitudinal plane. A first vertical end portion of the flange **370** (e.g., an upper end portion) includes a first protrusion, shown as mounting tab **372**. A vertical end portion of the flange **370** (e.g., a lower end portion) includes a second protrusion, shown as mounting tab **374**. The mounting tab **372** and the mounting tab **374** may each define a laterally-extending aperture, passage, or mounting hole **376** configured to receive a fastener. The mounting tab **372** is substantially centered longitudinally about a plane P.sub.1 that extends laterally and vertically. The mounting tab **374** is substantially centered longitudinally about a plane P.sub.2 that extends laterally and vertically. The plane P.sub.2 is offset longitudinally forward from the baseplate **360**. The plane P.sub.1 is offset a distance D longitudinally forward from the plane P.sub.2. Accordingly, the mounting tab **372** is offset the distance D longitudinally forward from the mounting tab **374**. A longitudinal width of the mounting tab **372** is approximately equal to a longitudinal width of the mounting tab **374**.

(191) The flange **380** extends within a substantially vertical and longitudinal plane that is offset laterally from the flange **370**. A first vertical end portion of the flange **380** (e.g., an upper end portion) includes a first protrusion, shown as mounting tab **382**. A vertical end portion of the flange **380** (e.g., a lower end portion) includes a second protrusion, shown as mounting tab **384**. The mounting tab **382** and the mounting tab **384** may each define a laterally-extending aperture, passage, or mounting hole **376** configured to receive a fastener. The mounting tab **382** is substantially centered longitudinally about the plane P.sub.2. The mounting tab **384** is substantially centered longitudinally about the plane P.sub.1. Accordingly, the mounting tab **384** is offset the distance D longitudinally forward from the mounting tab **382**. The mounting tab **382** is located at approximately the same longitudinal position as the mounting tab **374**. The mounting tab **384** is located at approximately the same longitudinal position as the mounting tab **372**. A longitudinal width of the mounting tab **382** is approximately equal to a longitudinal width of the mounting tab **384**.

(192) Referring to FIGS. **28** and **29**, the insert **350** is shown according to an exemplary embodiment. For ease of description, a coordinate system is defined relative to the insert in FIGS. **28** and **29**. The coordinate system includes a lateral A axis, a vertical B axis, and a longitudinal C axis.

(193) The insert **350** includes a base portion or main body, shown as base **390**, extending vertically between a pair of columns, shown as posts **392**. The posts **392** are vertically offset from one another and extend laterally outward from the base **390**. A space defined between the posts **392** is sized to receive the mounting tray **240**.

(194) Each of the posts **392** defines a recess, shown as mounting tab recess **396**, that extends laterally into the post **392**. Each mounting tab recess **396** is sized to receive one of the mounting tabs (e.g., the mounting tab **372**, the mounting tab **374**, the mounting tab **382**, the mounting tab **384**). Each of the posts **392** further defines a slot, groove, or recess, shown as mounting tab slot **398**, that extends laterally through the post **392** to the base **390**. In each post **392**, the mounting tab slot **398** is positioned adjacent the mounting tab recess **396**. Each mounting tab slot **398** is sized to receive one of the mounting tabs (e.g., the mounting tab **372**, the mounting tab **374**, the mounting tab **382**, the mounting tab **384**).

(195) Each of the posts **392** further defines a pair of slots, grooves, or recesses, shown as rail slot **400** and rail slot **402**. The rail slots **400** and the rail slots **402** are each sized and shaped to receive one of the rails **364**. In each post **392**, the rail slot **400** and the rail slot **402** are positioned on opposite sides of the mounting tab recess **396** and the mounting tab slot **398** (i.e., the mounting tab recess **396** and the mounting tab slot **398** are positioned between the rail slot **400** and the rail slot

402).

(196) As shown in FIG. 29, the mounting tab slot **398** of one of the posts **392** is aligned with (e.g., positioned in a common lateral and vertical plane with) the mounting tab recess **396** of the other post **392**. In some embodiments, the mounting tab recess **396** of a post **392** is offset from the mounting tab slot **398** of the same post **392** by the distance **D**. The rail slots **400** of the posts **392** are aligned with one another (e.g., positioned in a common lateral and vertical plane). The rail slots **402** of the posts **392** are aligned with one another (e.g., positioned in a common lateral and vertical plane).

(197) Referring to FIGS. 26-30, during the initial assembly of the door **30**, the insert **350** is inserted into the main controller recess **158** and fixedly coupled to the lock stile **46**. In some embodiments, the insert **350** interlocks with the lock stile **46** (e.g., a protrusion of the lock stile **46** engages a corresponding portion of the insert **350**) to limit movement of the insert **350**. In some embodiments, the insert **350** is fastened to the lock stile **46** (e.g., using one or more fasteners) and/or adhered to the lock stile **46** (e.g., using an adhesive, such as epoxy).

(198) The mounting tray **240** may be slidably received within the insert **350**. The insert **350** may be configured to receive the mounting tray **240** in multiple different orientations or configurations, as selected by the user. For example, it may be advantageous to have a single door **30** that can be reconfigured between a right hand configuration (e.g., in which the door **30** is coupled to a first side jamb **22**) and a left hand configuration (e.g., in which the door **30** is coupled to an opposing, second side jamb **22**). When reconfiguring the door **30** between the right hand configuration and the left hand configuration, it may be desirable to invert or flip the interactive unit **140** (e.g., between a forward orientation and a reverse orientation) such that the interactive unit **140** faces outward regardless of which face of the door **30** is the exterior side. By way of example, this may ensure that a camera (e.g., the camera **520**) of the interactive unit **140** faces outward from the door **30**. To accommodate an inversion of the interactive unit **140** without requiring a different door wiring harness **170** (e.g., due to a change in the distance between connectors on each module), it may be advantageous to invert or flip each of the modules when the interactive unit **140** is inverted. The arrangement of the mounting tray **240** and the insert **350** may facilitate such an inversion.

(199) The rails **364** of the mounting tray **240** can be received by the rail slots **400** or by the rail slots **402** of the insert **350**. When the rail slots **400** or the rail slots **402** receive the rails **364**, engagement between the rails **364** and the posts **392** limits vertical and longitudinal movement of the mounting tray **240** relative to the insert **350** while permitting free lateral movement (e.g., sliding) of the mounting tray **240** (e.g., to facilitate insertion or removal of the mounting tray **240**). When the rail slots **400** receive the rails **364**, the mounting tray **240** faces in a first direction (e.g., a forward direction). When the rail slots **402** receive the rails **364**, the mounting tray **240** faces in a second direction opposite the first direction (e.g., a rearward direction). Accordingly, the orientation of the mounting tray **240** may be inverted by changing which pair of the rails slots receive the rails **364**. This feature permits a user to easily invert the main control unit **134** when reconfiguring the door **30** between the right hand configuration and the left hand configuration.

(200) The mounting flanges (e.g., the mounting tab **372**, the mounting tab **374**, the mounting tab **382**, the mounting tab **384**) facilitate selectively limiting lateral movement of the mounting tray **240** relative to the insert **350**. By way of example, when the mounting tray **240** is in the desired position, a fastener may be inserted through the mounting hole **376** in one or more of the mounting flanges and engaged with (e.g., threaded into) one of the posts **392**. Contact between the mounting flange and the post **392** may limit lateral movement of the mounting tray **240** in a first direction, and contact between the mounting flange and the fastener may limit lateral movement of the mounting tray **240** in a second direction.

(201) The longitudinally offset positions of the mounting flanges, the mounting tab recesses **396**, and the mounting tab slots **398** facilitate installing the mounting tray **240** in a variety of different orientations. Specifically, this offset arrangement ensures that the two mounting flanges farthest

from the base **390** are received by the mounting tab recesses **396** and engage the posts **392**. The offset arrangement also prevents the two mounting flanges closest to the base **390** from interfering with insertion of the mounting tray **240**. Accordingly, the mounting tray **240** can be received by the insert **350** after the mounting tray **240** is inverted (e.g., rotated 180 degrees) about a vertical axis, a lateral axis, or a longitudinal axis. To illustrate this reconfigurability, four permitted configurations of the mounting tray **240** are now described by way of non-limiting example.

(202) In a first configuration (e.g., an upright, forward-facing configuration or forward orientation) shown in FIG. **26**, the rails **364** of the mounting tray **240** are received within the rail slots **402** of the insert **350**. The mounting tabs **382** and **384** are received within the mounting tab slots **398**, permitting the mounting tray **240** to be inserted into the insert **350** without interference. The mounting tabs **372** and **374** are received within the mounting tab recesses **396**, and the mounting tabs **372** and **374** engage the posts **392** to limit lateral movement of the mounting tray **240**.

(203) The mounting tray **240** is reconfigurable into a second configuration (e.g., an upright, rearward-facing configuration or rearward orientation) in which the mounting tray **240** is inverted about a vertical axis relative to the first configuration. In the second configuration, the rails **364** of the mounting tray **240** are received within the rail slots **400** of the insert **350**. The mounting tabs **372** and **374** are received within the mounting tab slots **398**, permitting the mounting tray **240** to be inserted into the insert **350** without interference. The mounting tabs **382** and **384** are received within the mounting tab recesses **396**, and the mounting tabs **382** and **384** engage the posts **392** to limit lateral movement of the mounting tray **240**.

(204) The mounting tray **240** is reconfigurable into a third configuration (e.g., an upside down, forward-facing configuration or forward orientation) in which the mounting tray **240** is inverted about a longitudinal axis relative to the first configuration. In the third configuration, the rails **364** of the mounting tray **240** are received within the rail slots **402** of the insert **350**. The mounting tabs **372** and **374** are received within the mounting tab slots **398**, permitting the mounting tray **240** to be inserted into the insert **350** without interference. The mounting tabs **382** and **384** are received within the mounting tab recesses **396**, and the mounting tabs **382** and **384** engage the posts **392** to limit lateral movement of the mounting tray **240**.

(205) The mounting tray **240** is reconfigurable into a fourth configuration (e.g., an upside down, rearward-facing configuration or rearward orientation) in which the mounting tray **240** is inverted about a lateral axis relative to the first configuration. In the fourth configuration, the rails **364** of the mounting tray **240** are received within the rail slots **400** of the insert **350**. The mounting tabs **382** and **384** are received within the mounting tab slots **398**, permitting the mounting tray **240** to be inserted into the insert **350** without interference. The mounting tabs **372** and **374** are received within the mounting tab recesses **396**, and the mounting tabs **372** and **374** engage the posts **392** to limit lateral movement of the mounting tray **240**.

(206) Referring to FIG. **31**, the power supply unit **130** utilizes a similar mounting arrangement to the main control unit **134**. Accordingly, the mounting tray **220** of the power supply unit **130** may be substantially similar to the mounting tray **240** of the main control unit **134** except as otherwise specified. An insert **350** is inserted into the power supply recess **150** and fixedly coupled to the hinge stile **42**. The insert **350** is configured to slidably receive the mounting tray **220** within the power supply recess **150**. The mounting tray **220** includes a baseplate **360** including a pair of rails **364**, a flange **370** including mounting tabs **372** and **374**, and a flange **380** including mounting tabs **382** and **384**. The rails **364** and the mounting tabs **372**, **374**, **382**, and **384** facilitate engagement of the mounting tray **220** with the insert **350** within the power supply recess **150** to secure the power supply unit **130** to the hinge stile **42**.

(207) Referring to FIGS. **32** and **33**, the window control unit **132** utilizes a similar mounting arrangement to the main control unit **134**. Accordingly, the mounting tray **230** of the window control unit **132** may be substantially similar to the mounting tray **240** of the main control unit **134** except as otherwise specified. The mounting tray **230** omits the flange **370** and the flange **380**.

Instead, the mounting tray **230** is flat, including a baseplate **360** having two rails **364**.

(208) The mounting tray **230** is received within an insert **420** that is inserted into the window controller recess **154**. The insert **420** may be substantially similar to the insert **350** except as otherwise specified. As shown in FIG. **33**, the insert **420** includes a series of longitudinal protrusions or fingers, shown as keys **422**, that extend longitudinally outward from a base **390**. The keys **422** engage the lock stile **46** to fixedly couple the insert **420** to the lock stile **46**. The insert **420** includes a pair of posts **392** extending from the base **390**. Each of the posts **392** defines a rail slot **400** and a rail slot **402** that receive the rails **364** of the mounting tray **230**. Because the mounting tray **230** omits the flanges **370** and **380**, the posts **392** can omit the mounting tab recesses **396** and the mounting tab slots **398** while still slidably receiving the mounting tray **230**.

(209) Wire Routing

(210) Referring to FIGS. **3-6** and **34-38**, the wire passage **172** that contains the door wiring harness **170** is shown according to an exemplary embodiment. The wire passage **172** includes a first groove, slot, notch, or passage, shown as wire groove **430**, and a second groove, slot, notch, or passage, shown as wire groove **432**, each defined by the subframe **40**. The wire groove **430** is defined on a first side of the subframe **40** (e.g., an exterior side) and extends into the subframe **40** toward an opposing second side of the subframe **40** (e.g., an interior side). The wire groove **432** is defined on the second side of the subframe **40** and extends into the subframe **40** toward the first side. Accordingly, the wire groove **430** and the wire groove **432** are positioned on opposite sides of the subframe **40**. The wire groove **430** is covered by the exterior skin **50** to define a first enclosed space that can receive the door wiring harness **170**. The wire groove **432** is covered by the interior skin **52** to define a second enclosed space that can receive the door wiring harness **170**. A user may position the door wiring harness **170** in the wire groove **430** or the wire groove **432**, depending upon the desired arrangement of the door **30** (e.g., as a right hand door or a left hand door).

(211) The wire groove **430** is defined by and extends through the hinge stile **42**, the top rail **44**, and the lock stile **46**. Specifically, the wire groove **430** includes a first vertical portion, shown as hinge portion **440**. The hinge portion **440** is defined by the hinge stile **42** and extends substantially vertically. The wire groove **430** includes a horizontal portion or top portion, shown as head portion **442**. The head portion **442** is defined by the top rail **44** and extends laterally through the top rail **44** from the hinge stile **42** to the lock stile **46**. The wire groove **430** includes a second vertical portion, shown as lock portion **444**. The lock portion **444** is defined by the lock stile **46** and extends substantially vertically. The hinge portion **440**, the head portion **442**, and the lock portion **444** connect to one another to form a continuous groove. The ends of each portion may be curved to facilitate a smooth, gradual transition between each portion. The top rail **44** further defines a third vertical portion of the wire groove **430**, shown as connecting portion **446**. The connecting portion **446** is connected to the head portion **442** and extends downward from the head portion **442**. The connecting portion **446** may facilitate connecting the door wiring harness **170** to the window **60**.

(212) The wire groove **432** may be substantially symmetrical with the wire groove **430** about a center plane of the door **30** (e.g., a vertical and lateral plane). Accordingly, the wire groove **432** may include a hinge portion **440**, a head portion **442**, a lock portion **444**, and a connecting portion **446**. In some embodiments, the hinge portion **440**, the head portion **442**, the lock portion **444**, and the connecting portion **446** of the wire groove **432** are the same shape and size as the hinge portion **440**, the head portion **442**, the lock portion **444**, and the connecting portion **446** of the wire groove **430**. This symmetrical arrangement of the wire grooves **430** and **432** may facilitate a single door wiring harness **170** being compatible with both the wire groove **430** and the wire groove **432**.

(213) The subframe **40** further defines a series of apertures, passages, or paths that extend longitudinally through the door **30** to connect the wire groove **430** and the wire groove **432**. These passages connect the wire groove **430** and the wire groove **432** to the recesses that receive the modules and other electrical components of the door **30**. Accordingly, the passages facilitate the door wiring harness **170** extending from the wire groove **430** or the wire groove **432** to the modules

and other electrical components of the door **30**.

(214) A first passage, shown as power supply passage **450**, is defined by the hinge stile **42**. The power supply passage **450** extends from the wire groove **430**, through the power supply recess **150**, to the wire groove **432**. As shown in FIG. **36**, the power supply passage **450** extends longitudinally through the entire hinge stile **42**, connecting the hinge portion **440** of the wire groove **430** to the hinge portion **440** of the wire groove **432**. The power supply passage **450** facilitates connecting the door wiring harness **170** to the power supply unit **130**.

(215) A second passage, shown as window control passage **452**, is defined by the lock stile **46**. The window control passage **452** extends from the wire groove **430**, through the window controller recess **154**, to the wire groove **432**. The window control passage **452** extends longitudinally through the entire lock stile **46**, connecting the lock portion **444** of the wire groove **430** to the lock portion **444** of the wire groove **432**. The window control passage **452** facilitates connecting the door wiring harness **170** to the window control unit **132**.

(216) A third passage, shown as deadbolt unit passage **454**, is defined by the lock stile **46**. The deadbolt unit passage **454** extends from the wire groove **430**, through a deadbolt recess **162**, to the wire groove **432**. The deadbolt unit passage **454** extends longitudinally through the entire lock stile **46**, connecting the lock portion **444** of the wire groove **430** to the lock portion **444** of the wire groove **432**. The deadbolt unit passage **454** facilitates connecting the door wiring harness **170** to one of the deadbolt units **138**. The subframe **40** may include one of the deadbolt unit passages **454** for each of the deadbolt units **138**. By way of example, a door **30** having two deadbolt units **138** may have two deadbolt unit passages **454**.

(217) A fourth passage, shown as main controller passage **456**, is defined by the lock stile **46**. The main controller passage **456** extends from the wire groove **430**, through the main controller recess **158**, to the wire groove **432**. The main controller passage **456** extends longitudinally through the entire lock stile **46**, connecting the lock portion **444** of the wire groove **430** to the lock portion **444** of the wire groove **432**. The main controller passage **456** facilitates connecting the door wiring harness **170** to the main control unit **134**.

(218) In some embodiments, the lock stile **46** further defines a pair of vertical grooves or passages, shown as interface grooves **458**. Similar to the wire grooves **430** and **432**, the interface grooves **458** are defined on an interior side and an exterior side of the subframe **40**, respectively. The interface grooves **458** extend substantially vertically through the lock stile **46**, connecting the main controller recess **158** with the interface bores **164**. The interface grooves **458** facilitate connecting the main control unit **134** with the interactive unit **140**. In other embodiments, the interface grooves **458** are extensions of the wire groove **430** and the wire groove **432**.

(219) Changing Between Left Hand Configuration and Right Hand Configuration

(220) In some embodiments, the door **30** is reconfigurable between a right hand configuration (e.g., in which the door **30** is a right hand door) and a left hand configuration (e.g., in which the door **30** is a left hand door). In the right hand configuration, the hinges **32** are coupled to a first side jamb **22**. In the left hand configuration, the hinges **32** are coupled to the other side jamb **22**.

Reconfiguring the door **30** between the right hand configuration and the left hand configuration exchanges the exterior side and the interior side of the door **30**, such that the interactive unit **140** is inverted (front-to-back, reversed). It may be advantageous to maintain a consistent orientation of the interactive unit **140** relative to the wall **12** in both the right hand configuration and the left hand configuration. In order to achieve this consistent orientation, the interactive unit **140** may be inverted relative to the door **30** when the door **30** is reconfigured between the right hand configuration and the left hand configuration.

(221) When the interactive unit **140** is inverted, the position of the connection between the door wiring harness **170** and the interactive unit **140** shifts within the door **30**. If the door wiring harness **170** were to remain in the same position, the connection between the door wiring harness **170** and the interactive unit **140** could be strained, or the door wiring harness **170** could be unable to reach

the interactive unit **140**. To ensure that the door wiring harness **170** can form all of the desired connections when the interactive unit **140** is inverted, the door wiring harness **170** may also be inverted within the door **30**. One or more of the modules may be inverted within the door **30** to place the modules into a better position to reach the door wiring harness **170**.

(222) Referring to FIGS. **34-38**, to invert the door wiring harness **170**, the exterior skin **50** and the interior skin **52** may be removed to expose the door wiring harness **170**. The door wiring harness **170** may be disconnected from each of the modules and removed from the door **30**. If the door wiring harness **170** was originally positioned in the wire groove **430**, the door wiring harness **170** may then be inserted into the wire groove **432**. If the door wiring harness **170** was originally positioned in the wire groove **432**, the door wiring harness **170** may then be inserted into the wire groove **430**. Accordingly, the door wiring harness **170** is inverted by changing which of the wire grooves receives the door wiring harness **170**.

(223) Referring to FIG. **26**, to invert the main control unit **134**, the plate **160** may be removed, and the main control unit **134** may be removed from the insert **350**. The main control unit **134** may be rotated about a vertical axis and replaced back into the insert **350**. The main control unit **134** may be coupled to the door wiring harness **170**, and the plate **160** may be coupled to the lock stile **46**.

(224) Referring to FIG. **31**, to invert the power supply unit **130**, the plate **152** may be removed, and the power supply unit **130** may be removed from the insert **350**. The power supply unit **130** may be rotated about a vertical axis and replaced back into the insert **350**. The power supply unit **130** may be coupled to the door wiring harness **170**, and the plate **152** may be coupled to the hinge stile **42**.

(225) Referring to FIG. **32**, to invert the window control unit **132**, the plate **156** may be removed, and the window control unit **132** may be removed from the insert **420**. The window control unit **132** may be rotated about a vertical axis and replaced back into the insert **420**. The window control unit **132** may be coupled to the door wiring harness **170**, and the plate **156** may be coupled to the lock stile **46**. With one or more of the modules and the door wiring harness **170** inverted, the exterior skin **50** and the interior skin **52** may be coupled to the subframe **40**.

(226) Interactive Unit

(227) Referring to FIGS. **39-47**, the interactive unit **140** is shown according to an exemplary embodiment. In some embodiments, the interactive unit **140** utilizes the bridge controller **270** as a controller for the interactive unit **140**. Additionally or alternatively, the interactive unit **140** may utilize the main controller **242** as a controller for the interactive unit **140**. By way of example, the bridge controller **270** and the main controller **242** may be operatively coupled to the various input devices (e.g., sensors, user inputs, microphones, etc.) and output devices (e.g., lights, speakers, displays, etc.) of the interactive unit **140**. As shown in FIGS. **44-47**, the bridge module **136** is positioned within the interactive unit **140**. In some embodiments, the main controller **242** is additionally or alternatively positioned within the interactive unit **140**. In some embodiments, the bridge module **136** is received within a recess of the door **30** (e.g., the window controller recess **154**, one of the rail slots **400** or **402** that is free, a separate bridge recess, etc.). The bridge controller **270** and the main controller **242** may receive information from the input devices and/or provide commands to the output devices. Alternatively, the interactive unit **140** may include a dedicated controller separate from the bridge controller **270** and the main controller **242**. However, for simplicity, the functions of the interactive unit **140** described herein are described as being performed or controlled by the bridge controller **270**.

(228) Structure

(229) As shown in FIGS. **40-47**, the interactive unit **140** includes a first structural component, chassis, frame, or base, shown as exterior plate **500**. The exterior plate **500** is positioned on the exterior side of the door **30** and coupled to the subframe **40**. The exterior plate **500** serves as a structure to support various exterior side components of the interactive unit **140**. A perimeter of the exterior plate **500** is covered by a trim piece or cover, shown as exterior trim **502**. The exterior trim **502** is removably coupled to the exterior plate **500** (e.g., to facilitate swapping for different types of

trim). The exterior trim **502** may facilitate sealing the exterior side of the interactive unit **140**.
(230) The interactive unit **140** includes a second structural component, chassis, frame, or base, shown as interior core plate **504**, and a third structural component, chassis, frame, or base, shown as interior top plate **506**. The interior core plate **504** and the interior top plate **506** are positioned on the interior side of the door **30** and coupled to the subframe **40**. The interior core plate **504** and the interior top plate **506** may be removably coupled to one another. The interior core plate **504** and the interior top plate **506** serve as a structure to support various interior side components of the interactive unit **140**. A perimeter of the interior core plate **504** and the interior top plate **506** is covered by a trim piece or cover, shown as interior trim **508**. As shown in FIG. **43**, the interior trim **508** surrounds the interior core plate **504** and the interior top plate **506**. The interior trim **508** is removably coupled to the interior core plate **504** and the interior top plate **506**. The interior trim **508** may facilitate sealing the interior side of the interactive unit **140**.

(231) As shown, the exterior plate **500** includes a series of protrusions, shown as standoffs **510**, that extend through the subframe **40** toward the interior side of the door **30**. The interior core plate **504** and the interior top plate **506** are each removably coupled to one or more of the standoffs **510**. By way of example, a fastener (e.g., a bolt) may pass through the interior core plate **504** and threadedly engage one of the standoffs **510** to couple the interior core plate **504** to the exterior plate **500**. Similarly, a fastener (e.g., a bolt) may pass through the interior top plate **506** and threadedly engage one of the standoffs **510** to couple the interior top plate **506** to the exterior plate **500**. When assembled, the exterior skin **50**, the subframe **40**, and the interior skin **52** are positioned between (e.g., sandwiched by) (a) the exterior plate **500** and (b) the interior core plate **504** and the interior top plate **506**.

(232) In some embodiments, the exterior trim **502** and the interior trim **508** may be removed and replaced with another piece of exterior trim **502** or interior trim **508**, respectively. By way of example, the exterior trim **502** and/or the interior trim **508** may be replaced when damaged or worn. By way of another example, the exterior trim **502** and/or the interior trim **508** may be replaced with a trim piece having a different aesthetic or appearance (e.g., having a different material, having a different surface finish, having a different color, etc.). As shown in FIGS. **44** and **45**, the exterior trim **502** and the interior trim **508** each include a series of protrusions, shown as tabs **512**, extending inward to couple the exterior trim **502** and the interior trim **508** to the subframe **40**. The tabs **512** of exterior trim **502** are captured between the exterior plate **500** and the exterior skin **50**. The tabs **512** of the interior trim **508** are captured between (a) the interior core plate **504** and the interior top plate **506** and (b) the interior skin **52**. The exterior trim **502** and the interior trim **508** may be removed by first removing the exterior plate **500**, the interior core plate **504**, and the interior top plate **506** from the door **30** to free the tabs **512**. Alternatively, the exterior trim **502** and the interior trim **508** may be removed by bending the exterior trim **502** and the interior trim **508** outward until the tabs **512** are freed.

(233) Camera

(234) The interactive unit **140** further includes an image capture device or image sensor, shown as camera **520**, coupled to the exterior plate **500** and positioned on the exterior side of the door **30**. In some embodiments, the camera **520** is separate from the interactive unit **140** and otherwise positioned and/or the door **30** includes a second camera. By way of example, the camera **520** or the second camera may be positioned along the door frame **20** (e.g., one of the side jambs **22**, the head jamb **24**, etc.). The camera **520** is configured to capture image data (e.g., still images, video, etc.) of an area within a line of sight of the camera **520**. The camera **520** may be positioned facing outward such that the line of sight captures the exterior surroundings of the door **30**. By way of example, the camera **520** may capture image data regarding activity outside the door **30** such as a user standing in front of the door **30**, an animal nearby the door **30**, and/or a package positioned in front of the door **30**. In some embodiments, the camera **520** includes an actuator (e.g., an electric motor) that moves the camera **520** to vary the field of view, a direction of view, zoom in, and/or zoom out.

(235) The camera **520** may capture image data based on received light in a variety of different wavelengths. In some embodiments, the camera **520** captures image data based on visible light (e.g., light of the wavelength range that is visible to the human eye). Such a configuration may be useful during the day when the field of view is illuminated by the sun or at night when the field of view is artificially illuminated. In some embodiments, the camera **520** captures image data based on light that is not visible to the human eye, such as infrared light. Such a configuration may be useful at night when visible light is scarce. To facilitate capturing image data based on infrared light, the interactive unit **140** includes a pair of lamps (e.g., light-emitting diodes (LEDs)), shown as infrared emitters **522**. The infrared emitters **522** may be operatively coupled to and controlled by the bridge controller **270**. The infrared emitters **522** may be positioned to emit infrared light outward toward the field of view of the camera **520**. This infrared light may be reflected back toward the camera **520** by objects within the field of view of the camera **520**. As infrared light is not visible to the human eye, the infrared light may be used to capture image data at night without generating light pollution that would be considered undesirable by a user. In some embodiments, the camera **520** can capture image data based on visible light and image data based on infrared light such that the camera **520** can be used throughout the entire day.

(236) As shown, the camera **520** and the infrared emitters **522** are coupled to the exterior plate **500** by a plate or cover, shown as camera cover **524**. The camera cover **524** may surround the camera **520** without covering the view of the camera **520**, thereby avoiding interference with the field of view of the camera **520**. The camera cover **524** may be made from a material (e.g., plastic) that is permeable to a predetermined range of light wavelengths (e.g., infrared permeable, permitting infrared light to pass through the camera cover **524**). In such an embodiment, the camera cover **524** may obscure the infrared emitters **522** from view without hindering the performance of the infrared emitters **522**. The camera cover **524** may protect the infrared emitters **522** from external contaminants and reduce visual clutter on the interactive unit **140**.

(237) Display and Touch Inputs

(238) The interactive unit **140** further includes a display screen, shown as display **530**, positioned on the interior side of the door **30**. The display **530** is coupled to the interior top plate **506**, and the interior top plate **506** defines a cutout or aperture such that the display **530** is visible. The display **530** is configured to reproduce image data, displaying the corresponding image to a user positioned on the interior side of the door **30**. In some embodiments, the display **530** is configured to reproduce the image data captured by the camera **520**. The display **530** may provide a live video feed of the field of view of the camera **520**, providing the user on the interior side of the door **30** with visual information regarding the surroundings of the door **30** without requiring the window **60** to be changed to the viewing state. The display **530** may provide a relatively large reproduction of the field of view of the camera **520** that is visible at a distance from the door **30**.

(239) In some embodiments, the camera **520** and the display **530** are operatively coupled to the main control unit **134**. By way of example, the camera **520** and/or the display **530** may be coupled to the main control unit **134** by the door wiring harness **170**. Processing image data may require significant processing power and relatively large amounts of data. The use of a wired connection may facilitate transferring large amounts of data more quickly than a wireless connection.

Additionally, the main controller **242** may have more processing power than the bridge controller **270**, such that the main controller **242** may be better suited to image processing than the bridge controller **270**. The main control unit **134** may receive image data from the camera **520**. The main control unit **134** may command the display **530** to display an image (e.g., representing the image data captured by the camera **520**, a logo, a time, a date, daily weather, a selectable menu for updating settings, an authentication interface to receive manually entered credentials to changes settings and/or to approve installation of a new module, etc.).

(240) Referring to FIGS. **39**, **44**, and **45**, the interactive unit **140** further includes an input device, shown as touch input **532**. The touch input **532** is configured to register a location of a contact with

a human (e.g., where contact with a user's finger occurs) and provide corresponding touch input data. In some embodiments, the touch input **532** utilizes a capacitive touch sensor that is configured to identify human contact based on a measured change in capacitance. As shown in FIG. **44**, the touch input **532** is positioned behind the display **530** and is in engagement with the display **530** such that the touch input **532** registers when a user contacts the display **530**. Accordingly, the touch input **532** and the display **530** may cooperate together form a touchscreen or touch panel. The touchscreen may act as a user interface, displaying information to the user (e.g., footage from the camera **520**, a graphical user interface, etc.) and receiving inputs (e.g., a selection of an object on the display **530**, a command, a settings change, a key code to facilitate authorizing a settings change, etc.) from the user.

(241) User Authentication Inputs

(242) The interactive unit **140** may require a user to successfully perform an authentication prior to enabling one or more functions of the door system **10**. By way of example, the door system **10** may require a user to be authenticated prior to changing the door **30** from the locked state to the unlocked state or from the unlocked state to the locked state. By way of another example, the door system **10** may require a user to be authenticated prior to changing certain settings of the door system **10**.

(243) The interactive unit **140** may include components that are able to receive various types of user authentication inputs. By way of example, the interactive unit **140** may receive a passcode, a biometric input, or a communication from a fob or a user device **190**. These user authentication inputs may be evaluated to determine if the associated user or device has permission to access the door **30**. A user or device may be granted access over a predetermined time period (e.g., for one week, from October 1.sup.st to December 14.sup.th, etc.). Alternatively, a user or device may have permission to access the door **30** until their permissions are manually revoked (e.g., by a homeowner user through a user device **190**, by an owner using the touchscreen of the interactive unit **140**). The process of permitted users and evaluating user authentication inputs may be performed by components of the door **30**, user devices **190**, and/or servers **200**. Further details regarding the user authentication functionality of the door system **10** may be found in U.S. Patent Publication No. 2022/0319264, filed Jun. 17, 2022, which is incorporated herein by reference in its entirety.

(244) Keypad

(245) Referring to FIGS. **40** and **41**, an additional touch input **532** (e.g., a keypad) is usable by a user positioned on the exterior side of the interactive unit **140**. In such a configuration, the touch input **532** may be utilized to authenticate a user (e.g., to confirm that the user is permitted to unlock the door **30**). The touch input **532** defines a series of touch zones **534** on the exterior plate **500**. The touch input **532** is capable of identifying when each touch zone **534** is contacted by a user. The touch inputs **532** may be used as a keypad to authenticate a user. By way of example, an authorized user may be associated with (e.g., provided with) a predetermined contact pattern of the touch inputs **532** (e.g., corresponding to a passcode). By contacting the touch inputs **532** in the predetermined contact pattern, the user confirms their identity as an authorized user. In response to such a confirmation, the control system **100** may permit user to access the door **30** (e.g., by unlocking the door **30**).

(246) As shown in FIG. **41**, each touch zone **534** has an identifier or symbol (e.g., letters, numbers, etc.), shown as a number. The identifiers may facilitate the user accurately identifying and contacting the touch zones **534** in the predetermined contact pattern. By way of example, an authorized user may be provided with a passcode (e.g., a series of symbols such as 8490, 0933, etc.) corresponding to the predetermined contact pattern. By pressing the touch zones **534** with the identifiers of the passcode in the order of the passcode, the user may contact the touch zones **534** in the predetermined contact pattern. Advantageously, the passcode may be easier to transmit to the user or easier for the user to remember than the shape of the predetermined contact pattern.

(247) In some embodiments, each touch zone **534** includes an associated light or illuminator that selectively illuminates the identifiers such that the numbers are selectively visible. By way of example, the lights may be disabled to put the interactive unit **140** into a “sleep mode” (e.g., as shown in FIG. **40**) in which the interactive unit **140** is minimally visible and consumes minimal energy. In response to a user interaction (e.g., motion sensed outside the door, the user touching the touch input **532**, etc.), the lights may illuminate to put the interactive unit **140** into a “wake mode” (e.g., as shown in FIG. **41**), in which the identifiers are visible. In other embodiments, the identifiers are printed, embossed, individual buttons, or otherwise permanently visible on the surface of the exterior plate **500**.

(248) NFC Radio

(249) Referring to FIGS. **39** and **44-47**, the interactive unit **140** further includes a short range communication module or radio, shown as near-field communication (NFC) radio **540**. The NFC radio **540** is configured to communicate wirelessly with one or more external devices (e.g., user devices **190**, key fobs, etc.). The NFC radio **540** may be configured to communicate through NFC. In other embodiments, the NFC radio **540** is configured to communicate through another type of communication protocol, such as another type of radio-frequency identification (RFID).

(250) The NFC radio **540** includes a controller, adapter, driver, or receiver, shown as NFC adapter **542**, and a broadcast device or antenna, shown as NFC antenna **544**. The NFC antenna **544** is operatively coupled to the NFC adapter **542** such that the NFC adapter **542** communicates through the NFC antenna **544**. The NFC adapter **542** may send and/or receive wireless signals through the NFC antenna **544**. The NFC adapter **542** may be operatively coupled to the bridge controller **270** such that the bridge controller **270** controls operation of the NFC radio **540**.

(251) The exterior plate **500** includes a protrusion or disc, shown as NFC cover **546**. The NFC radio **540** is positioned directly behind the NFC cover **546** and aligned with the NFC cover **546**. Accordingly, a user may align an external device (e.g., a fob, a user device **190**, etc.) with the NFC radio **540** by placing the external device adjacent the NFC cover **546**. The NFC cover **546** may be made from a material (e.g., plastic) that minimizes interference with the wireless communication of the NFC radio **540**.

(252) The NFC radio **540** may communicate with one or more external devices to authenticate a user. By way of example, the NFC adapter **542** may control the NFC antenna **544** to emit a first signal. The first signal may contain information identifying one or more components of the door **30** (e.g., a device identifier, a device key, etc.). In response to receiving the first signal, an external device (e.g., a fob, a user device **190**, etc.) may emit a second signal back to the NFC radio **540**. The second signal may contain information identifying the external device (e.g., a device indicator, a device key, etc.) and/or information identifying a user associated with the external device (e.g., a user indicator, a user key, etc.). The NFC antenna **544** may receive the second signal and transfer the second signal to the NFC adapter **542**. The NFC adapter **542** may transfer the information from the second signal to the bridge controller **270** and/or the main controller **242**. The user device **190**, the bridge module **136**, the main control unit **134**, and/or a server **200** may cooperate to authenticate the user and/or the external device. In response to successfully confirming that the user is an authenticated user and/or that the external device is an authenticated device, the control system **100** may permit the user to access the door **30** (e.g., by unlocking the door **30**).

(253) In some embodiments, the NFC radio **540** is configured to communicate with a key fob. By way of example, a first user (e.g., a homeowner) may provide a second user (e.g., a guest) with a key fob (e.g., an RFID tag) containing a device indicator. In some embodiments, the NFC radio **540** is configured to communicate with a user device **190**. By way of example, a first user (e.g., a homeowner) may authenticate their own user device **190** and/or a user device **190** of a second user (e.g., a guest). Each user device could be associated with a device indicator.

(254) Through a user device **190**, the first user may set a device indicator (e.g., associated with a key fob or user device **190**) as being an authorized device for a predetermined period of time (e.g.,

one day, one week, one month, etc.), after which the permission will expire. Such a configuration may be useful when the second user will be occupying the home for a short period of time (e.g., as a short term rental), after which the second user should not be able to access the home. Alternatively, the first user may set the device indicator as being an authorized device until the first user manually revokes the permission.

(255) Biometric Input

(256) Referring to FIG. 39, the interactive unit 140 includes a biometric input device or biometric device, shown as fingerprint scanner 550. The fingerprint scanner 550 is operatively coupled to the bridge module 136. The fingerprint scanner 550 is configured to acquire a biometric credential of a user (e.g., a biological characteristic or trait unique to a user). By way of example, the biometric credential may include a fingerprint scan (e.g., an image of a user's fingerprint or the relative positions of key portions of a user's fingerprint). In other embodiments, the fingerprint scanner 550 is replaced by another type of biometric sensor (e.g., a camera, a microphone, etc.) that receives a different type of biometric credential, such as a retinal scan (e.g., an image of a user's eye or the relative positions of key portions of a user's eye), a voice command, or a facial scan.

(257) The fingerprint scanner 550 may be used to authenticate a user. By way of example, the biometric credential retrieved by the fingerprint scanner 550 may be used to authenticate the user. As the biometric credential is unique to each user, the biometric credential may be used as a user identifier. The biometric credential may be compared to a database of biometric credentials of known users to determine if the biometric credential provided by the fingerprint scanner 550 matches a biometric credential of an authenticated user. This process may be performed by a user device 190, a server 200, the bridge module 136, the main control unit 134, and/or another controller. In response to successfully confirming that biometric credential matches an authenticated user, the control system 100 may permit the user to access the door 30 (e.g., by unlocking the door 30).

(258) Acoustic Components

(259) Referring to FIGS. 39-47, the interactive unit 140 includes various components that facilitate auditory, acoustic, or verbal communication through the door 30. By way of example, such components may facilitate communication between a user positioned on the interior side of the door 30 and a user positioned on the exterior side of the door 30. In such an example, the components permit two users to speak clearly to one another without having to open the door 30 (e.g., which may be undesirable due to security concerns). By way of another example, the components may facilitate communication between a remote user and a user positioned on the exterior side of the door 30. In such an example, the components may facilitate communication even if a user (e.g., a homeowner) is not nearby the door 30.

(260) The interactive unit 140 includes a pair of receivers or sound recorders, shown as external microphone 560 and internal microphone 562. The external microphone 560 is coupled to and positioned behind the exterior plate 500. Specifically, the external microphone 560 is positioned to be exposed to sounds originating on the exterior side of the door 30. The exterior plate 500 may define one or more apertures to facilitate the transfer of sound (e.g., voices, ambient sounds, etc.) to the external microphone 560. The external microphone 560 may record these sounds and generate a corresponding external audio signal representing the recorded sounds. The external microphone 560 may be operatively coupled to the bridge module 136 such that the external microphone 560 provides the external audio signal to the bridge controller 270.

(261) The internal microphone 562 is coupled to and positioned behind the interior core plate 504. Specifically, the internal microphone 562 is positioned to be exposed to sounds originating on the interior side of the door 30. The interior core plate 504 may define one or more apertures to facilitate the transfer of sound (e.g., voices, ambient sounds, etc.) to the internal microphone 562. The internal microphone 562 may record these sounds and generate a corresponding internal audio signal representing the recorded sounds. The internal microphone 562 may be operatively coupled

to the bridge module **136** such that the internal microphone **562** provides the internal audio signal to the bridge controller **270**.

(262) The interactive unit **140** further includes a pair of monitors, speakers, or sound generators, shown as external speaker **564** and internal speaker **566**. The external speaker **564** and the internal speaker **566** may be electrically coupled to an amplifier **568** that provides electrical energy to power the external speaker **564** and the internal speaker **566**. The external speaker **564** is coupled to and positioned behind the exterior plate **500** and operatively coupled to the bridge controller **270**. The external speaker **564** may receive an external sound output signal from the bridge controller **270**, the external sound output signal representing a desired sound. In response to receiving the external sound output signal, the external speaker **564** may generate the corresponding desired sound. The external speaker **564** may be positioned and oriented to direct the generated sound away from the door **30** on the exterior side of the door **30**. The exterior plate **500** may define one or more apertures to facilitate the transfer of sound from the external speaker **564** to the surrounding environment.

(263) The internal speaker **566** is coupled to and positioned behind the interior core plate **504** and operatively coupled to the bridge controller **270**. The internal speaker **566** may receive an internal sound output signal from the bridge controller **270**, the internal sound output signal representing a desired sound. In response to receiving the internal sound output signal, the internal speaker **566** may generate the corresponding desired sound. The internal speaker **566** may be positioned and oriented to direct the generated sound away from the door **30** on the interior side of the door **30**. The interior core plate **504** may define one or more apertures to facilitate the transfer of sound from the internal speaker **566** to the surrounding environment.

(264) In some embodiments, the exterior plate **500** includes one or more baffles or chambers that limit or prevent the sound generated by the external speaker **564** from reaching the external microphone **560**. In some embodiments, the interior core plate **504** includes one or more baffles or chambers that limit or prevent the sound generated by the internal speaker **566** from reaching the internal microphone **562**. Referring to FIGS. **44-47**, the interactive unit **140** includes a divider, shown as baffle **569**, that divides the internal volume of the interactive unit **140**. Specifically, the baffle **569** separates (a) the external microphone **560** and the external speaker **564** from (b) the internal microphone **562** and the internal speaker **566**. The baffle **569** may acoustically isolate the external components from the internal components, preventing the generated and received sounds from bleeding between components unintentionally.

(265) In a first mode of operation, the external microphone **560**, the internal microphone **562**, the external speaker **564**, and the internal speaker **566** can be utilized to facilitate verbal communication between a first user on the interior side of the door **30** and a second user on the exterior side of the door **30**. In this first mode, the external microphone **560** records the speech of the second user, and the bridge controller **270** uses the corresponding external audio signal to generate an internal sound output signal. The bridge controller **270** sends the internal sound output signal to the internal speaker **566**, which replicates the speech of the second user. Similarly, the internal microphone **562** records the speech of the first user, and the bridge controller **270** uses the corresponding internal audio signal to generate an external sound output signal. The bridge controller **270** sends the external sound output signal to the external speaker **564**, which replicates the speech of the first user. In this way, the first user and the second user may have a verbal conversation through the door **30**.

(266) In some embodiments, the interactive unit **140** includes an input device or user interface element (e.g., a button, a switch, etc.), shown as intercom button **570**. The intercom button **570** may be coupled to the interior core plate **504** and accessible from the interior side of the door **30**. In an embodiment including the intercom button **570**, the bridge controller **270** may be configured to prevent the replication of the speech of the first user by the internal microphone **562** and the external speaker **564** unless the intercom button **570** is pressed (e.g., by the first user). The first

user may therefore utilize the intercom button **570** to control when the second user is permitted to hear the speech of the first user.

(267) In a second mode of operation, the external microphone **560** and the external speaker **564** can be utilized to facilitate verbal communication between a first user positioned remotely from the door **30** and a second user on the exterior side of the door **30**. By way of example, this communication may occur through a network (e.g., Internet, cellular, etc.) connection between the bridge controller **270** and a user device **190**. In this second mode, the external microphone **560** records the speech of the second user, and the bridge controller **270** instructs the user device **190** to reproduce the speech of the second user for the first user. The user device **190** records the speech of the first user, and the bridge controller **270** commands the external speaker **564** to reproduce the speech of the first user. This second mode of operation may be useful when the first user is positioned in a different room of the home away from the door **30** or outside of or away from the home entirely.

(268) Referring to FIGS. **39-41**, the interactive unit **140** includes an input device or user interface element (e.g., a button, a switch, etc.), shown as doorbell button **572**. The doorbell button **572** may be coupled to the exterior plate **500** and accessible from the exterior side of the door **30**. The doorbell button **572** is operatively coupled to the bridge controller **270**. When pressed, the doorbell button **572** generates a doorbell signal indicating that a user wishes to provide a notification inside of the home. In response to receiving the doorbell signal, the bridge controller **270** may cause the internal speaker **566** to provide a notification (e.g., an auditory notification, such as a ringing bell or siren). Additionally or alternatively, the bridge module **136** may transmit the doorbell signal to the doorbell module **184**, and the doorbell module **184** may cause the doorbell speaker **186** to provide a notification. In some embodiments, pressing the doorbell button **572** initiates recording of sound by the external microphone **560** (e.g., for reproduction by the internal speaker **566** and/or the user device **190**). By way of example, in response to a user pressing the doorbell button **572**, the door system **10** may initiate a telephone call between the interactive unit **140** and a user device **190** of a homeowner. The doorbell button **572** may replace the external doorbell **182**, when included. In some embodiments, the interactive unit **140** does not include the doorbell button **572**.

(269) Referring to FIGS. **40-47**, the interactive unit **140** includes a pair of removable covers or speaker grills, shown as mesh cover **574** and mesh cover **576**, that protect the speakers and microphones of the interactive unit **140**. The mesh cover **574** and the mesh cover **576** may define a series of apertures or perforations that facilitate the flow of sound and/or air through the mesh cover **574** and the mesh cover **576**. The mesh cover **574** is removably coupled to the exterior plate **500**. The mesh cover **574** covers the external microphone **560** and the external speaker **564**, obscuring the external microphone **560** and the external speaker **564** from sight. The mesh cover **574** defines an aperture through which the doorbell button **572** extends. The mesh cover **576** is removably coupled to the interior core plate **504**. The mesh cover **576** covers the internal microphone **562** and the internal speaker **566**, obscuring the internal microphone **562** and the internal speaker **566** from sight. The mesh cover **576** defines an aperture through which the intercom button **570** extends.

(270) Privacy Button

(271) Referring to FIGS. **39, 42, 43**, and **45-47**, the interactive unit **140** includes an input device or user interface element (e.g., a button, a switch, etc.), shown as privacy button **580**. The privacy button **580** may be coupled to the interior core plate **504** and accessible from the interior side of the door **30**. The privacy button **580** is operatively coupled to the bridge controller **270**. When pressed by a user, the privacy button **580** provides an indication to the bridge controller **270** that the user wishes to change a state of the window **60** (e.g., to the privacy state, to the viewing state, etc.). In response to the indication, the bridge controller **270** may command the window control unit **132** to change a state of the window **60**. By way of example, the window control unit **132** may change the window **60** from the privacy state to the viewing state. By way of another example, the window

control unit **132** may change the window **60** from the viewing state to the privacy state. Accordingly, pressing the privacy button **580** multiple times may cycle the window **60** between the privacy state and the viewing state, or various states therebetween (e.g., partially tinted, partially opaque, etc.).

(272) External Charge Port

(273) Referring to FIG. **39**, the interactive unit **140** includes a port (e.g., a USB-C port, a micro-USB port, a power port, etc.), shown as external power port **582**. The external power port **582** may be defined along the exterior plate **500** or the exterior trim **502**. According to an exemplary embodiment, the external power port **582** is configured to facilitate connecting an external power source (e.g., a portable battery pack, bank, etc.) to the interactive unit **140**. By way of example, the power source **102** may be disconnected from the door **30** (e.g., a power outage, the door **30** has not been hardwired into the building grid, etc.) and the backup battery **222** may be dead. Accordingly, by connecting the external power source to the external power port **582**, the user can power the interactive unit **140** and, therefore, the unlocking capabilities of the door **30**. Therefore, the user can gain entry through the door **30** during a power outage even if the backup battery **222** is also dead.

(274) Status Lights

(275) Referring to FIGS. **39**, **40**, **42**, and **44-47**, the interactive unit **140** includes a pair of indicators, status lights, or output devices, shown as external status light **590** and internal status light **592**, that generate and emit visible light to visually convey information to a user. The external status light **590** is coupled to the exterior plate **500**. The external status light **590** is positioned on an exterior face of the interactive unit **140** such that the external status light **590** emits first, exterior light in an outward, external direction. Accordingly, the exterior light emitted by the external status light **590** may be visible to a user positioned on the exterior side of the door **30**. According to an exemplary embodiment, the external status light **590** is annular and surrounds the NFC cover **546**. Accordingly, the external status light **590** provides a visual indication guiding a user toward the NFC antenna **544**. The external status light **590** may illuminate one or more components on the exterior side of the interactive unit **140** (e.g., to facilitate visibility by a user).

(276) The internal status light **592** is coupled to the interior core plate **504**. The internal status light **592** is positioned on an interior face of the interactive unit **140** such that the internal status light **592** emits second, interior light in an inward, internal direction. Accordingly, the interior light emitted by the internal status light **592** may be visible to a user positioned on the interior side of the door **30**. According to an exemplary embodiment, the internal status light **592** is annular and surrounds the privacy button **580** and lock control interface **620**. Accordingly, the internal status light **592** provides a visual indication guiding a user toward the privacy button **580** and the lock control interface **620**. The internal status light **592** may illuminate one or more components on the interior side of the interactive unit **140** (e.g., to facilitate visibility by a user).

(277) The external status light **590** and the internal status light **592** may each include one or more light generators, such as LEDs or incandescent bulbs. In some embodiments, the external status light **590** and/or the internal status light **592** include multiple generators that are positioned in a ring to provide the annular shapes of the external status light **590** and the internal status light **592**. In some embodiments, the external status light **590** and/or the internal status light **592** include a diffuser or light guide that guides the exterior light and the interior light in annular patterns.

(278) The external status light **590** and the internal status light **592** are operatively coupled to the bridge controller **270**. The bridge controller **270** may control the operation of the external status light **590** and the internal status light **592**. By way of example, the bridge controller **270** may change a color, an on/off state, a flashing pattern, a flashing speed, or other characteristics of the exterior light and the interior light. The bridge controller **270** may control the external status light **590** and the internal status light **592** together (e.g., such that the external status light **590** and the internal status light **592** emit the same color light simultaneously). The bridge controller **270** may

control the external status light **590** and the internal status light **592** independent of one another (e.g., such that the external status light **590** emits light while the internal status light **592** is switched off, such that the external status light **590** and the internal status light **592** emit light of different colors, etc.).

(279) In some embodiments, the external status light **590** and/or the internal status light **592** are utilized to provide status information regarding a status of the door system **10**. In some embodiments, the external status light **590** and/or the internal status light **592** provide information regarding a locked status or unlocked status of the door **30**. By way of example, the external status light **590** and/or the internal status light **592** may provide a first visual indication (e.g., light of a certain color, turn on, turn off, flash in a certain pattern, etc.) to indicate that the door **30** is in a locked state and provide a different visual indication to indicate that the door is in the unlocked state. By way of another example, the external status light **590** and/or the internal status light **592** may provide a visual indication when the door **30** changes between the locked state and the unlocked state to provide a visual confirmation of the change. In some embodiments, the external status light **590** and/or the internal status light **592** provide information regarding an operational status of the door **30**. By way of example, the external status light **590** and/or the internal status light **592** may provide a first visual indication (e.g., light of a certain color, turn on, turn off, flash in a certain pattern, etc.) to indicate that the door **30** is in an operational state and provide a different visual indication to indicate that the door **30** has experienced an error. In some embodiments, the external status light **590** and/or the internal status light **592** provide information regarding the presence of a user on the opposite side of the door **30**. By way of example, the internal status light **592** may provide a visual indication in response to an input (e.g., from the camera **520**, from the doorbell button **572**, from contact with the touch zones **534**, from the NFC radio **540**, etc.) indicating that a user is positioned on the exterior side of the door **30**.

(280) Time of Flight Sensors

(281) Referring to FIGS. **39**, **41**, and **43-47**, the interactive unit **140** includes a pair of presence sensors, distance sensors, time of flight (TOF) sensors, or input devices, shown as external TOF sensor **594** and internal TOF sensor **596**. The external TOF sensor **594** is coupled to the exterior plate **500** and faces in an outward, exterior direction. The internal TOF sensor **596** is coupled to the interior core plate **504** and faces in an inward, interior direction. The external TOF sensor **594** and the internal TOF sensor **596** are operatively coupled to the bridge controller **270**. The external TOF sensor **594** and the internal TOF sensor **596** communicate sensor data to the bridge controller **270**.

(282) The external TOF sensor **594** and the internal TOF sensor **596** provide sensor data (e.g., a signal, information, etc.) indicating a distance between the sensor and an object in the environment. Specifically, the external TOF sensor **594** provides distance data indicating a distance between the external TOF sensor **594** and an object on an exterior side of the door **30** (e.g., a user, an animal, a box, a vehicle, etc.). The internal TOF sensor **596** provides distance data indicating a distance between the internal TOF sensor **596** and an object on an interior side of the door **30**.

(283) In some embodiments, the external TOF sensor **594** and the internal TOF sensor **596** are each time of flight sensors that measure a distance to an object by releasing a pulse of light, waiting for the light to be reflected by the object, and recording the time delay between when the initial pulse is released and the reflected light is received. Based on this measured delay, the TOF sensor calculates the distance. In other embodiments, the external TOF sensor **594** and/or the internal TOF sensor **596** utilize a different type of distance sensor, such as a LiDAR sensor or an ultrasonic sensor. In other embodiments, the external TOF sensor **594** and/or the internal TOF sensor **596** use a camera or another type of sensor to detect the presence, location, or motion of an object.

(284) The bridge controller **270** may utilize the sensor data from the external TOF sensor **594** and the internal TOF sensor **596** to identify the presence of a user. The bridge controller **270** may determine if a user is present on the exterior side of the door **30** based on sensor data from the external TOF sensor **594**. The bridge controller **270** may determine if a user is present on the

interior side of the door **30** based on sensor data from the internal TOF sensor **596**. By way of example, the bridge controller **270** may determine that a user is present near the door **30** in response to the external TOF sensor **594** or the internal TOF sensor **596** detecting an object within a threshold distance of the door **30**. By way of another example, the bridge controller **270** may determine that a user is present near the door **30** in response to the external TOF sensor **594** or the internal TOF sensor **596** detecting that an object is moving.

(285) In response to determining that an object is present, the bridge controller **270** may activate one or more parts of the door system **10**. This may facilitate automatic activation of certain functions as soon as a user approaches the door **30** rather than requiring the user to interact with the door (e.g., by pushing buttons). By way of example, in response to a user being detected on the exterior side of the door **30**, the bridge controller **270** may (a) activate the external status light **590** (e.g., illuminate, flash, etc.) to provide light and guide the user, (b) activate the camera **520** to begin recording images or video, (c) provide a notification to a user device **190**, (d) provide the recorded images or video for display on the user device **190** and/or the display **530**, (e) activate the external microphone **560** and the internal speaker **566** to begin transferring sound from the exterior side of the door **30** to the interior side of the door **30**, (f) activate the internal status light **592** to visually indicate the presence of someone or something outside the door **30**, and/or (g) control other functions. By way of another example, in response to a user being detected on the interior side of the door **30**, the bridge controller **270** may (a) activate the internal status light **592** to provide light and guide the user, (b) activate the camera **520** to begin recording images or video, (c) provide the recorded images or video for display on the display **530**, (d) activate the external microphone **560** and the internal speaker **566** to begin transferring sound from the exterior side of the door **30** to the interior side of the door **30**, and/or (e) control other functions.

(286) Accelerometer Unit

(287) Referring to FIGS. **39** and **44-47**, the interactive unit **140** includes a movement sensor or input device, shown as accelerometer unit **600**. The accelerometer unit **600** is positioned between the exterior plate **500** and the interior core plate **504** and coupled to the exterior plate **500**. The accelerometer unit **600** is operatively coupled to the bridge controller **270**. The accelerometer unit **600** provides sensor data to the bridge controller **270**. Specifically, the accelerometer unit **600** is configured to provide sensor data describing a movement of the accelerometer unit **600**. Due to the coupling of the accelerometer unit **600** to the exterior plate **500**, the sensor data from the accelerometer unit **600** describes the movement of the door **30**.

(288) The accelerometer unit **600** may measure movement along one or more axes (e.g., a lateral axis, a vertical axis, a longitudinal axis). The accelerometer unit **600** may measure rotation about one or more axes. The accelerometer unit **600** may measure or be used to determine position, velocity, and/or acceleration. The accelerometer unit **600** may include one or more accelerometers, gyroscopes, or inertial measurement units (IMUs).

(289) The bridge controller **270** may use the sensor data from the accelerometer unit **600** to determine if the door is being moved (e.g., opened or closed). By way of example, if the velocity of the door **30** is over a threshold velocity (e.g., 0 m/s, 10 degrees/s, etc.), the bridge controller **270** may determine that the door **30** is being moved. By way of example, if the acceleration of the door **30** is over a threshold acceleration (e.g., 0 m/s², 1 degree/s², etc.), the bridge controller **270** may determine that the door **30** is being moved. In response to a determination that the door **30** is being moved, the bridge controller **270** may provide a notification. By way of example, the bridge controller **270** may command a user device **190** to provide a push notification indicating that the door **30** is being moved. By way of another example, the bridge controller **270** may command the doorbell module **184** and/or a user device **190** to provide an alarm indicating that a break in may be occurring in response to a determination that the door **30** is moving while the door is in the locked state.

(290) Lux Sensor

(291) Referring to FIGS. 39 and 44-47, the interactive unit 140 includes a light sensor, luminosity sensor, lux sensor, or input device, shown as lux sensor 602. The lux sensor 602 is positioned between the exterior plate 500 and the interior core plate 504 and coupled to the exterior plate 500. The lux sensor 602 is operatively coupled to the bridge controller 270. The lux sensor 602 provides sensor data (e.g., light data) to the bridge controller 270. Specifically, the lux sensor 602 is configured to provide light data describing a light level (e.g., intensity, brightness, etc.) of light in the environment surrounding the door 30.

(292) In some embodiments, the lux sensor 602 is positioned to be exposed to the exterior side of the door 30, such that light from the exterior side of the door 30 reaches the lux sensor 602. Accordingly, the light data from the lux sensor 602 varies based on a light level of the environment surrounding the exterior side of the door 30. The lux sensor 602 may provide an indication of the level of visibility on the exterior side of the door 30. By way of example, when the light data indicates a light level below a threshold light level, the bridge controller 270 may determine that the exterior environment has low visibility conditions. By way of another example, when the light data indicates a light level above the threshold light level, the bridge controller 270 may determine that the exterior environment has high visibility conditions. In some embodiments, the light data indicates the time of day and/or the presence of a storm.

(293) The bridge controller 270 may control operation of the door system 10 based on the light data. By way of example, if the bridge controller 270 determines that the exterior environment has high visibility conditions based on the light data, the bridge controller 270 may disable the external status light 590. By way of another example, if the bridge controller 270 determines that the exterior environment has low visibility conditions based on the light data, the bridge controller 270 may activate the infrared emitters 522 whenever the camera 520 is activated. By way of another example, if the bridge controller 270 determines that the exterior environment has high visibility conditions based on the light data, the bridge controller 270 may change the window 60 to the privacy state to prevent people from looking into the home through the window 60.

(294) Ambient Condition Unit

(295) Referring to FIGS. 39 and 44-47, the interactive unit 140 includes a temperature sensor, humidity sensor, ambient condition sensor, or input device, shown as ambient condition unit 604. The ambient condition unit 604 is positioned between the exterior plate 500 and the interior core plate 504 and coupled to the exterior plate 500. The ambient condition unit 604 is operatively coupled to the bridge controller 270. The ambient condition unit 604 provides sensor data (e.g., ambient condition data) to the bridge controller 270. Specifically, the ambient condition unit 604 provides ambient condition data describing an ambient condition, such as temperature and/or humidity, in the environment surrounding the door 30. The ambient condition data may include temperature data, humidity data, or data relating to another ambient condition.

(296) The ambient condition unit 604 may be exposed to (e.g., fluidly coupled to) the environment surrounding the door 30. In some embodiments, the ambient condition unit 604 is fluidly coupled to the atmosphere on the exterior side of the door 30 (e.g., through an aperture defined by the exterior plate 500). In some embodiments, the ambient condition unit 604 is fluidly coupled to the interior atmosphere on the interior side of the door 30 (e.g., through an aperture defined by the interior core plate 504). Accordingly, the ambient condition data from the ambient condition unit 604 varies based on an ambient condition of the environment surrounding the exterior side of the door 30 or the interior side of the door 30. In some embodiments, the ambient condition unit 604 includes a temperature sensor configured to provide temperature data indicating an ambient temperature on the exterior side of the door 30. In some embodiments, the ambient condition unit 604 includes a humidity sensor configured to provide humidity data indicating a humidity of the atmosphere on the exterior side of the door 30. Additionally or alternatively, the ambient condition unit 604 may measure a temperature and/or a humidity on the interior side of the door 30. In some embodiments, the door system 10 utilizes data from the ambient condition unit 604 to determine

the current season (e.g., spring, summer, fall, winter).

(297) The bridge controller **270** may control operation of the door system **10** based on the ambient condition data. By way of example, if the bridge controller **270** determines that the ambient temperature on the exterior side of the door **30** is above a threshold temperature, the bridge controller **270** may change the window **60** to the privacy state to mitigate the a temperature increase on the interior side of the door **30**. By way of another example, if the bridge controller **270** determines that (a) the ambient temperature on the exterior side of the door **30** is above the ambient temperature on the interior side of the door **30** and (b) the ambient temperature on the interior side of the door **30** is below a user setpoint, the bridge controller **270** may change the window **60** to the viewing state to permit light to pass through the window **60** and increase the temperature within the home. By way of another example, if the bridge controller **270** determines that the humidity on the exterior side of the door **30** is above a threshold, the bridge controller **270** may provide a push notification to a user device **190** instructing the user to close their other windows and doors.

(298) UWB Sensors

(299) Referring to FIG. **39**, the interactive unit **140** includes a plurality of ultra-wideband (“UWB”) sensors (e.g., three UWB sensors), shown as UWB sensors **606**. According to an exemplary embodiment, the UWB sensors **606** are disposed within the housing of the interactive unit **140** and spaced about the camera **520**. In some embodiments, the UWB sensors **606** are otherwise positioned about the interactive unit **140**. In some embodiments, the UWB sensors **606** are additionally or alternatively positioned about and/or within the door frame **20** and/or the door **30**. According to an exemplary embodiment, the UWB sensors **606** are operatively coupled to the control system **100** (e.g., the bridge controller **270**, the main controller **242**, etc.).

(300) According to an exemplary embodiment, the UWB sensors **606** are configured to transmit and receive short-range wireless signals to and from the user devices **190** (e.g., a smartphone, a key fob, etc.). The short-range wireless signals may be UWB signals in the UWB spectrum, which may be generally between 3.1 GHz and 10.6 GHz. Based on the transmission and receipt of the UWB signals via the UWB sensors **606**, the control system **100** may configured to determine a distance of the user device **190** relative to the door **30** in three-dimensional coordinate space. More specifically, the control system **100** may be configured to provide angle of arrival technology and real-time locations services (“RTLS”) using the UWB sensors **606** to detect and triangulate the location of the user device **190** to permit “hands free” authentication and unlock functionality (e.g., the user may only have to touch a touch capacitive portion of the door **30**, the user does not need to touch any component of the door **30** for unlock function, etc.). By way of example, the control system **100** may be configured to analyze the UWB signals to determine whether the user device **190** is on the exterior side of the door **30**, on the interior side of the door **30**, to the side of the door **30** but inside a residence, to the side of the door **30** but outside of the residence, above the door **30** (e.g., on a second floor, on a balcony, on a deck, etc.), and/or below the door **30** (e.g., in a basement, on a lower level patio, etc.). The control system **100** may control operation of the door system **10** based on the UWB signals. By way of example, the control system **100** may unlock or lock the door **30** based at least partially on the UWB signals.

(301) Radar Sensor

(302) Referring to FIG. **39**, the interactive unit **140** includes at least one radar sensor, shown as radar sensor **608**. The radar sensor **608** may be coupled to or disposed within the exterior plate **500** of the interactive unit **140** and face in an outward, exterior direction. In some embodiments, the radar sensors **608** are additionally or alternatively positioned within and/or about the door frame **20** and/or the door **30**. According to an exemplary embodiment, the radar sensor **608** is operatively coupled to the control system **100** (e.g., the bridge controller **270**, the main controller **242**, etc.).

(303) The radar sensor **608** provides sensor data (e.g., a signal, information, etc.) indicating a distance between the radar sensor **608** and an object in the environment (e.g., a user/person, an animal, a box, a vehicle, etc.). Specifically, the radar sensor **608** provides (a) presence data

indicating a presence of the object on the exterior side of the door **30** and/or (b) distance data indicating a distance between the radar sensor **608** and the object. In some embodiments, the radar sensor **608** is a 60 GHz millimeter wave radar detector with three-dimensional detection capabilities.

(304) The control system **100** may control operation of the door system **10** based on the presence data and/or the distance data. By way of example, the control system **100** may active the camera **520** based on the presence data and/or the distance data. By way of another example, the control system **100** may active lights (e.g., the jamb lights **790**, the handle light **792**, etc.) based on the presence data and/or the distance data. By way of yet another example, the control system **100** may unlock the door **30** based at least partially on the presence data and/or the distance data.

(305) Sliding Lock Control Interface

(306) Referring to FIGS. **39** and **42-48**, the interactive unit **140** includes a user interface, slider, sliding interface, lock control interface, input device, or output device, shown as lock control interface **620**. The lock control interface **620** is coupled to the interior core plate **504** and accessible from the interior side of the door **30**. The lock control interface **620** includes an interface member or interior slider, shown as slider **622**, that is configured to move laterally between a locked position and an unlocked position offset from the locked position. The slider **622** is moved to the locked position to indicate that the door **30** is in the locked state. Similarly, the slider **622** is moved to the unlocked position to indicate that the door **30** is in the unlocked state.

(307) The lock control interface **620** includes a pair of input devices or sensors, shown as lock switches **624**, that detect the position of the slider **622**. The lock switches **624** are both operatively coupled to the bridge controller **270**. Specifically, a first lock switch **624** detects when the slider **622** is in the locked position. A second lock switch **624** detects when the slider **622** is in the unlocked position. A user positioned on the interior side of the door **30** may grab the slider **622** and push or drag the slider **622** between the locked position and the unlocked position. When the user moves the slider **622** to the locked position, the first lock switch **624** provides a signal to the bridge controller **270**, and the bridge controller **270** changes the door **30** to the locked state (e.g., by commanding the deadbolt units **138** to extend the deadbolts **252**). When the user moves the slider **622** to the unlocked position, the second lock switch **624** provides a signal to the bridge controller **270**, the bridge controller **270** changes the door **30** to the unlocked state (e.g., by commanding the deadbolt units **138** to retract the deadbolts **252**).

(308) In some embodiments, the door system **10** permits a user to change the door **30** between the locked state and the unlocked state remotely (e.g., while positioned away from the door **30**). By way of example, a user may choose to unlock the door **30** remotely using a user device **190** to permit a delivery service to deposit a package in the home. While locking or unlocking the door **30** can be accomplished through electronic control over the deadbolt units **138**, the position of the slider **622** may not be physically tied to the operation of the deadbolt units **138**. Accordingly, a situation could arise where the position of the slider **622** does not correspond to the current locked or unlocked state of the door **30**. To avoid this situation, the lock control interface **620** includes an actuator or output device, shown as slider actuator **626**, that can effect movement of the slider **622**. The slider actuator **626** is operatively coupled to the bridge controller **270** and controlled by the bridge controller **270**. When a command from a remote user changes the door **30** between the locked state and the unlocked state, the bridge controller **270** controls the slider actuator **626** to move the slider **622** to accurately reflect the current state of the door **30**. Beneficially, the slider actuator **626** can mimic the operation of a traditional mechanical deadbolt without having to mechanically tie or link the position of the slider **622** to the operation of the deadbolt units **138**.

(309) Referring to FIGS. **44-49**, the structure of the lock control interface **620** is shown according to an exemplary embodiment. The lock control interface **620** includes a frame or base, shown as slider plate **630**. The slider plate **630** is fixedly coupled to the interior core plate **504**. The slider plate **630** extends through an aperture defined by the interior core plate **504** and is exposed on the

interior face of the interactive unit **140**. The slider plate **630** defines a passage, shown as slot **632**, extending through the slider plate **630**. The slot **632** has a length oriented laterally. The slot **632** is sized to receive a protrusion **634** of the slider **622**, such that the protrusion **634** extends from the back of the slider **622** and through the slot **632**. The lateral orientation of the slot **632** permits the slider **622** to move laterally between the locked position and the unlocked position without interference between the protrusion **634** and the slider plate **630**. The slider plate **630** further defines a groove, notch, or slot, shown as rail groove **636**, extending partially into the slider plate **630** from the back side of the slider plate **630**. The rail groove **636** is substantially vertically centered about the slot **632**. A vertical height of the rail groove **636** is greater than a vertical height of the slot **632**.

(310) The slider plate **630** further includes a series of posts or protrusions, shown as locator posts **638**, that extend rearward from a back surface of the slider plate **630**. Each locator post **638** is sized to be received within a corresponding passage defined by one of the lock switches **624**. Each lock switch **624** defines two passages, such that two of the locator posts **638** are received by each lock switch **624**. The locator posts **638** facilitate fixedly coupling the lock switches **624** to the slider plate **630** and locating the lock switches **624** to detect when the slider **622** moves to the lock position and the unlock position.

(311) The lock control interface **620** further includes a fixture, shown as carriage **640**, that moves laterally with the slider **622**. The carriage **640** includes a base, shown as rail **642**, that extends laterally. The rail **642** is received within the rail groove **636**, slidably coupling the carriage **640** to the slider plate **630**. Specifically, the rail **642** is sized to slide laterally along the length of the rail groove **636** while the height of the rail groove **636** limits vertical movement of the rail **642**. The rail **642** defines a longitudinal recess **644** that is sized to receive the protrusion **634** of the slider **622**, coupling the slider **622** to the carriage **640**. Accordingly, the carriage **640** slidably couples the slider **622** to the slider plate **630**.

(312) Referring to FIGS. **48-52**, the lock control interface **620** includes a series of biasing elements or magnets that facilitate positioning the slider **622**. As shown, the slider plate **630** and the carriage **640** define a series of apertures, passages, or recesses, shown as magnet apertures **650**, each configured to receive a magnet. Specifically, the slider plate **630** defines three magnet apertures **650** that each receive either an end position magnet **652** or a center position magnet **654**. The carriage **640** defines a single magnet aperture **650** that receives a carriage magnet **656**.

(313) The magnets of the lock control interface **620** are positioned to bias the carriage **640** and the slider **622** away from a center position and toward either the locked position or the unlocked position. When in the locked position or the unlocked position, the magnets hold the slider **622** in place, resisting movement toward the center position. This magnet arrangement mimics the feel of a two position, spring-based switch to a user while still permitting the slider **622** to be moved by the slider actuator **626** when the door **30** is remotely locked or unlocked.

(314) Referring to FIG. **50**, the magnets of the lock control interface **620** are shown with the slider **622** in the locked position. In the configuration of FIG. **50**, the carriage magnet **656** is aligned with a first one of the end position magnets **652**. The magnetic fields of the carriage magnet **656** and the first end position magnet **652** are arranged to generate an attractive biasing force. This biasing force holds the slider **622** in the locked position and resists movement toward the center position.

(315) Referring to FIG. **51**, the magnets of the lock control interface **620** are shown with the slider **622** in the center position. In the configuration of FIG. **51**, the carriage magnet **656** is aligned with the center position magnet **654**. The magnetic fields of the carriage magnet **656** and the center position magnet **654** are arranged to generate a repulsive biasing force. This biasing force resists the slider **622** remaining in the center position, instead forcing the slider **622** toward the locked position or the unlocked position, whichever is closer.

(316) Referring to FIG. **52**, the magnets of the lock control interface **620** are shown with the slider **622** in the unlocked position. In the configuration of FIG. **52**, the carriage magnet **656** is aligned

with a second one of the end position magnets **652**. The magnetic fields of the carriage magnet **656** and the second end position magnet **652** are arranged to generate an attractive biasing force. This biasing force holds the slider **622** in the unlocked position and resists movement toward the center position.

(317) Referring to FIGS. **48** and **49**, the slider actuator **626** includes an electric motor, servomotor, actuator, or output device, shown as servo **660**. The servo **660** is fixedly coupled to the slider plate **630**. The slider actuator **626** further includes a lever assembly, shown as servo arm **662**, including a pair of arms, shown as levers **664**. The servo **660** is configured to cause the servo arm **662** to rotate about a substantially longitudinal axis. The carriage **640** includes a longitudinal protrusion, shown as post **666**, that is positioned to be engaged by the levers **664** to cause lateral movement of the carriage **640** when the servo arm **662** is rotated.

(318) The servo **660** is operatively coupled to the bridge controller **270** and controlled by the bridge controller **270**. The bridge controller **270** may cause the servo **660** to rotate in a first direction or an opposing second direction. In some embodiments, the servo **660** includes a position sensor (e.g., an encoder or potentiometer) that provides positional feedback for closed-loop control over the position of the servo arm **662**.

(319) Referring to FIGS. **53-55**, an example of the operation of the slider actuator **626** is shown according to an exemplary embodiment. As shown, the carriage **640** is moved from the locked position to the unlocked position. In other embodiments, the locked position and the unlocked position are switched (e.g., depending upon if the door is a right hand door or a left hand door), such that FIGS. **53-55** illustrate the carriage **640** moving from the unlocked position to the locked position.

(320) In FIG. **53**, the carriage **640** and the slider **622** are in an end position (e.g., the locked position) and held in place by the carriage magnet **656** and an end position magnet **652**. The servo arm **662** is in a center or neutral position. In this neutral position, the levers **664** are spaced from the post **666** and do not contact the post **666**. The servo arm **662** is shaped such that the carriage **640** can slide freely between the locked position and the unlocked position without the levers **664** contacting or otherwise interfering with the movement of the post **666**. Accordingly, with the servo arm **662** in the neutral position, a user can freely slide the slider **622** between the locked position and the unlocked position without interference from the slider actuator **626**.

(321) Referring to FIG. **54**, to move the carriage **640** and the slider **622** out of the locked position, the servo **660** rotates the servo arm **662** in a first direction. This rotation causes one of the levers **664** to engage the post **666**. With further rotation of the servo arm **662**, the lever **664** forces the carriage **640** toward, and subsequently beyond, the center position. As the carriage **640** moves beyond the center position, the magnetic biasing forces between the center position magnet **654** and the carriage magnet **656** move the carriage **640** to the unlocked position. The servo arm **662** is then moved back to the neutral position. To move the carriage **640** from the unlocked position to the locked position, the servo **660** may rotate the servo arm **662** in a second direction opposite the first direction. When the carriage **640** moves beyond the center position, the magnetic biasing forces between the center position magnet **654** and the carriage magnet **656** move the carriage **640** to the locked position.

(322) Reconfigurable Interactive Unit

(323) Referring to FIGS. **42** and **56**, the interactive unit **140** is reconfigurable between a first, expanded functionality, or elevated cost configuration (e.g., as shown in FIG. **42**) and a second, reduced functionality, light, or reduced cost configuration (e.g., as shown in FIG. **56**). The expanded configuration may include additional functionality relative to the reduced configuration. In contrast, the reduced configuration may remove one or more components relative to the expanded configuration, reducing the overall cost of the interactive unit **140**. The expanded configuration or the reduced configuration may be selected during the initial production of the door **30**. Additionally or alternatively, the interactive unit **140** may be reconfigurable between the

expanded configuration and the reduced configuration by the end user. By way of example, a user may initially purchase a door **30** having the reduced configuration. The user may later decide to upgrade the interactive unit **140** to the expanded configuration by purchasing and adding upgrade components to the interactive unit **140**.

(324) FIG. **56** illustrates a reduced configuration of the interactive unit **140**, according to an exemplary embodiment. In the reduced configuration, the interior top plate **506**, the display **530**, and the touch input **532** corresponding to the display **530** are omitted from the interactive unit **140**. Accordingly, the interactive unit **140** does not include the touch display functionality of the display **530** and the touch input **532**. However, by omitting the display **530** and the touch input **532**, the overall cost of the interactive unit **140** is reduced. Instead of displaying images and/or video from the camera **520** on the display **530**, the images and/or video may be transferred to the user device **190** and displayed by a screen of the user device **190**. Similarly, instead of controlling the interactive unit **140** directly through the touch input **532**, a user may instead provide commands to the interactive unit **140** through a user device **190**.

(325) To accommodate the loss of the interior top plate **506**, the reduced configuration may utilize a new trim piece, shown as interior trim **680**. The interior trim **680** is smaller than the interior trim **508** of the expanded configuration, facilitating a tight fit around the reduced footprint of the interactive unit **140**. Apart from the size, the interior trim **680** may have a similar construction to the interior trim **508**.

(326) In some embodiments, the interior core plate **504** and the interior top plate **506** are each coupled to the subframe **40** by a different set of fasteners, such that the interior top plate **506** can be added to or removed from the door **30** without removing the interior core plate **504**. As shown in FIG. **57**, the interior core plate **504** is coupled to the subframe **40** by a first set of fasteners, shown as fasteners **682**. The fasteners **682** extend longitudinally through the interior core plate **504** and engage the subframe **40**. The fasteners **682** may be removed to remove the interior core plate **504** from the subframe **40**. The interior top plate **506** is coupled to the subframe **40** by a second set of fasteners, shown as fasteners **684**. The fasteners **684** extend longitudinally through the interior top plate **506** and engage the subframe **40**. The fasteners **686** may be removed to remove the interior top plate **506** from the subframe **40**.

(327) Referring to FIGS. **57** and **58**, the interior portion of the interactive unit **140** includes an upper portion **690**, containing the components coupled to the interior top plate **506**, and a lower portion **692**, containing the components coupled to the interior core plate **504**. The components of the upper portion **690** (e.g., the display **530** and the touch input **532**) are connected to components within the door **30** (e.g., the door wiring harness **170**) to receive power and data. However, if a passage were to be formed directly through the interior skin **52** to route the wiring for the upper portion **690**, the passage would be exposed when the upper portion **690** is removed to put the interactive unit **140** into the reduced configuration.

(328) To avoid an exposed passage in the reduced configuration, the upper portion **690** connects with the internal components of the door **30** (e.g., the door wiring harness **170**) through the lower portion **692**. The wiring for both the upper portion **690** and the lower portion **692** may pass through a single passage through the interior skin **52** that is covered by the lower portion **692**. As the lower portion **692** is present in both the expanded configuration and the reduced configuration, this passage may be covered in all configurations of the door **30**.

(329) As shown in FIGS. **57** and **58**, the upper portion **690** includes a connector **694** extending downward from the upper portion **690**. A top surface of the lower portion **692** defines an aperture or passage, shown as connector aperture **696**, that receives the connector **694**. By way of example, the connector **694** may include a series of wires or cables that are routed through the connector aperture **696**. By way of another example, the connector **694** may include a male connector that is received within a female connector within the connector aperture **696**.

(330) As shown in FIG. **58**, the connector aperture **696** is positioned along a top surface of the

lower portion **692** (e.g., along a top surface of the interior core plate **504**). When the interactive unit **140** is in the expanded configuration, the upper portion **690** abuts the lower portion **692**, obscuring the connector **694** and the connector aperture **696** from view and preventing contaminants from entering the connector aperture **696**. When the interactive unit **140** is in the reduced configuration, the interior trim **680** covers the connector aperture **696**. The interior trim **680** obscures the connector **694** from view and prevents contaminants from entering the connector aperture **696**.

(331) Alternative Interactive Unit Configuration

(332) Referring to FIGS. **59** and **60**, another embodiment of the interactive unit **140** is shown. The interactive unit **140** of FIGS. **59** and **60** may be substantially similar to the interface unit of FIGS. **40** and **42**, except as otherwise specified herein. The interactive unit **140** of FIGS. **59** and **60** may be utilized with any embodiment of the door system **10** described herein.

(333) The interactive unit **140** of FIGS. **59** and **60** includes a fingerprint scanner **550** on the exterior side of the interactive unit **140**. Specifically, the fingerprint scanner **550** is accessed through the NFC cover **546**. The NFC cover **546** defines a dimple or recess above the fingerprint scanner **550** such that the fingerprint scanner **550** is positioned to read a user's fingerprint when the finger of the user is placed in the dimple.

(334) The interactive unit **140** of FIGS. **59** and **60** omits the privacy button **580** and instead includes a two-position switch, shown as privacy switch **700**, operatively coupled to the bridge controller **270**. In some embodiments, the privacy switch **700** is a rocker switch. The privacy switch **700** serves a similar function as the privacy button **580**, controlling the operating state of the window **60**. The privacy switch **700** is repositionable between a first position or privacy position and a second position or viewing position. When the privacy switch **700** is switched to the privacy position, the bridge controller **270** commands the window **60** to change to the privacy state. When the privacy switch **700** is switched to the viewing position, the bridge controller **270** commands the window **60** to change to the viewing state.

(335) The interactive unit **140** of FIGS. **59** and **60** omits the sliding lock control interface **620** and instead includes a two-position switch, shown as lock control interface **702**, operatively coupled to the bridge controller **270**. In some embodiments, the lock control interface **702** is a rocker switch. The lock control interface **702** serves a similar function as the lock control interface **620**, controlling the locked/unlocked state of the door **30**. The lock control interface **702** is repositionable between a first position or locked position and a second position or unlocked position. When the lock control interface **702** is switched to the locked position (e.g., by pressing on an upper portion of the lock control interface **702**), the bridge controller **270** changes the door **30** to the locked state. When the lock control interface **702** is switched to the unlocked position (e.g., by pressing on a lower portion of the lock control interface **702**), the bridge controller **270** changes the door **30** to the unlocked state.

(336) Interface on User Device

(337) Referring to FIG. **61**, a graphical user interface (GUI) **710** is shown on the user device **190**, according to an exemplary embodiment. The elements of the GUI **710** may perform some or all of the functionality of the interactive unit **140**. By way of example, the GUI **710** may provide information to a user and/or receive commands from a user. Operation of the GUI **710** may be controlled by the user device **190**, a server **200**, a module of the door system **10**, and/or any other device.

(338) The GUI **710** includes a first element, shown as camera feed **712**. The camera feed **712** may provide a visual representation of image data captured by the camera **520**. The camera feed **712** may display the image data in real time or image data that as previously captured by the camera **520**.

(339) The GUI **710** further includes a second element, shown as door status indicator **714**. The door status indicator **714** may indicate whether the door **30** is currently open or closed. The deadbolt controller **266** may determine if the door **30** is open or closed based on data from the door close

sensor **262**.

(340) The GUI **710** further includes a third element, shown as weather indicator **716**. The weather indicator **716** may indicate a current ambient condition on the exterior side of the door **30**. By way of example, the weather indicator **716** may indicate a current ambient temperature, a current humidity, a current light level (e.g., an outdoor light intensity), a current weather pattern (e.g., whether it is currently sunny, cloudy, rainy, etc.), and/or other ambient or weather conditions. The door system **10** may determine the current ambient conditions based on data from the lux sensor **602** or the ambient condition unit **604**. Additionally or alternatively, the door system **10** may retrieve the ambient conditions from another source (e.g., a third party source accessed via the Internet).

(341) The GUI **710** further includes a fourth element, shown as microphone button **718**. When pressed, the microphone button **718** may toggle between (a) a recording mode in which a microphone of the user device **190** records sound for playback by the external speaker **564** of the door **30** and (b) a silent/mute mode in which the microphone is disabled. The microphone button **718** may provide a visual indication (e.g., color, images, text, etc.) indicating whether the user device **190** is in the recording mode or the silent mode.

(342) The GUI **710** further includes a fifth element, shown as notification button **720**. When pressed, the notification button **720** may toggle between (a) a notification mode in which the user device **190** provides audible notifications regarding the door system **10** and (b) a no-notifications mode in which the audible notifications are disabled. Such notifications may be triggered by an interaction with the doorbell button **572**, the camera **520**, or the external TOF sensor **594** detecting movement, the door close sensor **262** detecting that the door **30** has been opened, or other conditions. The notification button **720** may provide a visual indication (e.g., color, images, text, etc.) indicating whether the user device **190** is in the notification mode or the no-notifications mode.

(343) The GUI **710** further includes a sixth element, shown as lock button **722**. When pressed, the lock button **722** may toggle the door **30** between the locked state and the unlocked state. The lock button **722** may provide a visual indication (e.g., color, images, text, etc.) indicating whether the door **30** is in the locked state or the unlocked state.

(344) Additional Door Features

(345) Referring to FIGS. **9B**, **62**, and **63**, the door system **10** may include various additional features that further enhance the user experience when operating the door **30**. Although these additional features may have been omitted from FIGS. **1** and **2** to avoid overcrowding the image, it should be understood that FIGS. **1**, **2**, **62**, and **63** may all represent a single embodiment. It should also be understood that the position and combination of features shown in FIGS. **62** and **63** is exemplary only and not intended to be limiting. The features of FIGS. **62** and **63** may be controlled by any of the controllers described herein (e.g., the bridge controller **270**, the main controller **242**, etc.). Additionally, the features of FIGS. **62** and **63** may be powered by the power source **102**, the backup battery **222** of the door **30**, or any other source of energy.

(346) Motorized Door Actuator

(347) Referring to FIGS. **9B** and **62**, the door system **10** includes an actuator (e.g., an electric actuator), shown as door actuator **750**. The door actuator **750** is coupled to the door **30** and the wall **12**. The door actuator **750** is configured to open and/or close the door **30** (e.g., move the door **30** between an open position and a closed position). By way of example, the door actuator **750** may include an electric motor that drives rotation of a lead screw to move the door **30**. By way of another example, the door actuator **750** may include an electric motor that drives rotation of a linkage to move the door **30**. The door actuator **750** may be used to remotely open or close the door **30** (e.g., in response to a user command through a user device **190**). The door actuator **750** may be used to automatically open or close the door in response to certain conditions. By way of example, the door actuator **750** may automatically close the door at a certain time of day or when the ambient

temperature (e.g., as measured by the ambient condition unit **604**) falls below or exceeds a set temperature threshold. By way of another example, the door actuator **750** may automatically open the door **30** when the camera **520** detects a user approaching the door **30** (and carrying a user device **190** or fob that provides access to the door **30**).

(348) Motorized Screen

(349) Referring to FIGS. **9B** and **63**, the door system **10** includes a door screen or motorized screen, shown as screen assembly **760**. The screen assembly **760** includes a base or actuator, shown as screen motor **762**, that is coupled to the wall **12**. The screen assembly **760** further includes a screen, mesh, or rolled member, shown as screen **764**, coupled to the screen motor **762**. The screen **764** is actuated by the screen motor **762** to move the screen **764** between a first, raised, or retracted position, shown in FIG. **63**, and a second, lowered, or extended position. In the retracted position, the screen **764** is moved away from the doorway **28** to permit free movement of users through the doorway **28**. In the extended position, the screen **764** moves across the doorway **28**. The screen **764** may be made from a perforated material that permits air to pass through the screen **764** but prevents insects and animals from passing through the screen **764**. The screen **764** may seal against the door **30**, the door frame **20**, and/or the wall **12** such that any material passing through the doorway **28** forced to pass through the screen **764**. Accordingly, with the screen **764** in the extended position, the screen **764** prevents undesirable organisms from entering the home while permitting the free flow of outdoor air through the doorway **28**.

(350) The screen motor **762** is configured to control movement of the screen **764**. In some embodiments, the screen motor **762** includes a roller driven by an electric motor. The electric motor causes the screen motor **762** to rotate and retract or pay out the screen **764**. In other embodiments, the screen **764** is otherwise moved. The screen motor **762** may be manually controlled (e.g., through a user device **190** or the touch input **532**). Additionally or alternatively, the screen motor **762** may be used to automatically reposition the screen **764** in response to certain conditions. By way of example, the screen motor **762** may automatically lower the screen **764** to the extended position in response to the door **30** remaining in the open position for a threshold period of time. By way of another example, the screen motor **762** may automatically raise the screen **764** to the retracted position when the camera **520** detects a user approaching the door **30**.

(351) In some embodiments, the screen **764** is positioned to facilitate movement of the door **30** between the open position and the closed position while the screen **764** is in the extended position. The screen **764** may be positioned to offset the screen **764** from the door **30**, such that there is no contact between the door **30** and the screen **764** when the screen **764** is in the extended position and the door **30** is moved.

(352) In some embodiments, the screen **764** is embedded within the wall **12** and/or the door frame **20** such that the screen **764** moves within a plane that is coincident with the door frame **20**. By way of example, a slot or groove may run vertically through the side jambs **22** and the head jambs **24**, providing a path through which the screen **764** may pass. A pair of vertical guides may be received within the wall **12**. The vertical guides may be slidably coupled to the screen **764** and may align the screen **764** with the grooves. In other embodiments, the screen **764** is positioned outside of the wall **12** (e.g., offset on the interior side or exterior side of the wall **12**). The screen **764** may be positioned opposing the direction of movement of the door **30**. By way of example, if the door **30** is configured to swing inward, the screen **764** may be positioned on the exterior side of the door **30**. By way of another example, if the door **30** is configured to swing outward, the screen **764** may be positioned on the interior side of the door **30**.

(353) Motorized Blinds

(354) Referring to FIGS. **9B** and **63**, the door system **10** includes a shade assembly or motorized blind, shown as blind assembly **770**. The blind assembly **770** includes a base or actuator, shown as blind motor **772**, that is coupled to the door **30**. The blind assembly **770** further includes a blind, shade, shutter or rolled member, shown as blind **774**, coupled to the blind motor **772**. The blind **774**

is actuated by the blind motor **772** to move the blind **774** between a first, raised, or retracted position, shown in FIG. **63**, and a second, lowered, or extended position. In the retracted position, the blind **774** is moved away from the window **60** to permit light to freely pass through the window **60**. In the extended position, the blind **774** covers the window **60** and obscures a line of sight through the window, limiting the passage of light through the window **60**. In some embodiments, the blind **774** is used in place of electrochromic material of the window **60**.

(355) The blind motor **772** is configured to control movement of the blind **774**. In some embodiments, the blind motor **772** is a roller driven by an electric motor. The electric motor causes the blind motor **772** to rotate and retract or pay out the blind **774**. In other embodiments, the blind **774** is a series of slats suspended by lift cords. The electric motor extends or retracts the lift cords to lower or raise the slats. In other embodiments, the blind **774** is otherwise moved. The blind motor **772** may be manually controlled (e.g., through a user device **190** or the touch input **532**). Additionally or alternatively, the blind motor **772** may be used to automatically reposition the blind **774** in response to certain conditions. By way of example, the blind motor **772** may automatically lower the blind **774** to the extended position in response to the light level detected by the lux sensor **602** exceeding a threshold. By way of another example, the blind motor **772** may automatically raise the blind **774** to the retracted position for a predetermined period of time each day (e.g., at nighttime).

(356) Adjustable Door Seals

(357) Referring to FIGS. **9B** and **62**, the door system **10** includes an adjustable door seal assembly, shown as door seal system **780**. The door seal system **780** includes a series of adjustable door seals, shown as seals **782**, and an actuator, shown as door seal actuator **784**. The seals **782** each seal a portion of the doorway **28** between the door frame **20** and the door **30**. Specifically, a first seal **782** extends between a first side jamb **22** and the door **30**. A second seal **782** extends between a second side jamb **22** and the door **30**. A third seal **782** extends between the head jamb **24** and the door **30**. A fourth seal **782** extends between the sill **26** and the door **30**. In other embodiments, one or more of these seals **782** are omitted. Together, the seals **782** prevent air, debris, and pests from passing between the door **30** and the door frame **20**.

(358) The door seal actuator **784** is configured to vary a thickness of the seals **782**. In some embodiments, the seals **782** are inflatable (e.g., define a flexible, sealed volume). In such embodiments, the door seal actuator **784** may add or remove a working fluid (e.g., air) from the seals **782** to adjust a thickness of the seals **782**. By way of example, the door seal actuator **784** may include a pump or compressor. In other embodiments, the seals **782** are flexible, and the door seal actuator **784** adjusts a thickness of the seals **782** by compressing the seals **782** in the longitudinal direction. Such a compression deflects material outward in the thickness direction. In such an embodiment, the door seal actuator **784** may include one or more electric motors that drive a press to compress the seals **782**.

(359) The door seal actuator **784** may be manually controlled (e.g., through a user device **190** or the touch input **532**). Additionally or alternatively, the door seal actuator **784** may be used to automatically change the thickness of the seals **782** in response to certain conditions. By way of example, as the ambient temperature or humidity changes, components of the door system **10** may grow or shrink. The door seal actuator **784** may automatically expand or contract the seals **782** to maintain a consistent seal based on the humidity data and temperature data received from the ambient condition unit **604**.

(360) Exterior Lighting

(361) Referring to FIGS. **9B** and **62**, the door system **10** includes a series of exterior lights, shown as jamb lights **790** and handle light **792**. The jamb lights **790** and the handle light **792** may include LEDs or other types of lights. The jamb lights **790** and the handle light **792** may have variable colors, brightnesses, patterns, on/off states, or other properties. The jamb lights **790** are positioned along an inner surface of the door frame **20** and directed to illuminate the door **30**. Alternatively,

the jamb lights **790** may be positioned on the door **30**. The handle light **792** is coupled to the door **30** and directed to illuminate the handle assembly **70** and the interactive unit **140**. The jamb lights **790** and the handle light **792** may increase the visibility of the door **30** in dark conditions, improving ease of use and aesthetically highlighting the door **30**. Additionally or alternatively, the jamb lights **790** and the handle light **792** may be used to signal information to a user.

(362) The jamb lights **790** and the handle light **792** may be manually controlled (e.g., through a user device **190** or the touch input **532**). Additionally or alternatively, the jamb lights **790** and the handle light **792** be operated automatically in response to certain conditions. By way of example, the jamb lights **790** and the handle light **792** may turn on automatically in response to the light level detected by the lux sensor **602** falling below a threshold. By way of another example, the jamb lights **790** and the handle light **792** may turn on automatically in response to sensors (e.g., the camera **520**, the radar sensor **608**, the external ToF sensor **594**, etc.) detecting a user approaching. By way of another example, the jamb lights **790** and the handle light **792** may change color to indicate that the door **30** is in the locked state. Further details regarding the exterior lighting features are described in greater detail herein with respect to FIGS. **92-98**.

(363) Control Strategies

(364) The door system **10** implements various control strategies to improve the performance and user experience of the door system **10**. The control strategies described herein may be performed by any of the controllers and the control system **100** described herein (e.g., the power supply unit **130**, the window control unit **132**, the main control unit **134**, the bridge module **136**, the deadbolt units **138**, the doorbell module **184**, the user devices **190**, the servers **200**, etc.) alone or in combination with one another. Accordingly, any control strategies described herein may be performed by any component of the door system **10**, unless specified otherwise.

(365) Strategies for Energy Savings

(366) In some situations, the door **30** may not receive a steady supply of electrical energy from the power source **102**. By way of example, a municipal power grid may experience outages due to storms or maintenance. By way of another example, the door **30** may be installed without a connection to the power source **102**. In such situations, the door **30** may be powered partially or entirely from the backup battery **222**. In order to prolong the life of the backup battery **222**, the door system **10** may change from a normal mode of operation to a power saving mode of operation in which various control processes reduce energy consumption. In some embodiments, the door system **10** automatically changes to the power saving mode in response to an indication that the power supply from the power source **102** is unavailable.

(367) In some embodiments, the camera **520** operates continuously (e.g., 24 hours per day, every day) while in the normal mode of operation. This continuous operation may permit the camera **520** to capture more comprehensive image data and permit the door system **10** to react to situations quickly (e.g., by continuously processing image data to identify the triggers for one or more conditions). However, continuous operation of the camera **520** may increase energy consumption.

(368) In the power saving mode, the camera **520** is shut off for periods of time. In some embodiments, the camera **520** runs on a predetermined schedule. By way of example, the camera **520** may operate for 10 seconds once every 30 seconds. In some embodiments, the camera **520** is activated in response to an input. By way of example, the camera **520** may be activated in response to the external TOF sensor **594**, the internal TOF sensor **596**, and/or the radar sensor **608** detecting motion. By way of another example, the camera **520** may be activated in response to a user interacting with the interactive unit **140** (e.g., by pressing the doorbell button **572**, by communicating with the NFC radio **540**, by contacting the fingerprint scanner **550**, etc.). By way of another example, the camera **520** may be activated in response to the accelerometer unit **600** detecting movement of the door **30**. By way of another example, the camera **520** may be activated in response to a user opening an application on a user device **190** that would utilize the image data.

(369) In some embodiments where the door **30** includes multiple deadbolt units **138** or a main

deadbolt of the handle assembly **70** and one or more deadbolt units **138**, all of the deadbolt units **138** and/or the main deadbolt may be operated simultaneously while in the normal mode of operation. By way of example, in an embodiment that includes two deadbolt units **138** or the main deadbolt and one or more deadbolt units **138**, both of the deadbolt units **138** or the main deadbolt and the one or more deadbolt units **138** may be in the same locked or unlocked state. Utilizing both of the deadbolt units **138** or the main deadbolt and one or more deadbolt units **138** increases the security of the door **30**, but increases the power consumption required to shift the deadbolts and/or hold the deadbolts in place. In the power saving mode of operation, one or more of the deadbolt units **138** may be disabled to conserve power. The disabled deadbolt units **138** may remain in the unlocked state throughout operation, while the enabled deadbolt units **138** and/or main deadbolt may continue to lock or unlock as necessary. By disabling a subset of the deadbolt units **138** or only maintaining operation of the main deadbolt, a power savings is achieved while still permitting the remaining enabled deadbolt units **138** and/or the main deadbolt to lock the door **30**.

(370) In some embodiments, activation of the window **60** is disabled to conserve energy when the door **30** is cutoff from the power source **102**. In some embodiments, the activation of window **60** is permitted until a state-of-charge threshold is reached and then activation is disabled. In some embodiments, activation of the door lighting features (e.g., the jamb lights **790**, the handle light **792**, etc.) is disabled to conserve energy when the door **30** is cutoff from the power source **102**. In some embodiments, the activation of door lighting features is permitted until a state-of-charge threshold is reached and then activation is disabled. In some embodiments, Wi-Fi capabilities of the door system **10** are disabled to conserve energy when the door **30** is cutoff from the power source **102** (i.e., so that the door system **10** does not continually try to reconnect to Wi-Fi with the power to the residence and, therefore, the Wi-Fi being disabled).

(371) In some embodiments, the user of the door system **10** can pre-define a hierarchy of which components they would like to remain active for the longest (e.g., the camera **520**, the lights, the deadbolt units **138**, etc.) and the door system **10** will begin disabling the components according to the predefined hierarchy. In some embodiments, the door system **10** may send a notification to the user device **190** requesting instructions from the user when the door **30** is cutoff from the power source **102**. The user may respond to the notification via the user device **190** with various instructions. By way of example, the instructions may include an on-demand hierarchy for the components. By way of another example, the instructions may additionally or alternatively include a time limit for the door system **10** to remain functional using the backup battery **222**. The door system **10** may then evaluate the remaining charge of the backup battery **222** and, based on the on-demand hierarchy and/or the user specified time limit, selectively disable certain functionality of the door **30** as necessary to achieve the user requested time of battery operation or favored component operation. For example, the user may know that the power will only be out for a certain period of time (e.g., an expected power outage). Rather than the door system **10** automatically disabling certain features when power is lost, the door system **10** can more intelligently determine if components even need to be disabled and whether they need to be disabled immediately or in the future to maintain complete or more complete functionality of the door system **10** for extended periods of time. In some embodiments, the door system **10** is configured to implement the predefined hierarchy until a response is received from the user device **190**.

(372) Camera Image Processing

(373) In some embodiments, the door system **10** performs image processing (e.g., image recognition) on the image data captured by the camera **520**. This processing may be performed locally by one or more edge devices (e.g., the power supply unit **130**, the window control unit **132**, the main control unit **134**, the bridge module **136**, the deadbolt units **138**, the doorbell module **184**, the user devices **190**, etc.). Additionally or alternatively, the processing may be performed by one or more cloud devices (e.g., the servers **200**). In some embodiments, the processing is performed as a hybrid edge/cloud process, where a first portion of the processing is performed by the edge

devices (e.g., low intensity processing) and a second portion of the processing is performed by the cloud devices (e.g., higher intensity processing).

(374) Using image processing, the door system **10** may identify one or more objects within the image data captured by the camera **520**. The door system **10** may utilize machine learning or other types of image processing to perform the image recognition. The door system **10** may react in predetermined ways to detecting certain types of objects. By way of example, the door system **10** may identify or otherwise differentiate between humans (e.g., specific humans), animals (e.g., wildlife, pets of the homeowner, etc.), vehicles (e.g., mail delivery vehicles, a vehicle known to belong to a specific user, etc.), packages, or other objects.

(375) In some embodiments, the door system **10** is configured to identify specific individuals. By way of example, the door system **10** may have a list of approved users, and facial recognition data regarding those approved users may be predetermined and stored (e.g., within the main control unit **134**, on a server **200**, etc.). Upon detecting a human within the image data, the door system **10** may perform facial recognition on the image data to determine if the image data matches an approved user. In response to such a determination, the door system **10** may automatically change the door **30** to the unlocked state, facilitating quick entry into the home. By way of another example, the door system **10** may have a list of blacklisted users that are barred from entry into the home. Facial recognition data regarding those blacklisted users may be predetermined and stored. Similarly, the door system **10** may use image recognition to determine if a human is intentionally obscuring their face (e.g., using a mask). Upon detecting a human within the image data, the door system **10** may perform facial recognition on the image data to determine if the image data matches a blacklisted user or a user that is obscuring their face. In response to such a determination, the door system **10** may automatically change the door **30** to the locked state, preventing unauthorized users from entering the home.

(376) In some embodiments, the door system **10** is configured to identify packages, vehicles, and/or uniforms worn by a human. Such an embodiment may be utilized during a package delivery. A delivery vehicle may park in front of the door **30** when executing a package delivery. If the door system **10** identifies a delivery vehicle, the door system **10** may attempt to identify a company or delivery service associated with the delivery vehicle. If successful, the identified company or delivery service may be stored. After parking, a mail carrier may exit the delivery vehicle and walk toward the door **30** carrying a package. If the door system **10** identifies a human, the door system **10** may attempt to identify a company or delivery service associated with the uniform of the mail carrier. If successful, the identified company or delivery service may be stored. While the package is carried toward the door **30** and/or after the package is delivered, the package may be visible to the camera **520**. If the door system **10** identifies a package, the door system **10** may attempt to read identifying information on the package, such as a shipping label or graphics on the exterior of the package.

(377) In response to detecting a delivery, the door system **10** may provide a notification to a user. The door system **10** may provide the notification on the display **530**, on a user device **190**, as an audible tone through the doorbell system **180**, or through another device. Along with the notification that a delivery has occurred, the door system **10** may provide additional information to the user that the door system **10** was able to identify from the image data. By way of example, the door system **10** may provide an image of the package. By way of another example, the door system **10** may provide the name of the company or delivery service that delivered the package, the time that the package was delivered, information contained in the shipping label, a name of a company that sent the package (e.g., based on the graphics on the exterior of the package), and/or other information.

(378) In some embodiments, the door system **10** is configured to identify wildlife captured in the image data. A user may wish to know what types of wildlife pass by their home. If the door system **10** identifies wildlife, the door system **10** may attempt to identify a species of the wildlife. The door

system **10** may designate different species of wildlife within different categories. By way of example, a low priority category may include animals, such as dogs or cats, that belong to nearby neighbors. By way of another example, a low priority category may include animals common to that particular area, such as squirrels, chipmunks, or birds. By way of another example, a high priority category may include predators that might be desirable to have near a residence, such as bears, wolves, or mountain lions. By way of another example, a high priority category may include animals that are rare (e.g., endangered) or uncommon to the nearby area. By way of another example, a user may designate a certain species of animal as being high priority. When a high priority animal is detected, the door system **10** may provide a notification to the user. The notification may include a time stamp indicating when the sighting occurred, as well as an image of the animal.

(379) Data Sharing

(380) In some embodiments, the door system **10** is configured to communicate with other systems. The door system **10** may send information generated by the door system **10** to the other systems or receive information from the other systems. By way of example, a user may wish to share image data captured by the camera **520** on social media. In response to such a command, the door system **10** may send the image data to a third party social media website. By way of another example, the user may utilize a home security system that operates outside of the door system **10**. The door system **10** may share image data from camera **520** with the home security system, or vice versa to replace or supplement the functionality of the camera **520**. By way of example, the door system **10** may receive weather data describing the current and projected weather from a weather monitoring organization. The weather data may supplement or replace data supplied by the ambient condition unit **604**.

(381) Motion Activation

(382) In some embodiments, the motion detection capabilities of the door system **10** can be up to twenty feet from the door **30**. In some embodiments, the door system **10** is configured to control the window **60** based on motion detection (e.g., via the camera **520**, the TOF sensors **594** and **596**, the lux sensor **602**, the radar sensor **608**, etc.). By way of example, when motion is detected proximate the door **30** (on the inside and/or the outside thereof), the door system **10** may be configured to transition the window **60** from a first state to a second state. In some embodiments, the first state is the privacy state (e.g., a translucent state, an opaque state, a tinted state, etc.) and the second state may be the viewing state (e.g., a transparent state) such that a person can see through the window **60** when approaching the door **30** (e.g., from the inside). In some embodiments, the first state is the viewing state and the second state may be the privacy state such that a person cannot see through the window **60** when approaching the door **30** (e.g., from the outside). In some embodiments, the transition from the viewing state to the privacy state is automatically engaged in response to motion detection and determining that the person is unrecognized by the door system **10** or if the person is portraying odd behaviors such as peeping through the window **60**.

(383) In some embodiments, the door system **10** is configured to control lighting features of the door system **10** (e.g., external status light **590**, internal status light **592**, jamb lights **790**, handle light **792**, etc.) based on motion detection. By way of example, the door system **10** may be configured to activate one or more lights when a person approaches the door **30** (from the inside and/or the outside) to help guide the person to and interact with the door **30**.

(384) In some embodiments, the door system **10** is configured to control display features of the door system **10** (e.g., the display **530**, etc.) based on motion detection. By way of example, the door system **10** may be configured to activate the camera **520** and provide a live display via the display **530** as a person approaches the door **30** from the inside (so a person on the inside can see to the other side of the door **30** if the window **60** is in a privacy state) and/or from the outside (to show a live view to a person inside showing a person is at/outside the door **30**).

(385) In some embodiments, the user can configure settings that may be implemented by the door system **10** in response to motion activation. By way of example, the configurable settings may include door light activation, brightness settings, and/or color settings; level of opacity or tint of the window **60**; auto-activation of the camera **520** and/or the display **530**; auto-lock or unlock; door auto-open and/or close (via the door actuator **750**); auto-deployment or auto-retraction of the screen **764**; and/or auto-deployment or auto-retraction of the blind **774**; among other possible door functions and settings. Multiple different user profiles or user preferences for door settings may be pre-stored or preset.

(386) User Identification/Authentication and Settings

(387) In some embodiments, the door system **10** is configured to implement user specific settings that are preset or pre-stored (e.g., locally at the door **30**, remotely at a server, etc.) based on user identification/authentication. In some embodiments, the door system **10** is configured to perform user identification/authentication by detecting an electronic device (e.g., a smartphone, a smartwatch, a key fob, the user device **190**, etc.) carried by or on a user as the user approaches or reaches the door **30**. By way of example, the electronic device may have a device identifier associated therewith or the electronic device may have a specific user identifier stored thereon that is detectable by the door system **10**. The device identifier and/or the user identifier can be associated with the user specific settings that are preset or pre-stored such that they are implemented when the device identifier and/or the user identifier is acquired by the door system **10**. In some embodiments, the door system **10** is configured to perform user identification/authentication using facial recognition (e.g., via image processing based on data acquired with the camera **520**) or other biometric detection techniques (e.g., fingerprint scanning, voice analysis, etc.). By way of example, the user specific settings that may be implemented by the door system **10** in response to user identification may include door light activation, brightness settings, and/or color settings, activation and/or level of opacity or tint of the window **60**, auto-lock or unlock, door auto-open and/or close (via the door actuator **750**), auto-deployment or auto-retraction of the screen **764**, and/or auto-deployment or auto-retraction of the blind **774**, among other possible door functions and settings. Multiple different user profiles or user preferences for door settings may be pre-stored or preset.

(388) Gesture/Presence Controls

(389) As described above with respect to motion activation, various features of the door system **10** may be activated or controlled in response to detecting the presence of a person proximate the door **30**. In some embodiments, the control system **100** is configured to activate or control various features of the door system **10** in response to motion/presence detection and gestures performed by the person (e.g., detected using one or more of the camera **520**, the radar sensor **608**, the external ToF sensor **594**, etc.). By way of example, the gestures may include various hand/arm motions such as swiping left-to-right, swiping right-to-left, swiping left-right-left, swiping right-left-right, swiping down-up, swiping up-down, swiping down-up-down, swiping up-down-up, waving overhead back-and-forth, rotating in a clockwise direction, rotating in a counter-clockwise direction, etc. The gestures may be predefined or user customizable. The gestures may be used to perform various functions including light controls (e.g., on/off, color, pattern, etc.), lock controls (e.g., unlock, lock, etc.), glass controls (e.g., activate, deactivate, level of opacity/tint/privacy, etc.), door open/close functions (via the door actuator **750**), deployment/retraction of the screen **764**, deployment/retraction of the blind **774**, among other possible functions.

(390) In some embodiments, the control system **100** is configured to activate or control various features of the door system **10** in response to motion/presence detection, gesture detection, and user identification/authentication (as described herein) to provide user specific or personalized gesture controls. By way of example, a first user of the door system **10** (e.g., an adult, a parent, etc.) may be able to define first gestures that when performed by the first person causes the door system **10** to perform first functions and a second user of the door system (e.g., a spouse, a child, etc.) may be

able to define second gestures that when performed by the second person causes the door system **10** to perform second functions. The first functions and the second functions may be the same, but he first gestures and the second gestures may be different.

(391) Message Center

(392) In some embodiments, the door system **10** and the control system **100** are configured to provide a message center via the interactive unit **140** (e.g., the display **530**, the external speaker **564**, the internal speaker **566**, etc.), the window **60** (e.g., if the window **60** is a transparent display as described herein), and/or the user device **190**. By way of example, the control system **100** may be configured to provide a message when a user opens the door **30**. The user message may be generic or user customizable. For example, the message may be a text message and/or an audio message providing a salutation, a greeting, or an informative tidbit (e.g., “Welcome home.”, “Have a good day.”, “The weather outside today is 70 degrees and sunny.”, etc.) provided via the interactive unit **140**, the window **60**, and/or the user device **190**. The text message and/or the audio message may differ when entering through the door versus when exiting through the door. In some embodiments, the message is adaptive based on the user accessing the door (e.g., based on detecting the user device **190** on the user's person, facial recognition, tailored for a specific user, etc.). For example, a parent may leave a message for a child returning from school such as a list of tasks to complete before the parent returns from work (e.g., “clean your room”), reminders (e.g., “remember you have soccer at 5:00 pm today”), and the like such that when the child enters through the door, the message from the parent is provided to them. In some embodiments, the message is sent as a notification to a user that is not entering or exiting the door **30**. By way of example, the message may be an application notification (e.g., associated with the door **30**), a SMS message, or the like sent to the user device **190** of the user indicting that another person has come home and indicating who that person is (e.g., “Mom is home”, tailored based on the user entering the door **30**, etc.).

(393) Haptics

(394) In some embodiments, the handle assembly **70** includes haptic components (e.g., a vibration element, a motor, etc.) that facilitate providing haptic feedback (e.g., vibration patterns, profiles, etc.) to a user that grabs or grasps the doorknob or door handle of the handle assembly **70**. The control system **100** may be configured to active the haptic components according to one or more haptic profiles to provide feedback or information to the user based on the current situation (e.g., sensors of the door system **10** detecting the presence of persons or animals proximate the door). By way of example, the control system **100** may be configured to cause the interior doorknob or handle to vibrate according to a first haptic feedback profile in response to a user attempting to open the door **30** from the inside with someone standing on the exterior side of the door **30** (e.g., without the doorbell having been rung recently such as within the last minute). Such functionality can, therefore, provide advanced warning to the user that someone is standing outside of the door **30** (and, possibly unbeknownst to the user). By way of another example, the control system **100** may be configured to cause the exterior doorknob or handle to vibrate according to a second haptic feedback profile in response to a user attempting to open the door **30** from the outside with some standing on the interior side of the door **30**. Such functionality can, therefore, provide advanced warning to the user that someone is standing inside of the door **30** and to be cautious so that they do not hit the person on the inside. The first haptic feedback profile and the second haptic feedback profile may have the same or different vibratory patterns. By way of yet another example, the control system **100** may be configured to cause the interior doorknob or handle to vibrate according to a third haptic feedback profile in response to a user attempting to open the door **30** from the inside without their associated user device **190** nearby (i.e., on their person) to indicate that the user may have forgotten their user device **190**. The third haptic feedback profile may have a different vibratory pattern than the first haptic feedback profile and the second haptic feedback profile. While only the first haptic feedback profile, the second haptic feedback profile, and the third haptic

feedback profile have been described herein, it would be understood that additional or alternative haptic feedback profiles may be used when a different set of conditions apply (e.g., to indicate weather outside, etc.). Further, the user may be able to select which vibratory pattern applies to which situation, and define the parameters (e.g., selected from a predefined list, manually defined parameters, etc.) for which a respective haptic feedback profile applies/activates.

(395) Access Control using UWB Sensors and/or Radar Sensors

(396) Referring to FIG. **100**, a structure (e.g., a residence, a house, an apartment, a condominium, a commercial building, etc.), shown as building **1300**, has perimeter walls, shown as exterior walls **1302**, that define an exterior of the building **1300**, shown as outside **1304**, and an interior of the building **1300**, shown as inside **1306**. As shown in FIG. **100**, the building **1300** includes the door system **10** (or the door system **1100**) to permit selective access to the inside **1306** of the building **1300** from the outside **1304**. The building **1300** may include one or more above-ground levels or stories (e.g., a first/main floor, a second floor, etc.) and/or one or more below-ground levels (e.g., a basement, a walkout basement, etc.).

(397) According to an exemplary embodiment, the control system **100** is configured to (a) detect the presence of a user and/or a user device **190** proximate the building **1300**, (b) determine a distance of the user and/or the user device **190** relative to the door **30**, (c) determine whether the user and/or the user device **1900** is on the outside **1304** and/or on the inside **1306** of the building **1300**, (d) determine user access intent, and/or (e) determine whether to permit “hands free” unlock the door **30** based at least on the presence, the distance, whether the user and/or the user device **190** is on the inside **1306** or on the outside **1304**, and/or the user access intent.

(398) According to an exemplary embodiment, the control system **100** is configured to facilitate distinguishing between whether a user is inside of the door **30** or outside of the door **30**, or whether the user actually intends to open the door **30** or whether the user is just passing by the door **30**. Without such capabilities, the control system **100** may otherwise inadvertently or unnecessarily activate “hands free” door unlocking events (e.g., not having to enter a code, not having to select unlock on a user device, etc.). For example, a user may enter the building **1300** from another entry point (e.g., a garage door, a rear door, a side door, etc.) and walk past the door **30**. If the control system **100** does not distinguish between inside or outside, the control system **100** may otherwise recognize the user's returned presence (e.g., based on the user device **190** of the user being detected) and unlock the door **30** even though the user is on the inside **1306**. As another example, a user may be in the garage or living room of the building **1300**, which in FIG. **100** extends at least partially past an exterior side of the door **30** of the door system **10**, or the user may be on a lower level or upper level of the building **1300** that extends past the exterior side of the door **30** of the door system **10**. If the control system **100** were not capable of distinguishing between a user in front of the door **30** but inside of the building **1300**, or identify position in a three-dimensional space, the control system **100** may unlock the door **30** even though the user is not outside of the building **1300** or in a location to access the door **30**. As yet another example, a user may be outside of the building **1300** but not actually intend to enter the building **1300** through the door **30** (e.g., walking past the door **30** as they are doing yardwork, etc.). If the control system **100** does not take access intent into account, the control system **100** may simply unlock when the user's presence is detected nearby and, therefore, unlock in instances when the user is passing by the door **30** but not actually attempting to enter.

(399) As described above, the control system **100** is configured to receive the UWB signals from the UWB sensors **606** to triangulate the user's position in two-dimensional or three-dimensional coordinate space. As such, the control system **100** can determine whether the user is on the inside **1306** or on the outside **1304** based on the UWB signals. In some embodiments, the control system **100** is configured to determine whether the user is inside **1306** or outside **1304** using different sensors (e.g., the camera **520**, the radar sensor **608**, the ToF sensor(s), etc.). For example, if the user is not detected outside **1306** with such sensor(s), the user may be determined to be inside **1304**. In

some embodiments, the control system **100** can additionally determine whether the user is above the door **30**, below the door **30**, and/or on the same level as the door **30** based on the UWB signals (or other sensor signals). In some embodiments, the perimeter of the exterior walls **1302** of the building **1300** are pre-defined within the control system **100** and/or the control system **100** is trained (e.g., by a user walking around the building **1300** with the user device **190** and setting a border) to understand which regions are considered eligible for unlock operations. Accordingly, the control system **100** can prevent inadvertent unlocking of the door **30** when the user is on the inside **1306**, above the door **30**, or below the door **30**.

(400) Though, when the control system **100** determines that the user is in front of the door **30** and on the outside **1304** of the building **1300**, there still may be instances like described above where the user does not actually intend to access the building **1300** through the door **30**. The control system **100** may be configured to determine whether the user intends to access the building **1300** through the door **30** based on a signal received from a capacitive touch component of the door **30** and/or the sensor data received from the radar sensor **608**. By way of example, the doorknob or handle of the handle assembly **70** may have a capacitive touch sensing function that facilitates detecting a user's hand when grasped. Accordingly, the control system **100** may be configured to unlock the door **30** in response to detecting the user on the outside **1304** and in response to receiving the signal from the handle assembly **70** indicating a user has grabbed/grasped the doorknob or handle (i.e., indicating user access intent). By way of another example, the control system **100** may be configured to unlock the door **30** in response to the signals from the radar sensor **608** (and/or the camera **520**) indicating that the user is reaching for the doorknob or handle (i.e., indicating user access intent).

(401) Building Integration

(402) In some embodiments, the door system **10** integrates or communicates with other building control systems within a building (e.g., a home, a residence, etc.) within which the door **30** is installed. The other building control systems may include systems such as a lighting control system, an alarm system, a heating, ventilation, and air conditioning (“HVAC”) system (e.g., a thermostat), smart home systems (e.g., Google Home, Amazon Alexa, etc.), home water monitoring and control systems, and the like.

(403) By way of example, the door system **10** may be set to an away mode (e.g., by a user or occupant, automatically, etc.). In the away mode, the door system **10** may not only lock the door **30**, but may communicate with one or more other building control systems to control them, provide instructions thereto, and/or receive information therefrom. As an example, the door system **10** may communicate with the lighting controls of the building to instruct the lighting control system to turn off one or more lights within the building that may be on at the time of the away mode being activated. As another example, the door system **10** may communicate with the alarm system of the building to instruct the alarm system to enter into an active mode in response to the away mode being activated. As yet another example, the door system **10** may communicate with the HVAC system of the building to instruct the thermostat to enter the HVAC system into an away mode (e.g., operate the HVAC system to provide a certain preset temperature set by the user, engage away temperature control operations, etc.). As still yet another example, the door system **10** may communicate with the water monitoring and control system of the building to instruct the water monitoring and control system to enter the home water system into an away mode (e.g., engage away water monitoring and control operations such as closing a water supply valve to the building, entering the water monitoring and control system into a high alert state where water usage within the building sets off an alarm and/or pushes notifications to the user, etc.).

(404) Advanced Controls and Artificial Intelligence

(405) According to an exemplary embodiment, the control system **100** is configured to provide advanced control capabilities whether through advanced programming logic or through artificial intelligence (“AI”) and machine learning. The advanced control capabilities may, for example,

relate to weather/environment-based controls, schedule-based controls, and/or presence detection controls. According to various implementations, the control system **100** may be able to provide such advanced capabilities using any kind of AI/machine learning model or architecture, such as neural networks/advanced neural networks, reinforcement learning, Bayesian hierarchical analysis, generative AI models such as generative adversarial networks (“GANs”) and/or transformer models, or any other type of machine learning model/AI model.

(406) In some embodiments, the control system **100** is configured to detect or acquire weather/environmental characteristics (e.g., temperature, sky/cloud darkness, precipitation, directly with sensors, through weather reports accessed through a weather system, etc.) and provide various notifications and/or perform various functions based on the weather/environmental characteristics. By way of example, the control system **100** may detect rain and/or dark clouds approaching (e.g., via the camera **520**, via the ambient condition unit **604**, via weather reports from a weather system). The control system **100** may be configured to provide an alert or notification via the door system **10** and/or through the user device **190** to close any open screens or windows. In some instances, the door system **10** may have an internal or external camera to detect open windows of the building and/or a screened storm door, and provide such information to the user to close such windows. The control system may be trained using images of different weather conditions (e.g., sunny, rainy, snowy, etc.), images of open/closed windows, data that indicates inclement weather (e.g., pressure readings, temperature readings, etc.) and then be capable of evaluating whether current conditions indicate inclement weather and/or open windows. By way of another example, the control system **100** may detect or acquire external, ambient temperature (e.g., via the ambient condition unit **604**, via weather reports from a weather system, etc.). In some embodiments, the control system **100** is configured to (a) provide an alert or notification via the door system **10** and/or through the user device **190** to shutoff a water supply to external water spickets in response to the temperature falling below a first temperature threshold and/or (b) communicate with the home water system to automatically turn off the water supply in response to the temperature falling below a second temperature threshold. The second temperature threshold may be the same or less than the first temperature threshold. In some embodiments, the control system **100** is configured to automatically transition the window **60** (and/or the panel windows **62** described herein below) from the transparent state to the opaque state, and vice versa. By way of example, the control system **100** may be configured to transition between the transparent state and the opaque state to prevent heat loss or maximize heat transfer based on the external temperature, the internal temperature, position of the sun relative to the door **30**, etc.

(407) In some embodiments, the control system **100** is configured to learn occupant schedules and/or schedules of external events over time and provide alerts when a schedule deviates from what is expected (e.g., by monitoring location of the user device **190** to identify that the user is entering/exiting the door **30**, using facial recognition during entry and exit events, via sensors, via cameras, etc.). By way of example, the control system **100** may be configured to monitor the days and times at which a child typically leaves and returns from school (e.g., using facial recognition, monitoring the child's devices, etc.). For example, if the child does not leave by a certain time on a day of school (e.g., the camera feed has not captured the child leaving), the control system **100** may be configured to provide a notification to the user device **190** of the child and/or the parent of the child, sound the doorbell as an “alarm,” or the like. As another example, if the child does not return by a certain time, the control system **100** may be configured to provide a notification to the user device **190** of the parent notifying them that the child has not returned home according to their usual schedule. By way of another example, the control system **100** may be configured to monitor events external to the door **30** (e.g., via the camera **520**) such as refuse pickup schedules. For example, if the occupant puts refuse at the curb according to the learned refuse pickup schedule and the refuse is not picked up according to historical pickup times, the control system **100** may be configured to provide an alert to the occupant and/or contact the refuse pickup organization that the

refuse pickup was missed. The control system **100** may be trained using images of refuse vehicle in various states of refuse collection and then be capable of evaluating whether refuse collection occurred by evaluating a camera feed over a range of time around the typical pickup schedule. As another example, if a child catches a school bus at a particular time and the school bus does not arrive according to historical pickup times, the control system **100** may be configured to provide an alert to the parent and/or contact the school that the child's bus stop was missed. The control system **100** may be trained using images of school buses and child pickup events and then be capable of evaluating whether a school pickup event occurred by evaluating a camera feed over a range of time around the typical pickup schedule. By way of still another example, the control system **100** may be configured to monitor snow removal timing by a snow removal service during snowfall and provide a notification to the occupant and/or contact the snow removal service if the timing deviates from historical response timing. The above examples may similarly apply to a cleaning service schedule, a lawn care schedule, a mail delivery schedule, and the like. The control system **100** may similarly be trained using images related to each scenario similar to outlined above.

(408) In some embodiments, the control system **100** is configured to provide advanced presence detection capabilities. By way of example, the control system **100** may be configured to implement various filters when analyzing motion proximate the door **30** and eliminate false movements from activating door features (e.g., activation of the camera **520**, the exterior lighting, etc.). For example, the control system may be configured to detect an object is swaying in the wind (e.g., a tree limb, a hanging light fixture, etc.) and prevent such movement from activating the features of the door **30**. Such detection may be performed using data from the camera **520**, the radar sensor **6008**, the external ToF sensor **594**, and/or other sensors.

(409) By way of another example, the control system **100** may be configured to provide various advanced delivery features. For example, the control system **100** may be configured to detect that a delivery person is approaching the door **30** with a package or mail and perform one or more features. The control system **100** may be trained using images of delivery persons approaching a door with various types of packages and then be capable of evaluating whether a package delivery is occurring by evaluating a camera feed when presence at the door **30** is detected. One feature may include sending a notification to the user device **190** of the occupant regarding delivery. The notification may include various information regarding the time, the type of delivery service (e.g., UPS, USPS, DHL, FEDEX, Amazon, etc.) based on detected characteristics of the delivery person, their vehicle, and/or identifiers on the package, and/or characteristics of the packages (e.g., identifiers, dimensions, etc.). Another feature may include disabling the doorbell functionality and only sending a notification to the user device **190** of the occupant when a delivery is detected (e.g., so not to disturb the occupant, since the occupant may not need to go down immediately to get the package, etc.). The control system **100** may, however, detect that the delivery person has a signature device indicating that the occupant needs to sign for the delivery and, in such instance, may keep the doorbell function active. In such instances, the control system **100** may be trained using various images that do and do not include package deliveries where the delivery person has a signature device and, therefore, learn to detect such a scenario. Still another feature may include providing a "Do Not Disturb" visual on the outside of the door **30** to prevent the delivery person from ringing the doorbell or knocking. Yet another feature may include providing an indicator on the inside of the door **30** that a package and/or a person is on the outside of the door **30**.

(410) By way of still another example, the control system **100** may be configured to provide advanced pet functionality. For example, the control system **100** may be configured to detect that a pet (e.g., a dog) is at the door **30** and perform one or more features. One feature may include sending a notification to the user device **190** of the occupant regarding the pet being at the door. Another feature may be providing an alternative doorbell tone specific to the pet being present at the door. Still another feature may include opening the door **30** (e.g., via the door actuator **750**). Yet another feature may include unlocking or opening a pet door within the door **30**. In some

embodiments, the control system **100** is configured to determine that the pet has an intent to come inside before performing the features outlined above. For example, the control system **100** may analyze audio data (e.g., via audio sensors of the door **30**) to identify whether the pet has barked, whined, or emitted a certain tone consistent with wanting to be let in while standing next to the door **30**. As another example, the control system **100** may analyze accelerometer data (e.g., from the accelerometer unit **600**) to identify if the pet is scratching at the door **30**. The control system **100** may also be configured to learn which animal is the pet associated with the building within which the door **30** is installed (e.g., based on frequently being captured by the camera **520**, based on an identification device on the pet, etc.) and prevent all such pet functionality for other animals that are not the pet (e.g., wild animals, neighbors pets, etc.). In some embodiments, the control system **100** is configured to analyze visual data (e.g., from the camera **520**) to detect whether a person other than the occupant can be seen in the frame. If a person is detected, the control system **100** may prevent the opening the door **30** or pet door thereof to prevent unwanted entry by such person.

(411) By way of yet another example, the control system **100** may be configured to provide advanced lighting functionality. For example, the control system **100** may be configured to perform one or more lighting control features. One feature may include determining the intent of a person at the door and changing lighting features based on such intent. For example, the control system **100** may be configured to distinguish between a good intent (e.g., soft knocking) or a bad intent (e.g., hard knocking, shaking of the door, characteristics of a false entry attempt) based on accelerometer data (e.g., from the accelerometer unit **600**). If a bad intent is identified, the control system **100** may be configured to change lighting color (e.g., from white to red, etc.), flash the lights, etc. to deter such bad intent. Another feature may include controlling the external lights to provide external notifications. For example, if the building alarm system has been tripped (e.g., fire alarm, home alarm, occupant fall alert, etc.), the lights may flash and/or change color(s) to catch the eye of response personnel.

(412) By way of still yet another example, the control system **100** may be configured to provide automatic activation of the away mode. For example, the control system **100** may be configured to monitor building occupancy and ingress and egress through the door **30** by monitoring the location of the user devices **190**, accessing data from in-building sensors (e.g., water use sensors, occupancy sensors, motion sensors, light sensors, vibration sensors, etc.), and/or accessing data acquired from the sensors of the door **30**. If the control system **100** determines that all occupants are out of the building, the control system **100** may be configured to engage the away mode (or send a notification to the user device **190** to request activation of the away mode). In some instances, the control system **100** may active the away mode when it is apparent that the occupants will be gone for an extended period of time (e.g., more than a few hours, for a few days, etc.). For example, the occupants may leave with suitcases indicating that they are going on a trip. As another example, the occupants may leave with backpacks, briefcases, etc. indicating that they are going to school and work for the day.

(413) With respect to the away mode and the control of the home water monitoring and control system, the control system **100** may be configured to vary the functions it performs based on confidence level. For example, if the control system **100** has a lower confidence that the building is completely empty when the away mode is active, the control system **100** may send a leak detection notification or alert to the user device **190** if water is being used. If, however, the confidence level is higher that the building is unoccupied, the control system **100** may shut of the water to the building if water use is detected (i.e., a leak).

(414) By way of another example, the control system **100** may be configured to provide various advanced notification suppression features. For example, the control system **100** may detect continued or increased activity at the door **30** (e.g., indicating a party). One feature may include preventing continuous presence detection and/or doorbell notifications on the user device **190**.

Another feature may include suppressing critical water usage alert notifications and false shutoffs (until the guest leave), understanding that water usage will be higher than usual while more occupants are in the building.

(415) Perimeter Wire Routing Channels

(416) Referring to FIG. 64, a door 30 is shown as another embodiment to the door 30 of FIG. 3. The door 30 of FIG. 64 may be substantially similar to the door 30 of FIG. 3, except as otherwise specified herein. Accordingly, any description of the door 30 of FIG. 3 may apply to the door 30 of FIG. 64.

(417) Referring to FIG. 64, the door 30 has a series of faces, surfaces, or sides. The door 30 has a front surface 800 and a rear surface 802 facing opposite directions. The door 30 may be configured such that the front surface 800 defines an exterior surface of the door 30, and the rear surface 802 defines an interior surface of the door 30. Alternatively, the door 30 may be configured such that the front surface 800 defines an interior surface of the door 30, and the rear surface 802 defines an exterior surface of the door 30. The door 30 has a top surface 804 facing upward and a bottom surface 806 facing downward. The door 30 has a hinge surface 808 along a hinge side of the door 30. The hinges 32 may be coupled to the hinge surface 808. The door 30 has a lock surface 810 opposite the hinge surface 808.

(418) Referring still to FIG. 64, the door 30 includes a module, interface, controller, section, or control unit, shown as interactive unit 820. The interactive unit 820 may incorporate some or all of the functionality of the modules of the door 30 of FIG. 3. By way of example, the interactive unit 820 may include the components and/or functionality of the power supply unit 130, the window control unit 132, the main control unit 134, the bridge module 136, the deadbolt units 138, and/or the interactive unit 140. The interactive unit 820 may have a similar position to the interactive unit 140.

(419) As shown, the door 30 includes a plurality of (e.g., two) deadbolt units 138 positioned along the lock surface 810. As shown, the deadbolt units 138 are electrically coupled to the interactive unit 820 through a secondary door bus, shown as secondary door wiring harness 822. The secondary door wiring harness 822 may pass power and/or data between the deadbolt units 138 and the interactive unit 820. In some embodiments, the deadbolt controllers 266 are omitted from the deadbolt units 138, and the interactive unit 820 directly controls operation of the deadbolt units 138 (e.g., actuation of the deadbolt actuators 260). In some embodiments, the deadbolt actuators 260 are omitted and the interactive unit 820 or a single deadbolt actuator is mechanically coupled to the deadbolt units 138 (e.g., by a linkage), such that the interactive unit 820 or the single deadbolt actuator provides mechanical energy to actuate each of the deadbolts 252 of the deadbolt units 138.

(420) The door wiring harness 170 may supply electrical energy to power the interactive unit 820 and the deadbolt units 138. As shown, the interactive unit 820 is electrically coupled to the power converter 104 through the power transfer assembly 106, the door wiring harness 170, the uppermost deadbolt unit 138, and the secondary door wiring harness 822. Specifically, the door wiring harness 170 is directly connected to the power transfer assembly 106 and the uppermost deadbolt unit 138, and the secondary door wiring harness 822 is directly connected to the uppermost deadbolt unit 138 and the interactive unit 820. In other embodiments, the door wiring harness 170 is directly connected to the interactive unit 820 instead of electrically passing to the interactive unit 820 indirectly through the uppermost deadbolt unit 138 and the secondary door wiring harness.

(421) The door 30 includes a series of grooves, recesses, or notches, or perimeter wire routing channels, shown as perimeter channels 830. Specifically, the perimeter channels 830 may be defined by the subframe 40 of the door 30. The perimeter channels 830 together define a continuous channel between the power transfer assembly 106 and the interactive unit 820 that receives the door wiring harness 170. The perimeter channels 830 facilitate containing the door wiring harness 170 within the door 30. The perimeter channels 830 extend around a periphery or

perimeter (e.g., one or more sides) of the door **30** between (a) the power transfer assembly **106** and (b) the interactive unit **820** and/or a deadbolt unit **138**. In some embodiments, the perimeter channels **830** are formed by removing material from (e.g., machining away) a portion of the door **30** (e.g., the subframe **40**).

(422) The perimeter channels **830** include (a) a first channel or hinge-side channel, shown as hinge channel **832**, (b) a second channel or top-side channel, shown as top channel **834**, and/or (c) a third channel or lock-side channel, shown as lock channel **836**. The hinge channel **832** extends substantially vertically along and inward from the hinge surface **808**. The top channel **834** extends substantially horizontally along and inward from the top surface **804**. The lock channel **836** extends substantially vertically along and inward from the lock surface **810**. The top channel **834** extends from the hinge channel **832** to the lock channel **836**, forming a single continuous channel. In other embodiments, one or more of the hinge channel **832**, the top channel **834**, or the lock channel **836** are omitted from the door **30**. By way of example, the lock channel **836** or a portion thereof may be omitted. In such an implementation, the lock channel **836** may be entirely omitted or the lock channel **836** may only extend to the location of the uppermost deadbolt unit **138**. In such an embodiment, the door wiring harness **170** may pass from the top channel **834** to the uppermost deadbolt unit **138** with or without use of the lock channel **836**, and the secondary door wiring harness **822** may pass from the uppermost deadbolt unit **138** to the interactive unit **820** without the use of the lock channel **836**.

(423) Referring to FIGS. **65** and **66**, the top channel **834** is shown according to an exemplary embodiment. In some embodiments, the hinge channel **832** and/or the lock channel **836** have a similar shape and/or size to the top channel **834**. The top channel **834** includes a first portion or outer portion, shown as cover portion **840**, and a second portion or inner portion, shown as wire portion **842**. The cover portion **840** extends between the wire portion **842** and the exterior surface of the door **30** (e.g., the top surface **804**). The cover portion **840** has a first width $W_{sub.1}$ measured perpendicular to the front surface **800**, and the wire portion **842** has a second width $W_{sub.2}$ measured perpendicular to the front surface **800**. The first width W_i is greater than the second width $W_{sub.2}$ such that the cover portion **840** extends beyond the wire portion **842**.

(424) FIGS. **67-77** illustrate various embodiments of covers that can be utilized with the door **30**. The covers can be inserted within the cover portions **840** and/or wire portions **842** of the perimeter channels **830** such that the covers extend over the wire portions **842**. The covers protect the door wiring harness **170** from contact with debris or other objects from outside of the door **30** and facilitate containing the door wiring harness **170** within the wire portions **842**. In some embodiments, the covers are removably coupled to the subframe **40**. When maintenance or upgrading of the door **30** is desired, a user may remove the cover to permit access to the door wiring harness **170**. Placing the perimeter channels **830** along the exterior of the door **30** and making the covers removable facilitates assembly of the door **30** at a different time and/or location than when and where the subframe **40** and other components of the door **30** are produced.

(425) The door **30** may include separate covers for each perimeter channel **830**, or a single, continuous cover may extend along multiple of the perimeter channels **830**. By way of example, a first cover may extend across the hinge channel **832** and the top channel **834**, and a second cover may extend across the lock channel **836**. Additionally, the door **30** may utilize any combination of the covers shown and described herein. By way of example, the door **30** may utilize the cover **850** of FIG. **68** to cover the hinge channel **832**, the cover **880** of FIG. **61** to cover the top channel **834**, and the cover **846** of FIG. **67** to cover the lock channel **836**.

(426) As shown in FIG. **65**, the door **30** may include one or more fasteners **844** that engage (e.g., thread into, nail into, etc.) the subframe **40** of the door **30**. The fasteners **844** may be screws, nails, bolts, or other types of fasteners. The fasteners **844** may facilitate coupling one or more of the covers (e.g., the cover **850**, the cover **880**, the raceway **890**, the tubular cover **920**, etc.) to the subframe **40**. The fasteners **844** may extend through apertures defined by the covers and engage

with the subframe **40** to removably couple the covers to the subframe **40**. When maintenance or upgrading of the door **30** is desired, a user may remove the fasteners **844** to release the cover and permit access to the door wiring harness **170**. The door **30** may include various features described herein that separate the fasteners **844** from the door wiring harness **170**, preventing contact that could otherwise damage the door wiring harness **170**.

(427) Referring to FIG. **67**, the door **30** includes a cover, plate, or bracket, shown as cover **846**. As shown, the cover **846** is received within the cover portion **840** of the lock channel **836**. The cover **846** is elongate and flat, having a rectangular cross section that fits within the cover portion **840**. The cover **846** may be held in place without the use of the fasteners **844**. By way of example, the cover **846** may interlock with the subframe **40**, may magnetically couple to the subframe **40**, may be adhered to the subframe **40**, or may be otherwise coupled to the subframe **40**. As the fasteners **844** are omitted from this portion of the door **30**, the door wiring harness **170** may be contained within the wire portion **842** of the lock channel **836** without the potential for contact between fasteners **844** and the door wiring harness **170**.

(428) Referring to FIG. **68**, a cover or bracket, shown as cover **850**, can be inserted within the cover portions **840** of the perimeter channels **830** to protect and contain the door wiring harness **170**. The cover **850** is flat, elongated, and has a substantially rectangular cross section. The cover **850** defines one or more apertures, shown as fastener apertures **852**, extending entirely through the cover **850**. Specifically, FIG. **68** illustrates the cover **850** as defining two fastener apertures **852**. In other embodiments, the cover **850** defines more or fewer fastener apertures **852**. The fastener apertures **852** are each sized to receive a fastener **844** to couple the cover **850** to the subframe **40**. As shown, the fastener apertures **852** are substantially laterally centered on the cover **850**. Accordingly, when the cover **850** is inserted into a cover portion **840** of a perimeter channel **830**, the fastener apertures **852** align with the wire portion **842**. When installed, the fasteners **844** extend through the fastener apertures **852** and the wire portion **842** to reach the subframe **40**.

(429) Referring to FIG. **69**, the door **30** may include one or more barriers or inserts **860** that prevent contact between the door wiring harness **170** and the fasteners **844** that couple the cover **850** to the subframe **40**. The insert **860** includes a base or baseplate, shown as plate **862**. The plate **862** is rectangular and may have a width that is slightly smaller than the width $W_{sub.2}$ to facilitate insertion of the insert **860** into a wire portion **842**. When inserted, the plate **862** may extend longitudinally along the wire portion **842**. A rounded protrusion, shown as protrusion **864**, extends outward from the plate **862**. As shown, the protrusion **864** is elongated in the same longitudinal direction as the plate **862**, facilitating routing of the door wiring harness **170** around the protrusion **864**. The protrusion **864** may have a substantially elliptical cross section. In some embodiments, the protrusion **864** and the plate **862** are integrally formed as a single, continuous piece. An aperture or passage, shown as fastener aperture **866**, extends entirely through the protrusion **864** and the plate **862**. The fastener aperture **866** is sized to receive a fastener **844** therethrough.

(430) To assemble the door **30**, the insert **860** is inserted within a wire portion **842** approximately where a user intends to install a fastener **844**. The insert **860** may be installed with the protrusion **864** facing outward from the perimeter channel **830** or inward into the perimeter channel **830**, although an outward-facing orientation may facilitate installation of the door wiring harness **170**. The door wiring harness **170** is then routed through the wire portion **842** and around the protrusion **864**. The cover **850** is installed within the cover portion **840**. A fastener **844** is inserted through both a fastener aperture **852** and the fastener aperture **866** and engaged with the subframe **40**, coupling the cover **850** and the insert **860** to the subframe **40** and to one another. In an installed configuration, the door wiring harness **170** is positioned between the cover **850**, the protrusion **864**, the plate **862**, and a wall of the wire portion **842**. The insert **860** (e.g., the plate **862** and the protrusion **864**) acts as a barrier, preventing contact between the fastener **844** and the door wiring harness **170**. The door **30** may include one insert **860** for each of the fastener apertures **852**.

(431) Referring to FIG. **70**, an insert or barrier, shown as insert **870**, is as another embodiment of

the insert **860**. The insert **870** may be substantially similar to the insert **860** except as otherwise specified herein. The insert **870** further includes a series of protrusions, shown as wire guides **872**, extending upward from the plate **862**. The wire guides **872** are substantially cylindrical and spaced apart from one another. The wire guides **872** facilitate routing the door wiring harness **170**. Specifically, a user can select one or more gaps between the wire guides **872** and route the door wiring harness **170** through those gaps, at which point the wire guides **872** will hold the door wiring harness **170** in that desired orientation.

(432) Referring to FIGS. **71** and **72**, a cover or bracket, shown as cover **880**, can be inserted within the perimeter channels **830** to protect and contain the door wiring harness **170**. The cover **880** includes a cover portion, shown as plate **882**. The plate **882** is flat, elongated, and has a substantially rectangular cross section. The plate **882** defines one or more apertures, shown as fastener apertures **884**, extending entirely through the plate **882**. Specifically, FIG. **71** illustrates the plate **882** as defining four fastener apertures **884**. In other embodiments, the plate **882** defines more or fewer fastener apertures **884**. The fastener apertures **884** are each sized to receive a fastener **844** to couple the cover **880** to the subframe **40**. As shown, the fastener apertures **884** are laterally offset from the center of the plate **882**. Specifically, the fastener apertures **884** are aligned with the portions of the subframe **40** where the perimeter channel **830** narrows from the width $W_{sub.1}$ to the width $W_{sub.2}$ to form a pair of inset shoulders. The cover **880** further includes a pair of protrusions, panels, or barriers, shown as flanges **886**, that are fixedly coupled to the plate **882** and extend outward from the plate **882**. The flanges **886** are laterally offset from one another to define a space that is sized to receive the door wiring harness **170**. The flanges **886** and the plate **882** may be integrally formed as a single, continuous piece.

(433) To install the cover **880**, the door wiring harness **170** is inserted into the space between the flanges **886**, and the cover **880** is inserted into the perimeter groove **330**. Fasteners **844** are inserted through the fastener apertures **884** and engage the subframe **40**. When the fasteners **844** are installed perpendicular to the plate **882**, the fasteners **844** remain within the subframe **40** and do not enter the wire portion **842**. However, if the user is imprecise when installing the fasteners **844**, the fasteners **844** may extend inward at an angle and penetrate into the wire portion **842**. The flanges **886** act as barriers in this situation, extending along the walls of the wire portion **842** and preventing the fasteners **844** from contacting the door wiring harness **170**.

(434) Referring to FIGS. **73-75**, a cover or bracket assembly, shown as raceway **890**, can be inserted within the perimeter channels **830** to protect and contain the door wiring harness **170**. As shown, the raceway **890** includes a first, removable portion, shown as cover **892**, and a second, fixed portion, shown as body **894**. The body **894** is fixedly coupled to the subframe **40** and positioned within the wire portion **842**. The cover **892** is removably coupled to the body **894** and positioned within the cover portion **840**. A space, volume, or void, shown as wire volume **896**, is defined between the body **894** and the cover **892** of the raceway **890**. The wire volume **896** receives the door wiring harness **170**.

(435) The cover **892** includes a plate **900** that is flat, elongated, and has a substantially rectangular cross section. The cover **892** further includes a pair of protrusions, panels, or barriers, shown as flanges **902**, that are fixedly coupled to the plate **900** and extend vertically outward from the plate **900**. The flanges **902** are laterally offset from one another. The flanges **902** and the plate **900** may be integrally formed as a single, continuous piece. Each flange **902** defines a recess, shown as engagement groove **904**, that extends along a length of the flange **902** and faces laterally outward. Each engagement groove **904** may have a tapered cross section.

(436) The body **894** includes a plate **910** and a pair of walls **912** extending upward from the plate **910**. The plate **910** defines one or more fastener apertures **914** that are each configured to receive a fastener **844** to couple the body **894** to the subframe **40**. In some embodiments, the fastener apertures **914** are countersunk or counterbored to inset the heads of the fasteners **844**. In other embodiments, the body **894** is otherwise coupled to the subframe **40** (e.g., adhered, press fit, etc.).

(437) Positioned near a distal end of each of the walls **912** is a protrusion, shown as engagement flange **916**. Each engagement flange **916** extends longitudinally along the corresponding wall **912** and laterally inward from the corresponding wall **912**. The engagement flanges **916** may have tapered shapes that correspond to the engagement grooves **904**. When installed, the engagement flanges **916** are each received within a corresponding engagement groove **904**. The cover **892** may be made of an elastically-deformable material, such as plastic or steel, such that the flanges **902** can bend laterally inward. To install the cover **892**, the cover **892** may be pressed down such that the tapered shapes of the engagement flanges **916** bend the flanges **902** inward, and the engagement flanges **916** are received within the engagement grooves **904**. This engagement holds the cover **892** in place until an upward force is applied that is sufficient to reverse this process.

(438) Referring to FIGS. **76** and **77**, a cover or bracket, shown as tubular cover **920**, can be inserted within the perimeter channels **830** to protect and contain the door wiring harness **170**. The tubular cover **920** includes a first portion, shown as tube **922**, coupled to a second portion, shown as mounting flange **924**. The tube **922** includes a series of walls **926** that are arranged in a rectangular (e.g., square) pattern to form a tubular member. The walls **926** define a passage or aperture, shown as wire passage **928**, therebetween, that receives the door wiring harness **170**. The wire passage **928** extends longitudinally through the entirety of the tube **922**, such that the door wiring harness **170** can extend beyond both ends of the tube **922**.

(439) The mounting flange **924** extends laterally outward from the tube **922**. As shown, the mounting flange **924** extends laterally outward in only one direction. In other embodiments, the tubular cover **920** includes two mounting flanges **924** that extend in opposing lateral directions. The mounting flange **924** defines a series of apertures, shown as fastener apertures **930**, that each extend through the entirety of the mounting flange **924**. Each fastener aperture **930** is sized to receive a fastener **844**.

(440) To install the tubular cover **920**, the door wiring harness **170** is inserted through the wire passage **928**, and the tubular cover **920** is inserted into the perimeter channel **830**. Once inserted, the wire portion **842** receives the tube **922**, and the cover portion **840** receives the mounting flange **924**. The fasteners **844** may be inserted through the fastener apertures **930** and engage the subframe **40**, removably coupling the tubular cover **920** to the subframe **40**. One of the walls **926** extends between the door wiring harness **170** and the fasteners **844**, acting as a barrier to protect the door wiring harness **170** from the fasteners **844**. Another of the walls **926** and the mounting flange **924** act as a cover to protect the door wiring harness **170** from contact with outside objects.

(441) In some embodiments, the door wiring harness **170** and the cover are preassembled into a subassembly that can later be installed on the door **30** or removed from the door **30** as a single component. By way of example, one tubular cover **920** may be sized to extend from the power transfer assembly **106** to the uppermost deadbolt unit **138**. The door wiring harness **170** may be inserted through the tubular cover **920** and fixedly coupled to the tubular cover **920** (e.g., by potting the door wiring harness **170** with a rubber or epoxy). In an embodiment where the door wiring harness **170** and the cover are formed as a subassembly, a user can easily install the subassembly by coupling the subassembly to the door **30** and electrically coupling the door wiring harness **170** to the power transfer assembly **106** and the deadbolt unit **138**. This simplifies the installation process relative to an embodiment where the door wiring harness **170** is separate from the cover. The ends of the door wiring harness **170** may be provided with quick disconnect electrical connectors to facilitate installation and removal of the subassembly.

(442) Referring to FIGS. **78** and **79**, a mounting configuration for the hinges **32** is shown according to an exemplary embodiment. Each hinge **32** is received within a recess, shown as hinge recess **940**, that is defined by the subframe **40**. The hinge recess **940** defines the vertical position of the hinge **32** on the door **30**. The hinges **32** are coupled to the subframe **40** by a series of fasteners, shown as hinge bolts **942**. Each hinge **32** may define one or more apertures or passages, shown as fastener apertures **944**, that each receive one of the hinge bolts **942**. The power transfer assembly

106 may be coupled to the subframe **40** using a similar mounting configuration.

(443) As shown, the hinges **32** are positioned along the hinge surface **808**, such that a portion of the hinge channel **832** passes directly beneath at least one of the hinges **32**. Due to the position of the hinges **32**, one or more of the hinge bolts **942** may pass in proximity to the door wiring harness **170**. Accordingly, it is desirable to protect the door wiring harness **170** from contact with the hinge bolts **942** and prevent damage to the door wiring harness **170**. To accomplish this, the door **30** includes an insert or spacer, shown as hinge insert **950**, that extends between the hinge **32** and the subframe **40** and guides the door wiring harness **170** around the hinge bolts **942**.

(444) As shown, the subframe **40** defines a recess, shown as insert recess **952**, that receives the hinge insert **950**. The insert recess **952** is positioned beneath the hinge recess **940**, such that the subframe **40** captures the hinge insert **950** in a position directly beneath the hinge **32**. The hinge insert **950** defines a series of passages or apertures, shown as fastener apertures **954**, through which the hinge bolts **942** may pass. The fastener apertures **954** may be positioned to align with the fastener apertures **954** of the hinge **32**.

(445) The hinge insert **950** further defines a passage, shown as wire passage **956**, through which the door wiring harness **170** passes. The wire passage **956** is positioned to be continuous with the hinge channel **832** when the hinge insert **950** is installed. The wire passage **956** guides the door wiring harness **170** around the fastener apertures **954** such that the hinge insert **950** acts as a barrier between the door wiring harness **170** and the hinge bolts **942**.

(446) FIG. **80** illustrates a hinge insert **960** an exemplary embodiment of the hinge insert **950**. The wire passage **956** of the hinge insert **960** is a channel that extends through the hinge insert **960**. The channel is inset into the hinge insert **960**, such that the hinge insert **960** separates the hinge **32** away from the door wiring harness **170**. The wire passage **956** oscillates throughout the thickness of the door **30** to avoid intersecting the wire passage **956** with the fastener apertures **954**.

(447) FIG. **81** illustrates a hinge insert **970** as another exemplary embodiment of the hinge insert **950**. The hinge insert **970** includes a base member, shown as base plate **972**. The base plate **972** defines a series of passages or apertures, shown as fastener apertures **974**, that are each configured to receive a fastener to couple the hinge insert **970** to the subframe **40**. A series of protrusions, shown as hinge standoffs **976**, extend upward from the base plate **972**, spacing the hinge **32** away from the base plate **972**. The hinge standoffs **976** are each substantially cylindrical. The hinge standoffs **976** are each substantially centered about one of the fastener apertures **954**. Accordingly, the hinge standoffs **976** act as a barrier between the hinge bolts **942** and the door wiring harness **170**. The wire passage **956** of the hinge insert **970** is defined between the hinge standoffs **976**.

(448) Window Connection

(449) As shown in FIGS. **82-91**, the door **30** includes a plurality of components that can be assembled to form an assembly, shown as door assembly **1000**. Any description of the door **30** of FIGS. **3** and **64** may apply to the door **30** and the door assembly **1000** of FIGS. **82-91**. The door assembly **1000** includes the subframe **40** (i.e., the hinge stile **42**, the top rail **44**, the lock stile **46**, and the bottom rail **48**), the exterior skin **50**, the interior skin **52**, the window **60**, a first wiring portion of the door wiring harness **170** including a first portion, shown as upper window wiring **174**, and a second portion, shown as lower window wiring **176**, an electrical connector, shown as window connector **1020**, connecting the upper window wiring **174** to the lower window wiring **176**, and a plug, shown as wire grommet **1030**. In some embodiments, the door assembly **1000** does not include the window connector **1020** and the upper window wiring **174** and the lower window wiring **176** are integrally connected (i.e., continuous wiring).

(450) As shown in FIGS. **82** and **90**, the top rail **44** includes (a) the raceway **890** including the cover **892** and the body **894** and (b) a support, shown as top rail support **1010**. While shown herein as including the raceway **890**, in other embodiments, the top rail **44** alternatively includes the cover **850**, the cover **880**, or the tubular cover **920**. Further, any description of the top rail **44** disclosed herein may similarly apply to the hinge stile **42** and the lock stile **46**. As shown in FIGS. **83** and **90**,

the top rail support **1010** has an elongated base, shown as support base **1012**, with protrusions, shown as side supports **1014**, extending along and upward from opposing lateral edges of the support base **1012**. The support base **1012** and the side supports **1014** cooperatively define a pair of recesses, shown as slots **1016**, along each lateral side thereof. The top rail support **1010** defines a cavity, shown as raceway cavity **1018**, positioned between the top rail support **1010** and the side supports **1014**. The support base **1012** defines an aperture, shown as window wiring aperture **1019**. (451) As shown in FIGS. **82**, **84**, and **90**, the window connector **1020** includes a first component, shown as connector base **1022**, a second component, shown as upper quick connect **1024**, and a third component, shown as lower quick connect **1026**. As shown in FIG. **90**, the upper quick connect **1024** is configured to couple to the upper window wiring **174**, the lower quick connect **1026** is configured to couple to the lower window wiring **176**, and the connector base **1022** is configured to receive the upper quick connect **1024** and the lower quick connect **1026** such that the upper quick connect **1024** and the lower quick connect **1026** releasably engage and electrically couple the upper window wiring **174** and the lower window wiring **176** together and, thereby, electrically couple the window **60** to the door wiring harness **170**.

(452) As shown in FIGS. **85-88**, the inner facing side or surface of each of the exterior skin **50** and the interior skin **52** defines (a) one or more first protrusions, shown as top rail guide **1040**, extending therefrom laterally along and proximate the upper or top end thereof, (b) one or more second protrusions, shown as hinge stile guide **1042**, extending therefrom vertically along and proximate the hinge end/side thereof, (c) one or more third protrusions, shown as bottom rail guide **1044**, extending therefrom laterally along and proximate the lower or bottom end thereof, (d) one or more fourth protrusions, shown as lock stile guide **1048**, extending therefrom vertically along and proximate the lock end/side thereof, and (e) central framing, shown as window framing **1050**, extending therefrom and defining an aperture therein, shown as window aperture **1051**.

(453) As shown in FIG. **90**, (a) the top rail **44** is disposed between the exterior skin **50** and the interior skin **52** where the slots **1016** of the top rail support **1010** engage with the top rail guides **1040** to secure the top rail **44** in position vertically between the exterior skin **50** and the interior skin **52** and (b) the raceway **890** is received within the raceway cavity **1018** of the top rail support **1010**. As shown in FIG. **91**, the bottom rail **48** is disposed between the exterior skin **50** and the interior skin **52** where a pair of recesses, shown as slots **49**, thereof engage with the bottom rail guides **1044** to secure the bottom rail **48** in position vertically between the exterior skin **50** and the interior skin **52**. While not shown, the hinge stile **46** and the lock stile **48** may define slots similar to the slots **1016** and the slots **49**, and similarly engage with the hinge stile guides **1046** and the lock stile guides **1048**, respectively, to secure the hinge stile **46** and the lock stile **48** in position horizontally between the exterior skin **50** and the interior skin **52**.

(454) As shown in FIGS. **85-88**, the window framing **1050** of each of the exterior skin **50** and the interior skin **52** includes a first portion, shown as inward tapering framing **1052**, extending along a first portion (e.g., a first half, a left half, a right half, a lower half, an upper half, etc.) of the window aperture **1051** and a second portion, shown as outward tapering framing **1056**, extending along a second portion (e.g., a second half, a right half, a left half, an upper half, a lower half, etc.) of the window aperture **1051**. More specifically, (a) the inward tapering framing **1052** has a substantially flat interior surface and a tapered exterior surface having a first profile, shown as exterior stepped profile **1054**, and (b) the outward tapering framing **1056** has a substantially flat exterior surface and a tapered interior surface having a second profile, shown as interior stepped profile **1058**. As shown in FIGS. **90** and **91**, the window framing **1050** of the exterior skin **50** and the interior skin **52** engage with each other to secure the window **60** therebetween such that the window **60** extends across the window apertures **1051** thereof. According to an exemplary embodiment, (a) the exterior stepped profile **1054** of the inward tapering framing **1052** of the exterior skin **50** is configured to engage with the interior stepped profile **1058** of the outward tapering framing **1056** of the interior skin **52** and (b) the exterior stepped profile **1054** of the inward tapering framing **1052** of the interior

skin **52** is configured to engage with the interior stepped profile **1058** of the outward tapering framing **1056** of the exterior skin **50** to provide a secure fit and engagement between the window framing **1050** of the exterior skin **50** and the interior skin **52**.

(455) As shown in FIGS. **85-88** and **90**, the upper portions of the window framing **1050** of the exterior skin **50** and the interior skin **52** cooperatively define an aperture (i.e., each at least partially defines a portion of the aperture), shown as window wiring aperture **1060**, that receives the wire grommet **1030**. As shown in FIG. **90**, the body **894** of the raceway **890** defines an aperture, shown as window wiring aperture **918**. As shown in FIGS. **89** and **90**, the door assembly **1000** defines (a) a first chamber, cavity, or passageway, shown as upper insulation chamber **1062**, between the top rail support **1010** of the top rail **44**, the window framing **1050** of the exterior skin **50** and the interior skin **52**, and the inner facing sides or surfaces of the exterior skin **50** and the interior skin **52** and (b) a second chamber, cavity, or passageway, shown as lower insulation chamber **1064**, between the bottom rail **48**, the window framing **1050** of the exterior skin **50** and the interior skin **52**, and the inner facing sides or surfaces of the exterior skin **50** and the interior skin **52**. Similar chambers or passages (e.g., a hinge stile insulation chamber, a lock stile insulation chamber, etc.) may be defined between the hinge stile **42**, the lock stile **46**, the window framing **1050** of the exterior skin **50** and the interior skin **52**, and the inner facing sides or surfaces of the exterior skin **50** and the interior skin **52**.

(456) According to an exemplary embodiment, the arrangement/configuration of the top rail **44** (e.g., the raceway **890**, the top rail support **1010**, etc.), the exterior skin **50** and the interior skin **52** (e.g., the window framing **1050** thereof, etc.), and the door wiring harness **170** facilitates electrically connecting the window **60** to the door wiring harness **170** and, therefore, the power transfer assembly **106**, with relative ease during manufacture, during assembly, and/or during on-site installation. As shown in FIG. **90**, the upper window wiring **174** extends (a) through the wire volume **896** of the raceway **890** and (b) through the window wiring aperture **918** of the body **894** of the raceway **890** where the upper window wiring **174** terminates with the upper quick connect **1024** of the window connector **1020**. The window connector **1020** is at least partially received by and extends through the window wiring aperture **1019** of the support base **1012** of the top rail support **1010**. In some embodiments, the connector base **1022** is received by and secured within the window wiring aperture **1019** of the support base **1012**. As shown in FIG. **90**, the lower window wiring **176** extends (a) from the lower quick connect **1026** of the window connector **1020**, (b) through the upper insulation chamber **1062**, (c) through the wire grommet **1030** and the window wiring aperture **1060** of the window framing **1050**, and (d) to the window **60** to electrically connect the window **60** to the door wiring harness **170**.

(457) As shown in FIGS. **90** and **91**, the door assembly **1000** includes an insulating material (e.g., foam, injection foam, etc.), shown as insulation **1070**, disposed within the upper insulation chamber **1062** and the lower insulation chamber **1064**. According to an exemplary embodiment, the connector base **1022** and the lower quick connect **1026** substantially seal the window wiring aperture **1019** of the support base **1012** and the wire grommet **1030** and the lower window wiring **176** substantially seal the window wiring aperture **1060** of the window framing **1050** such that, when assembled, the interior chambers or passages (e.g., the upper insulation chamber **1062**, the lower insulation chamber **1064**, the hinge stile insulation chamber, the lock stile insulation chamber, etc.) of the door assembly **1000** may be foam filled with a flowable and/or expandable insulation or foam.

(458) Door System with Access Panel and Lights with Integrated Control Circuitry

(459) Referring to FIGS. **92-99**, a door system **1100** is shown as another embodiment to the door system **10**. The door system **1100** may be substantially similar to the door system **10**, except as otherwise specified herein. Accordingly, any description of the door system **10** may apply to the door system **1100**. As shown in FIG. **92**, the door system **1100** includes the door frame **20**, the door **30**/the door assembly **1000**, the power source **102**, the doorbell system **180**, the jamb lights **790**, a

pair of side panels, shown as panels **1002**, an external panel, shown as access panel **1110**, and a second power distribution system or bus, shown as wiring harness **1120**.

(460) As shown in FIG. **92**, the panels **1002** are disposed along the right side and the left side of the door **30**. In other embodiments, the door system **1100** includes only one of the panels **1002** along the right side of the door **30** or the left side of the door **30**. In still other embodiments, the door system **1100** does not include the panels **1002**. As shown in FIG. **92**, each of the panels **1002** includes a pane of transparent material (e.g., glass, polycarbonate, etc.) or privacy window, shown as panel window **62**, embedded within the panels **1002**. In some embodiments, the panel windows **62** are operable (e.g., electrically) to vary a visibility level through the panel windows **62**. By way of example, the panel windows **62** may be made from and/or coated with an electrochromic material that changes opacity in response to an applied voltage. In some embodiments, the visibility level of the panel windows **62** is varied in response to a user input, permitting the user to control the visibility level of the panel windows **62**. Additionally or alternatively, the panel windows **62** include privacy elements embedded therein or integrated therewith. In some embodiments, the privacy elements are or include electrically-controllable shades/blinds disposed within the panel windows **62** or disposed along an interior side of the panel windows **62**. In some embodiments, the window **60** and/or the panel windows **62** are transparent displays. Such transparent displays can be controlled to provide messages, pictures, videos, etc. thereon and provide similar privacy functionality as described elsewhere herein (e.g., transition from a transparent to an opaque state).

(461) As shown in FIG. **92**, the access panel **1110** includes a housing, electric box, etc. that receives the power converter **104** (which is coupled to the power source **102**), the power supply unit **130** (e.g., power management unit, power management controller, main door controller, etc.), the window control unit **132**, the backup battery **222**, a wireless communications device, shown as wireless transceiver **1112**, a reset interface, shown as hard reset interface **1114**, and one or more switches, shown as electronic hardware disconnect switches **1116**. The power supply unit **130** is coupled to the power converter **104**, the window control unit **132**, the doorbell system **180**, the backup battery **222**, the wireless transceiver **1112**, the hard reset interface **1114**, and the electronic hardware disconnect switches **1116**.

(462) As shown in FIG. **92**, the wiring harness **1120** includes (a) a first portion, shown as access panel wiring harness **1122**, extending from power supply unit **130** of the access panel **1110** to an electrical connector, shown as frame connector **1124**, and (b) a second portion, shown as frame wiring harness **1126**, extending from the frame connector **1124** into and through the door frame **20** (e.g., the head jamb **24**, the side jambs **22**, etc.) of the door system **1100**, thereby electrically coupling the access panel **1110** to the door frame **20**. The frame connector **1124** may include a quick connect coupled to each of the access panel wiring harness **1122** and the frame wiring harness **1126** to facilitate easily connecting the two together during installation of the door system **1100**. According to an exemplary embodiment, the access panel wiring harness **1122** is configured to facilitate positioning the access panel **1110** external from and remote from the door frame **20** and the door **30** (e.g., up to 100 feet away, in a different room of a residence or building such as a garage or a basement or an attic, etc.). According to an exemplary embodiment, the access panel wiring harness **1122** and the frame wiring harness **1126** are configured to facilitate providing power and data communication (e.g., serial communication) from the access panel **1110** to the door frame **20** and, ultimately, the door **30**.

(463) The wireless transceiver **1112** may be configured to communicate via one or more short-range and/or long-range wireless communications protocols including, but not limited to, Wi-Fi, Bluetooth, NFC, cellular, and the like. In some embodiments, the wireless transceiver **1112** is configured to facilitate wireless communication between (a) the door **30** and the access panel **1110** (e.g., the power supply unit **130**) and (b) external devices. By way of example, the external devices may include a wireless router within the building communicating over a building Wi-Fi network.

By way of another example, the external devices may include a user device **190** communicating via a short-range communications protocol (e.g., Bluetooth, NFC, etc.) and/or a long-range communications protocol (e.g., cellular). In some embodiments, at least a portion of the data communication between the access panel **1110** (e.g., the power supply unit **13**, the wireless transceiver **1112**, etc.) and the door **30** (e.g., the interactive unit **140**, the deadbolt units **138**, the window **60**, the panel windows **62**, the handle assembly **70**, etc.) is performed wirelessly, rather than through hardwired data communication.

(464) As shown in FIG. **92**, the frame wiring harness **1126** includes first portions, shown as jamb light wiring **1128**, second portions, shown as upper panel window wiring **1132**, and a third portion, shown as power transfer assembly wiring **1138**. The jamb light wiring **1128** is coupled to the jamb lights **790** via electrical connectors (e.g., pluggable connections, quick connectors, etc.), shown as jamb light connectors **1130**. The upper panel window wiring **1132** is coupled to lower panel window wiring **1136** of the panels **1002** and, thereby, the panel windows **62**, via electrical connectors (e.g., pluggable connections, quick connectors, etc.), shown as panel window connectors **1134**. The power transfer assembly wiring **1138** is coupled to the power transfer assembly **106** via an electrical connector (e.g., a pluggable connection, a quick connector, etc.), shown as power transfer assembly connector **1140**. In some embodiments (e.g., embodiments where the camera **520** is separate from the interactive unit **140** and disposed along the door frame **20**), the frame wiring harness **1126** includes camera wiring extending through the door frame **20** to the camera **520**. An electrical quick connector may be disposed between the camera **520** and the camera wiring.

(465) As shown in FIG. **92**, the door wiring harness **170** is coupled to the power transfer assembly **106** via an electrical connector (e.g., a pluggable connection, a quick connector, etc.), shown as power transfer assembly connector **1142**. The door wiring harness **170** is coupled to (a) the window **60** via the upper window wiring **174**, the window connector **1020**, and the lower window wiring **176** (as described above) and (b) the various door components including the interactive unit **140**, the handle assembly **70**, and/or the deadbolt units **138** via third portions, shown as door component wiring **178**. In some embodiments, the door component wiring **178** only provides power to the interactive unit **140**, and not data. In such embodiments, the interactive unit **140** may perform data communications wirelessly with the power supply unit **130** of the access panel **1110** (e.g., directly, through the wireless transceiver **1112**, etc.). In some embodiments, the window **60** and the panel windows **62** operate at a higher voltage than the other electrically-operated components of the door system **1100** (e.g., the jamb lights **790**). In such embodiments, (a) the jamb light wiring **1128** and the door component wiring **178** provide electricity to the jamb lights **790** and the door components (e.g., the interactive unit **140**, the handle assembly **70**, and/or the deadbolt units **138**), respectively, at a first, lower voltage (e.g., 5V, 12V, etc.), (b) the upper panel window wiring **1132** and the upper window wiring **174** provide electricity to the panel windows **62** and the window **60**, respectively, at a second, higher voltage (e.g., 24V, 50V, etc.), and (c) the power transfer assembly wiring **1138** provides electricity to the power transfer assembly **106** at both the first, lower voltage and the second, higher voltage.

(466) The doorbell system **180** may be configured to supplement or replace the doorbell functionality provided by the interactive unit **140**. By way of example, when the doorbell button **572** of the interactive unit **140** is pressed, the interactive unit **140** may (a) activate the internal speaker **566** thereof to provide a doorbell chime function, (b) send a signal to the doorbell module **184** to activate the doorbell speaker **186**, and/or (c) send a notification to a user device **190**. The signal provided to the doorbell module **184** may be either (a) sent using wireless communication from the interactive unit **140** directly to the doorbell module **184**, (b) sent using wireless communication from the interactive unit **140** to the power supply unit **130** and then sent using wireless communication from the power supply unit **130** to the doorbell module **184**, (c) sent using wireless communication from the interactive unit **140** to the power supply unit **130** and then sent

using wired communication from the power supply unit **130** to the doorbell module **184**, (d) sent using wired communication from the interactive unit **140** to the power supply unit **130** and then sent using wireless communication from the power supply unit **130** to the doorbell module **184**, or (e) sent using wired communication from the interactive unit **140** to the power supply unit **130** and then sent using wired communication from the power supply unit **130** to the doorbell module **184**. (467) According to an exemplary embodiment, the hard reset interface **1114** and the electronic hardware disconnect switches **1116** facilitate initiating a hard reset process for one or more electronic hardware components coupled to or integrated into the door **30**, the door frame **20**, and/or the access panel **1110** (e.g., the power supply unit **130**, the window control unit **132**, the interactive unit **140**, an electronic lock of the handle assembly **70**, the deadbolt units **138**, the camera **520**, the window **60**, the jamb lights **790**, the video doorbell **1118**, etc.). The hard reset interface **1114** may include one or more buttons disposed within the access panel **1110** and coupled to the power supply unit **130**. In one embodiment, the hard reset interface **1114** includes a master reset button that facilitates resetting all of the connected electronic hardware components. In some embodiments, the hard reset interface **1114** includes a reset button associated with individual electronic hardware components or sets of electronic hardware components and, therefore, may include more than one reset button. Each of the electronic hardware disconnect switches **1116** may be positioned and configured to disconnect one or more of the electronic hardware components from power (e.g., the backup battery **222**, the power source **102**, etc.) to initiate a hard reset of the associated electronic hardware components connected thereto. In some embodiments, all of the electronic hardware components of the door **30** and the door frame **20** are connected to a single electronic hardware disconnect switch **1116**. In some embodiments, individual electronic hardware components or sets of electronic hardware components of the door **30** and the door frame **20** are connected to respective electronic hardware disconnect switches **1116** (e.g., the door components are connected to a first switch, the door frame components are connected to a second switch, etc.). In some embodiments, the power supply unit **130** has one of the electronic hardware disconnect switches **1116** integrated therewith or coupled thereto to facilitate disconnecting the power supply unit **130** from power.

(468) A user can engage with the hard reset process by applying a momentary (e.g., less than one second, less than two seconds, etc.) button press to the hard reset interface **1114**. Then, the power supply unit **130** may be configured to send a disconnect signal to one or more of the electronic hardware disconnect switches **1116** associated with the reset button pressed by the user. In response to the disconnect signal, the one or more electronic hardware disconnect switches **1116** are configured to open (i.e., stop power flow therethrough). According to an exemplary embodiment, the electronic hardware disconnect switches **1116** include a time delay relay circuit that is configured to maintain the electronic hardware disconnect switches **1116** in the open state for a certain period of time (e.g., ten second, fifteen seconds, twenty seconds, thirty seconds, etc.) and then automatically close once the certain period of time has elapsed to reinstate power to the electronic hardware components. Accordingly, the user can perform a hard reset on one or more electronic hardware components of the door system **1100** with a simple button press (e.g., rather than having to manually unplug components for a period of time and then plugging them back in, rather than having to disconnect the entire door system **1100** from the power source **102**, etc.).

(469) As shown in FIGS. **93** and **94**, the jamb lights **790** are disposed within and along a longitudinal channel defined by and extending along substantially an entire longitudinal length the side jambs **22**. The door assembly **1000** includes first jamb covers, shown as side jamb light covers **794**, positioned to enclose the channels and the jamb lights **790** of the side jambs **22**. The side jamb light covers **794** may be at least partially transparent such that light emitted by the jamb lights **790** shine therethrough. In some embodiments, the side jamb light covers **794** are partially frosted to obscure or hide the jamb lights **790** from being visible (e.g., when off). As shown in FIGS. **93** and **94**, the door assembly **1000** includes a second jamb cover, shown as head jamb cover **796**,

positioned to enclose a channel defined in the head jamb **24**. In some embodiments, the head jamb **24** does not include the jamb lights **790**. Rather, the head jamb cover **796** is included in the door assembly **1000** to provide a seamless appearance along the door frame **20** when the jamb lights **790** are off. In some embodiments, the head jamb **24** includes the jamb lights **790**.

(470) As shown in FIGS. **95**, **96**, and **98**, the jamb lights **790** include a base or substrate, shown as light strip **1200**, having a first side, shown as side **1202**, and an opposing second side, shown as side **1204**, with a respective jamb light connector **1130** disposed at an end thereof to facilitate electrically coupling the light strip **1200** to the jamb light wiring **1128**. According to an exemplary embodiment, the light strip **1200** is a flexible printed circuit board. In some embodiments, the flexible printed circuit board includes a stiff or rigid backer element (e.g., polyethylene terephthalate (“PET”), a plastic material, etc.). As shown in FIGS. **95**, **96**, and **98**, the light strip **1200** has a first or upper portion, shown as control circuitry portion **1210**, and a second or lower portion, shown as light portion **1220**.

(471) As shown in FIG. **95**, (a) the control circuitry portion **1210** of the light strip **1200** includes control circuitry (e.g., resistors, transistors, switches, capacitors, diodes, integrated circuits, fuses, etc.), shown as light control circuitry **1230**, disposed and spaced along the side **1202** thereof and (b) the light portion **1220** includes a plurality of light elements, shown as lights **1240**, disposed and spaced along the side **1202** thereof. Because the control circuitry portion **1210** does not include any of the lights **1240**, the control circuitry portion **1210** of the light strip **1200** will be dark when the lights **1240** of the light strip **1200** are on or activated. Accordingly, to prevent a dark spot from being visible through the side jamb light covers **794**, the control circuitry portion **1210** may be inserted and retained within the head jamb **24** (e.g., about 4 to 8 inches, 6 inches, etc.) until the dark spot is no longer visible. Then, to maintain the light strip **1200**, a user would need to pull thereon until the jamb light connector **1130** becomes accessible to disconnect the light strip **1200** from the frame wiring harness **1126**.

(472) As shown in FIG. **95**, (a) the control circuitry portion **1210** of the light strip **1200** includes the light control circuitry **1230** disposed and spaced along the side **1204** thereof and the lights **1240** disposed and spaced along the side **1202** thereof and (b) the light portion **1220** includes the lights **1240** disposed and spaced along the side **1202** thereof. Accordingly, with the lights **1240** disposed along both the control circuitry portion **1210** and the light portion **1220**, the dark spot provided by the light strip **1200** of FIG. **95** is substantially eliminated. Accordingly, the control circuitry portion **1210** may only need to be minimally inserted and retained within the head jamb **24** (e.g., about 1 to 2 inches).

(473) As shown in FIGS. **97** and **98**, the control circuitry portion **1210** is bent at an angle (e.g., 90 degrees) relative to the light portion **1220** at a flexion point, shown as hinge point **1250**. While shown with the light control circuitry **1230** disposed along the side **1202** of the light strip **1200** (similar to the light strip **1200** shown in FIG. **95**), in some embodiments, the light control circuitry **1230** is disposed along the side **1204** of the light strip **1200**. In such embodiments, the light strip **1200** may include the lights **1240** disposed along the side **1202** of the control circuitry portion **1210** (similar to the light strip **1200** shown in FIG. **96**). As shown in FIG. **97**, the control circuitry portion **1210** extends along the head jamb **24** such that the jamb light connector **1130** is readily accessible through the head jamb cover **796** for easy installation, maintenance, and replacement.

(474) As shown in FIG. **99**, the door assembly **1000** does not include at least the exterior portion of the interactive unit **140**. Rather, the exterior portion of the interactive unit **140** is replaced with a second user interface, a second operator interface, a second interface module, or a second interactive unit, shown as video doorbell **1118**. The video doorbell **1118** may include be a smart doorbell device having a doorbell button, an integrated camera, a speaker, a microphone, and various sensors (e.g., a light sensor, a motion sensor, a temperature sensor, radar sensor, UWB sensors, etc.). As shown in FIG. **99**, in embodiments that include the panels **1002**, the video doorbell **1118** can be installed along the side jamb **22** of one of the panels **1002**. In embodiments

that do not include the panels **1002**, the video doorbell **1118** can be installed along the side jamb **22** of the door frame **20**. Alternatively, the video doorbell **1118** may be mounted to the door **30** similar to the interactive unit **140**. As shown in FIG. **99**, the frame wiring harness **1126** includes a fourth portion, shown as video doorbell wiring **1144**. The video doorbell wiring **1144** is coupled to the video doorbell **1118** via an electrical connector (e.g., pluggable connections, quick connectors, etc.), shown as video doorbell connector **1146**. The video doorbell **1118** may be configured to communicate with the doorbell system **180** similar to the interactive unit **140** described herein.

(475) Manufacture/Assembly/Install Process

(476) According to an exemplary embodiment, the door systems, components, and assemblies disclosed herein (e.g., the door system **10**, the door **30**, the door assembly **1000**, the door system **1100**, etc.) are manufactured, assembled, and installed into a building to provide “smart door” functionalities in a seamless manner. The processes by which the door systems, components, and assemblies are manufactured, assembled, and installed are described in greater detail herein.

(477) As step A1, a plurality of components of a door system (e.g., the door system **10**, the door system **1100**) are manufactured and/or acquired. By way of example, the plurality of components may include the door frame **20** (i.e., the side jambs **22**, the head jamb **24**, and the sill **26**), the hinges **32**, the subframe **40** (i.e., the hinge stile **42**, the top rail **44**, the lock stile **46**, and the bottom rail **48**), the exterior skin **50**, the interior skin **52**, the window **60**, the handle assembly **70**, the power converter **104**, the power transfer assembly **106**, the power supply unit **130**, the window control unit **132**, the deadbolt unit(s) **138**, the interactive unit **140**, the door wiring harness **170**, the doorbell system **180**, the backup battery **222**, the jamb lights **790**, the side jamb light covers **794** and/or the head jamb cover **796**, the raceway **890**, the access panel **1110**, the wireless transceiver **1112**, the hard reset interface **1114**, the electronic hardware disconnect switches **1116**, the video doorbell **1118**, the access panel wiring harness **1122**, and/or the frame wiring harness **1126**.

(478) Now, the assembly of the door **30** (e.g., the door assembly **1000**) will be described. As step B1, the exterior skin **50** and the interior skin **52** undergo further processing. More specifically, the window wiring aperture **1060** is machined into the window framing **1050**. As step B2, the window **60**, the wire grommet **1030**, and the lower window wiring **176** of the door wiring harness **170** are installed between the exterior skin **50** and the interior skin **52** such that the window **60** is disposed within the window aperture **1051**, the wire grommet **1030** is received by the window wiring aperture **1060**, and the lower window wiring **176** extends through the wire grommet **1030**. As step B3, the subframe **40** is installed between the exterior skin **50** and the interior skin **52**, along the periphery thereof. As step B4, the exterior skin **50**, the interior skin **52**, and the subframe **40** undergo further processing. More specifically, various bores, apertures, and peripheral channels (e.g., the raceway cavity **1018**, the hinge channel **832**, the top channel **834**, the lock channel **836**, the handle bore **74**, the interface bores **164**, etc.) are machined into the exterior skin **50**, the interior skin **52**, and the subframe **40** to accommodate the hinges **32**, the handle assembly **70** (e.g., the interior doorknob or door handle, the exterior doorknob or door handle, the main latch, the main deadbolt, etc.), the power transfer assembly **106**, the deadbolt units **138**, the interactive unit **140**, the door wiring harness **170**, the raceway **890**, and/or the video doorbell **1118**. As step B5, the body **894** of the raceway **890** is installed into one or more of the peripheral channels machined into the subframe **40**. As step B6, the upper window wiring **174** of the door wiring harness **170** is routed through the body **894** of the raceway **890** and connected to the lower window wiring **176** through the window wiring aperture **1019** of the top rail support **1010** of the top rail **44** of the subframe **40** and the window wiring aperture **918** of the body **894** of the raceway **890** via the window connector **1020**. As step B7, the door component wiring **178** of the door wiring harness **170** is installed and routed through the body **894** of the raceway **890**. As step B8, the cover(s) **892** are installed to enclose the body/bodies **894** of the raceway **890**. As step B9, the interior chambers of the assembled door between (a) the exterior skin **50** and the interior skin **52** and (b) the subframe **40** are filled with the insulation **1070** (e.g., foam injection through a fill port or gap). In some

embodiments, the insulation **1070** is installed prior to the subframe **40** (e.g., a non-foam injection insulation). Further, it should be understood that the order of steps B1-B9 is only for explanation purposes. One or more of steps B1-B9 may be performed in a different order or simultaneously. Further, steps B1-B9 described herein refer to a door assembly including the subframe **40**, the exterior skin **50**, and the interior skin **52**, however, in some embodiments, the door **30** is configured as a solid core door.

(479) Now, the assembly of the door frame **20** will be described. As step C1, the side jambs **22**, the head jamb **24**, and the sill **26** are coupled together to provide the door frame **20**. As step C2, the door frame **20** undergoes further processing. More specifically, the channels of the side jambs **22** (i.e., for the jamb lights **790**) and, in some embodiments, the channel of the head jamb **24** are machined into the side jambs **22** and the head jamb **24**, respectively. As step C3, various bores, apertures, and channels are machined into the door frame **20** to accommodate the frame wiring harness **1126**. As step C4, the frame wiring harness **1126** is routed through the bores, apertures, and channels of the door frame **20**. It should be understood that the order of steps C1-C4 is only for explanation purposes. One or more of steps C1-C4 may be performed in a different order or simultaneously.

(480) Now, the assembly of the door frame **20** and the door **30** will be described. As step D1, portions of the hinges **32** are coupled to each of (a) the hinge stile **42** of the door **30** and (b) one of the side jambs **22** of the door frame **20**. As step D2, portions of the power transfer assembly **106** are coupled to each of (a) the hinge stile **42** of the door **30** and (b) one of the side jambs **22** of the door frame **20**. While coupling a first portion of the power transfer assembly **106** to the door **30**, the power transfer assembly **106** is connected to the door wiring harness **170** (e.g., the upper window wiring **174**, the door component wiring **178**). While coupling a second portion of the power transfer assembly **106** to the door frame **20**, the power transfer assembly **106** is connected to the frame wiring harness **1126** (e.g., the upper window wiring **174**, the door component wiring **178**). As step D3, the door **30** is pivotably mounted to the door frame **20** through engagement of the portions of the hinges **32**, and the door wiring harness **170** is electrically coupled the frame wiring harness **1126** through engagement of the first and second portions of the power transfer assembly **106**.

(481) Now, the installation and assembly of the access panel **1110** will be described. As step E1, a location within a building for installation/mounting of the access panel **1110** is prepared (e.g., drywall cut to provide a hole into which the access panel **1110** can be inserted). The location may be, for example, up to one hundred feet from the rough opening (“RO”) within which the door frame **20** and door **30** will be installed. As step E2, the power source **102** is routed to the location. As step E3, the access panel wiring harness **1122** is routed from the location for the access panel **1110** to a location proximate the RO for the door frame **20** and the door **30**. As step E4, the access panel **1110** is installed at the location, and the power source **102** and the access panel wiring harness **1122** are connected thereto. As step E5, the power converter **104** is inserted into the access panel **1110** and coupled to the power source **102**. As step E6, the power supply unit **130** is inserted into the access panel **1110**, and connected to the power converter **104** and the access panel wiring harness **1122**. As step E7, the window control unit **132**, the wireless transceiver **1112**, and the backup battery **222** are inserted into the access panel **1110** and connected to the power supply unit **130**. As step E8, the doorbell system **180** is installed within the building and connected to the access panel **1110** (e.g., wirelessly, with a wired connection, etc.). In some embodiments, the doorbell system **180** is an existing doorbell system within the building and does not need to be installed. It should be understood that the order of steps E1-E8 is only for explanation purposes. One or more of steps E1-E8 may be performed in a different order or simultaneously.

(482) Now, the installation of the assembly of the door frame **20** and the door **30** into the RO of the building will be described. As step F1, a bore hole is bored in the RO. As step F2, a free end of the frame wiring harness **1126** is inserted through the bore hole. As step F3, the assembly of the door

frame **20** and the door **30** are inserted into the RO and the door frame **20** is secured thereto (e.g., shimmed, nailed, caulked, etc.). As step F4, the door finishes and electronic devices are installed. More specifically, (a) the handle assembly **70** is installed within and to the door **30** and coupled to the door component wiring **178**, (b) the deadbolt units **138** are installed in the door **30** and coupled to the door component wiring **178**, (c) the interactive unit **140** is installed on the door **30** and coupled to the door component wiring **178** and/or the video doorbell **1118** is installed on the door frame **20** are coupled to the video doorbell wiring **1144** of the frame wiring harness **1126**, (d) the jamb lights **790** are installed within the channels of the side jambs **22** and/or the head jamb **24**, and coupled to the jamb light wiring **1128** of the frame wiring harness **1126**, and (e) the side jamb light covers **794** and/or the head jamb cover **796** are installed. As step F5, the access panel wiring harness **1122** and the frame wiring harness **1126** are coupled together to couple the assembly of the door frame **20** and the door **30** to the access panel **1110**. It should be understood that the order of steps F1-F5 is only for explanation purposes. One or more of steps F1-F5 may be performed in a different order or simultaneously. Further, one or more of the components outlined in step F4 may be installed earlier in the process described herein (e.g., such as during the assembly of the door frame **20**, during the assembly of the door **30**, etc.).

(483) As utilized herein with respect to numerical ranges, the terms “approximately,” “about,” “substantially,” and similar terms generally mean $\pm 10\%$ of the disclosed values. When the terms “approximately,” “about,” “substantially,” and similar terms are applied to a structural feature (e.g., to describe its shape, size, orientation, direction, etc.), these terms are meant to cover minor variations in structure that may result from, for example, the manufacturing or assembly process and are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(484) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(485) The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(486) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(487) The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit

(ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

(488) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(489) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(490) It is important to note that the construction and arrangement of the door system **10** and the door system **1100** as shown in the various exemplary embodiments are illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. For example, any element of the door system **10** can be incorporated into the door system **1100**, or vice versa. Although only one example from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

Claims

1. An electronic door system comprising: a door frame including a first side jamb, a second side jamb, a head jamb, and a sill, and wherein at least one of the first side jamb or the second side jamb defines a light channel extending substantially along an entire longitudinal length thereof; a frame wiring harness extending through the door frame; a light strip disposed within and extending along the light channel, a portion of the light strip extending into the head jamb and connected to the frame wiring harness; and a light cover enclosing the light strip within the light channel; a door hingedly coupled to the door frame, the door defining a channel extending along at least a portion of a peripheral edge thereof; an electrically-controllable component coupled to or disposed within the door; a door wiring harness including a wire extending along and within the channel, the wire coupled to the electrically-controllable component; a power transfer assembly positioned between the door frame and the door, the power transfer assembly electrically coupling the frame wiring harness to the door wiring harness; and a cover extending along the peripheral edge and covering the channel.
2. The electronic door system of claim 1, wherein the door defines a window aperture, further comprising a window disposed within the window aperture, wherein the window is the electrically-controllable component, and wherein the window is a controllable privacy glass or a transparent display.
3. The electronic door system of claim 2, wherein the channel defines a wire aperture that connects the channel to the window aperture, and wherein the wire extends through the wire aperture to the window.
4. The electronic door system of claim 3, further comprising a raceway disposed within the channel, wherein the raceway includes a base secured within the channel and the cover, wherein the wire aperture is a first wire aperture, wherein the base defines a wire volume and a second wire aperture that aligns with the first wire aperture, wherein the wire extends through the wire volume, the second wire aperture, and the first wire aperture to the window, and wherein the cover releasably engages with the base to enclose the wire volume.
5. The electronic door system of claim 1, further comprising: one or more sensors configured to facilitate determining a location of a user relative to the door; and a controller configured to make a door access decision at least partially based on the location of the user indicating that the user is outside and not inside of a building within which the electronic door system is installed.
6. The electronic door system of claim 5, wherein the one or more sensors include a plurality of ultra-wideband sensors configured to facilitate triangulating a device location of a user device carried by the user and, thereby, the location of the user.
7. The electronic door system of claim 5, wherein the controller is configured to make the door access decision based on (a) the location indicating that the user is outside and not inside of the building and (b) an access intent of the user.
8. The electronic door system of claim 7, further comprising at least one of (a) a radar sensor or (b) a capacitive touch doorknob or handle, wherein the controller is configured to determine the access intent of the user based on a signal received from the at least one of (a) the radar sensor or (b) the capacitive touch doorknob or handle.
9. The electronic door system of claim 1, further comprising: a doorknob or handle coupled to the door, the doorknob or handle including a haptic component; and a controller configured to activate the haptic component according to one or more haptic profiles to provide feedback or information to a user based on a current situation proximate the door.
10. The electronic door system of claim 1, further comprising: one or more sensors; and a controller configured to: detect a presence of a user proximate the door via the one or more sensors; detect a gesture performed by the user via the one or more sensors; and control a function of the

electronic door system based at least on the presence and the gesture.

11. The electronic door system of claim 1, wherein the electrically-controllable component is a first electrically-controllable component, further comprising: a second electrically-controllable component coupled to or disposed within the door frame or the door; a backup battery configured to power the first electrically-controllable component and the second electrically-controllable component in response to a mains power outage; and a controller configured to: provide a notification to a user device associated with the electronic door system in response to the mains power outage; receive a response from the user device based on the notification, the response including at least one of (a) a hierarchy for the first electrically-controllable component and the second electrically-controllable component or (b) a time limit for the electronic door system to remain at least partially functional using the backup battery; and evaluate a remaining charge of the backup battery; and selectively deactivate the first electrically-controllable component and the second electrically-controllable component based on the at least one of the hierarchy or the time limit.

12. The electronic door system of claim 1, wherein: the electronic door system is configured to integrate with a building system of a building within which the electronic door system is installed; the building system includes at least one of a lighting control system, an alarm system, a heating, ventilation, and air conditioning (HVAC) system, or a water monitoring and control system; and in response to the electronic door system entering an away mode, a controller of the electronic door system is configured to at least one of: communicate with the lighting control system to turn off one or more lights within the building that are on when the away mode is activated; communicate with the alarm system to enter the alarm system into an active mode; communicate with the HVAC system to engage away temperature control operations; or communicate with the water monitoring and control system to engage away water monitoring and control operations.

13. The electronic door system of claim 1, further comprising a controller configured to control the electrically-operated component to provide a visual or audible message tailored for a specific user when the specific a user walks through the door.

14. The electronic door system of claim 1, further comprising a controller configured to provide a notification to a user device associated with a user when another user walks through the door where the notification is tailored based on the other user.

15. The electronic door system of claim 1, further comprising a controller configured to: acquire temperature data regarding an ambient environment; and at least one of (a) provide a notification a user to turn off a water supply to water spickets of a building in which the electronic door system is installed or (b) automatically turn off the water supply in response to the temperature being less than a temperature threshold.

16. The electronic door system of claim 1, further comprising a controller configured to: detect inclement weather; and provide a notification to a user to close any open windows of a building in which the electronic door system is installed.

17. The electronic door system of claim 1, further comprising a controller configured to: monitor and learn a schedule of a user; and provide a notification to the user or another user in response to the schedule deviating.

18. The electronic door system of claim 1, further comprising a controller configured to: monitor an event external to the door over time to learn a schedule for the event; and at least one of (a) provide a notification to a user regarding a deviation from the schedule or (b) contact an organization associated with the event in response to the deviation.

19. The electronic door system of claim 1, further comprising a controller configured to: monitor an area outside of the door; detect a delivery person delivering a package; and disable a doorbell chime function in response to detecting the delivery person delivering the package.

20. The electronic door system of claim 19, wherein the controller is configured to maintain the doorbell chime function in response to detecting the delivery person with a signature device

indicating that the package needs to be signed for.

21. The electronic door system of claim 1, further comprising a controller configured to open the door, open a pet door of the door, or provide a notification to an owner of a pet in response to detecting the pet at the door and determining that the pet desires to be let inside through an action other than being present at the door.

22. The electronic door system of claim 1, further comprising: external lights; and a controller configured to: detect knocking at the door or movement of the door; and alter a lighting characteristic of the external lights based on a characteristic of the knocking or the movement.

23. The electronic door system of claim 1, further comprising a control configured to: identify that all occupants of a building within which the electronic door system is installed have left the building; and automatically engage an away mode in response to the identification.

24. An electronic door system comprising: a door frame; a frame wiring harness extending through the door frame; a door hingedly coupled to the door frame, the door defining a channel extending along at least a portion of a peripheral edge thereof; an electrically-controllable component coupled to or disposed within the door; a door wiring harness including a wire extending along and within the channel, the wire coupled to the electrically-controllable component; a power transfer assembly positioned between the door frame and the door, the power transfer assembly electrically coupling the frame wiring harness to the door wiring harness; a cover extending along the peripheral edge and covering the channel; an access panel configured to be positioned in a location spaced from the door and the door frame; a power converter disposed within the access panel and configured to receive AC power from a mains power source and convert the AC power to DC power; a wireless transceiver disposed within the access panel; a backup battery disposed within the access panel; and a controller disposed within the access panel, the controller coupled to the power converter, the wireless transceiver, the backup battery, and the frame wiring harness.

25. The electronic door system of claim 24, further comprising a user interface (a) coupled to the door frame and electrically coupled to the frame wiring harness or (b) coupled to the door and electrically coupled to the door wiring harness.

26. The electronic door system of claim 25, wherein the user interface includes an external charge port to facilitate powering the user interface if the mains power source is unavailable and the backup battery is depleted.

27. The electronic door system of claim 24, further comprising: a reset interface disposed within the access panel; and one or more switches positioned to disconnect at least one of the controller or the electrically-controllable component from the mains power source and the backup battery, the one or more switches configured to open for a period of time in response to receiving an open signal; wherein the controller is configured to send the open signal to the one or more switches in response to a user engaging the reset interface.

28. An electronic door system comprising: a door frame including a first side jamb, a second side jamb, a head jamb, and a sill, wherein at least one of the first side jamb or the second side jamb defines a light channel extending substantially along an entire longitudinal length thereof; a frame wiring harness extending through the door frame; a light strip disposed within and extending along the light channel, a portion of the light strip extending into the head jamb and connected to the frame wiring harness; a light cover enclosing the light strip within the light channel; a door hingedly coupled to the door frame, the door defining a wire channel extending along at least a portion of a peripheral edge thereof; an electrically-controllable component coupled to or disposed within the door; a door wiring harness including one or more wires extending along and within the wire channel, the door wiring harness coupled to the frame wiring harness and the electrically-controllable component; a channel cover extending along the peripheral edge and covering the channel; an access panel configured to be positioned in a location spaced from the door and the door frame; a power converter disposed within the access panel, the power converter configured to receive AC power from a mains power source and convert the AC power to DC power; a backup

battery disposed within the access panel; a power management controller disposed within the access panel, the power management controller coupled to the power converter and the backup battery; and an access panel wiring harness coupled to the power management controller and the frame wiring harness.
